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(10) **Pub. No.: US 2025/0276953 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **SMALL MOLECULE SIRTUIN INHIBITORS
AND USES THEREOF**(71) Applicant: **The Regents of the University of
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29, 2022.**Publication Classification**(51) **Int. Cl.****C07D 277/42** (2006.01)**A61K 47/55** (2017.01)**C07D 277/48** (2006.01)**C07F 9/6539** (2006.01)(52) **U.S. Cl.**CPC **C07D 277/42** (2013.01); **A61K 47/55**(2017.08); **C07D 277/48** (2013.01); **C07F****9/6539** (2013.01)

(57)

ABSTRACT

This invention is in the field of medicinal chemistry. In particular, the invention relates to a new class of small-molecules having a 2-hydroxybenzoic acid structure which function as sirtuin (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7) inhibitors and/or degraders which function as effective therapeutic agents for treating, ameliorating, and preventing disorders associated with sirtuin activity (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer). In addition, this invention also relates to a new class of proteolysis-targeting chimeras (PROTACs) (as defined herein) which function as sirtuin inhibitors and/or degraders within cancer and/or immune cells. Pharmaceutical compositions comprising said compounds are also within the scope of the present invention.

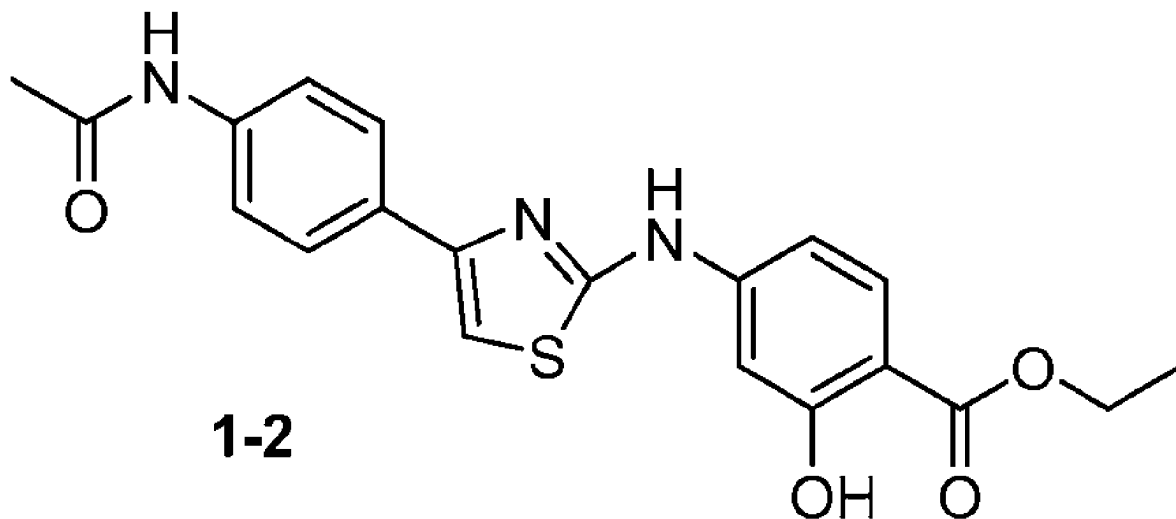


FIG. 1A

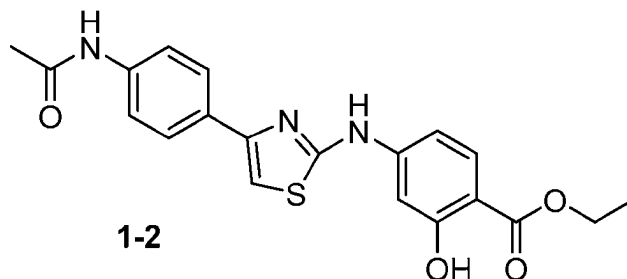


FIG. 1B

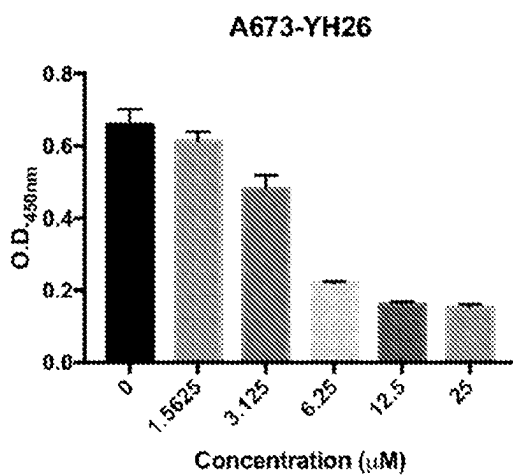
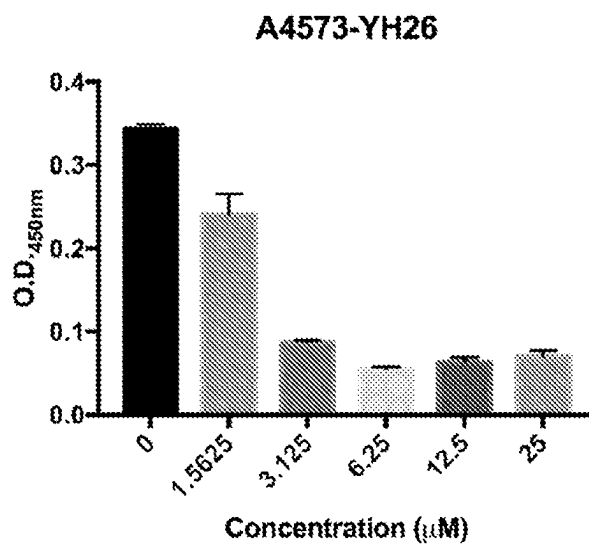


FIG. 2A

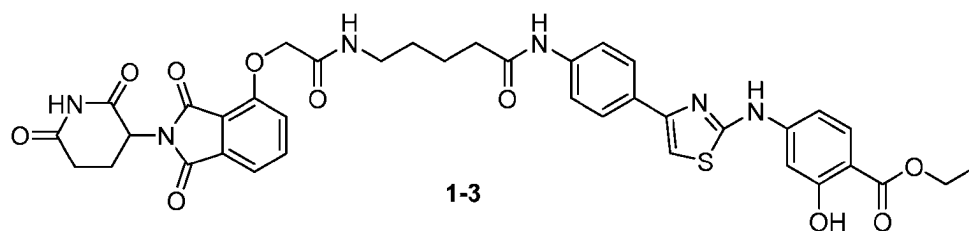


FIG. 2B

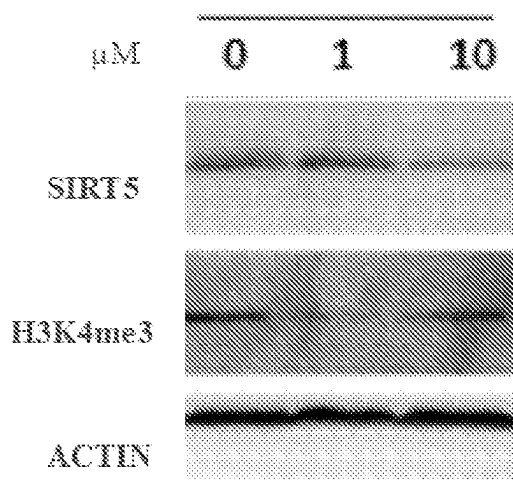
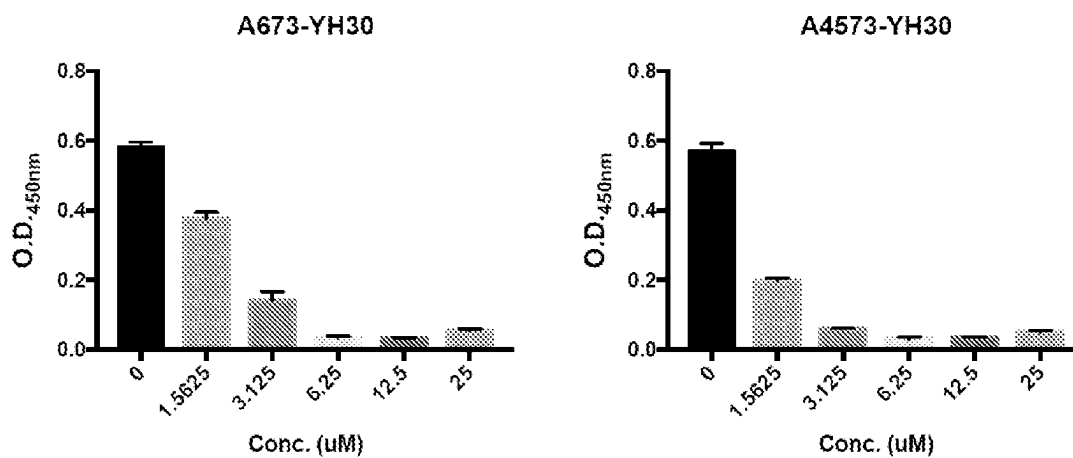


FIG. 2C



SMALL MOLECULE SIRTUIN INHIBITORS AND USES THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present invention claims the priority benefit of U.S. Provisional Patent Application 63/336,666, filed Apr. 29, 2022, which is hereby incorporated by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] This invention was made with government support under W81XWH-18-1-0567 awarded by the U.S. Army Medical Research and Development Command. The government has certain rights in the invention.

FIELD OF THE INVENTION

[0003] This invention is in the field of medicinal chemistry. In particular, the invention relates to a new class of small-molecules having a 2-hydroxybenzoic acid structure which function as sirtuin (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7) inhibitors and/or degraders which function as effective therapeutic agents for treating, ameliorating, and preventing disorders associated with sirtuin activity (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer). In addition, this invention also relates to a new class of proteolysis-targeting chimeras (PROTACs) (as defined herein) which function as sirtuin inhibitors and/or degraders within cancer and/or immune cells. Pharmaceutical compositions comprising said compounds are also within the scope of the present invention.

INTRODUCTION

[0004] Sirtuins are known as nicotinamide adenine dinucleotide (NAD⁺)-dependent protein deacylases which are involved in many important biological functions, including the regulation of life span, transcription, DNA repair, protein secretion, and metabolism. There are seven sirtuin members encoded in mammalian genomes, SIRT1 through SIRT7, which are distinguished from another with respect to catalytic activities, subcellular localization, protein targets, and biological functions.

[0005] Sirtuin-family proteins are NAD⁺-dependent protein deacylases regulating metabolism and other diverse aspects of cell biology (Giblin et al., 2014). Three sirtuins (SIRT3, SIRT4, and SIRT5) reside primarily or exclusively in the mitochondrial matrix (Lombard et al., 2012). SIRT5 is an inefficient deacetylase, instead removing succinyl, malonyl, and glutaryl groups from lysines of its target proteins (referred to herein as Ksucc, Kmal, and Kglu, respectively) (Du et al., 2011; Park et al., 2013; Peng et al., 2011; Rardin et al., 2013; Tan et al., 2014). SIRT5 remains a poorly understood sirtuin, in part because most phenotypes of SIRT5 deficiency in normal cells and tissues are fairly modest. SIRT5 knockout (KO) mice are grossly unremarkable, fertile, and healthy until at least 18 months of age (Lombard et al., 2007). Germline SIRT5 KO mice show mild biochemical defects in response to a prolonged 48-hour fast (Nakagawa et al., 2009; Nishida et al., 2015; Rardin et al., 2013; Yu et al., 2013), and mild cardiac defects with advancing age (Sadhukhan et al., 2016). In contrast, SIRT5

KO mice actually show modest protection from the negative consequences of high fat diet (Yu et al., 2013).

[0006] Like other sirtuins, SIRT5 has been implicated in neoplasia, as both an oncogene and a tumor suppressor, in a context-specific manner (Kumar et al., 2018). As an oncogene, SIRT5 promotes folate metabolism via activation of mitochondrial serine hydroxymethyltransferase (SHMT2), facilitating cancer cell growth in vitro and in vivo (Yang et al., 2018). Folate metabolism is a target of several approved chemotherapy drugs. Likewise, SIRT5 inhibits pyruvate kinase muscle isozyme 2 (PKM2), resulting in accumulation of glycolytic intermediates, driving xenograft growth (Xi-angyun et al., 2017). In colorectal cancer, SIRT5 promotes entry of glutamine into the TCA cycle by activating glutamate dehydrogenase 1 (GLUD1) (Wang et al., 2018). Additionally, SIRT5 desuccinylates citrate synthase (CS), the rate-limiting enzyme in the TCA cycle, promoting its activity (Ren et al., 2020). CS hyl ersuccinylation inhibits its function and suppresses colorectal cancer cell proliferation and migration (Ren et al., 2020). In breast cancer, SIRT5 regulates glutamine metabolism by desuccinylating glutaminase (GLS), protecting it from ubiquitin-mediated degradation (Greene et al., 2019). In melanoma, SIRT5 is required to maintain histone acetylation and methylation to promote expression of key genes, including MITF, a lineage-specific oncogene, and c-MYC (Giblin et al., 2021). Likewise, recent studies have documented oncogenic roles for SIRT5 in breast cancer and in AML (Yan et al., 2021; Abril et al., 2021).

[0007] Conversely, as a tumor suppressor, SIRT5 maintains fatty acid oxidation and redox homeostasis by inhibiting dimerization of acyl-CoA oxidase 1 (ACOX1), attenuating its function (Chen et al., 2018). Consistently, in hepatocellular carcinoma, low SIRT5 expression is associated with increased ACOX1 succinylation and activity (Chen et al., 2018). In AML, GBM, and certain other cancer types, isocitrate dehydrogenase (IDH) gain-of-function mutants convert α -ketoglutarate (α -KG) into the oncometabolite R-2-hydroxyglutarate (R-2HG), which in turn inhibits α -KG-dependent enzymes, including DNA and histone demethylases, thereby inducing epigenetic dysregulation (Kumar et al., 2018). Ectopic expression of SIRT5 reverses R-2HG-induced resistance to apoptosis in IDH1 mutant glioma cells, impairing their growth (Li et al., 2015). Recently, Hu and colleagues characterized a novel tumor suppressor function of SIRT5 in PDAC (Hu et al., 2021).

[0008] Proteolysis targeting chimeras (PROTACs) are a class of heterobifunctional molecules with two recruiting ligands connected via a linker. One ligand can recruit an E3 ubiquitin ligase while the other ligand specifically recruits the protein of interest (PoI). Upon ternary complex formation, PROTACs can recruit the E3 ligase to PoI in a spatially favorable presentation for substrate poly-ubiquitination, leading to subsequent degradation of PoI by the proteasome. In contrast to occupancy-based pharmacology, PROTACs technology offers several advantages. For example, it needs lower concentration than those occupancy-based inhibitors to achieve equivalent therapeutic effects. Besides, PROTACs can be used as a novel approach to target so called "undruggable" proteins, such as transcription factors, scaffolding proteins, and non-enzymatic proteins (Toure and Crews, 2016).

[0009] Therefore, SIRT5 is a new target for cancer therapy. SIRT5 inhibitors or degraders can be used as a single

agent and in combination to treat various cancers as well as other diseases related to SIRT5 dysregulation.

[0010] New compounds capable of inhibiting SIRT5 activity or degrading SIRT5 are needed.

[0011] The present invention address this need.

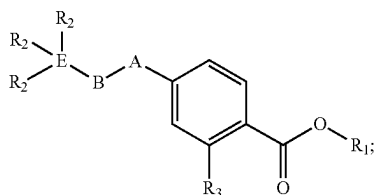
SUMMARY

[0012] Experiments conducted during the course of developing embodiments for the present invention involved design and discovery of novel 2-hydroxybenzoic acid derivatives as inhibitors and degraders of sirtuins. These compounds may be useful in the treatment of disorders associated with sirtuins, for example, cancer (melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer).

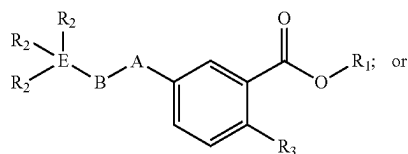
[0013] Accordingly, the present invention provides a new class of small-molecules having a 2-hydroxybenzoic acid structure which function as sirtuin (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7) inhibitors and/or degraders which function as effective therapeutic agents for treating, ameliorating, and preventing disorders associated with sirtuin activity (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer). In addition, this invention also relates to a new class of proteolysis-targeting chimeras (PROTACs) (as defined herein) which function as sirtuin inhibitors and/or degraders within cancer and/or immune cells. Pharmaceutical compositions comprising said compounds are also within the scope of the present invention.

[0014] In a particular embodiment, compounds encompassed within Formula (1), Formula (2), or Formula (3) are provided:

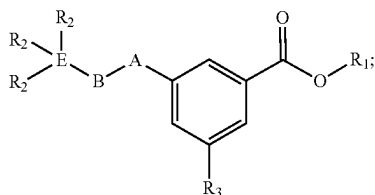
(Formula 1)



(Formula 2)



(Formula 3)



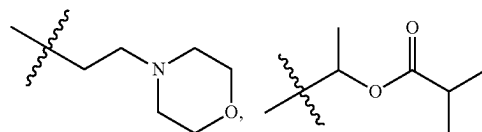
including pharmaceutically acceptable salts, solvates, PROTACs, and/or prodrugs thereof.

[0015] Formulas (1), (2), and (3) are not limited to a particular chemical moiety for R1, R2, R3, A, B and E. In some embodiments, the particular chemical moiety for R1,

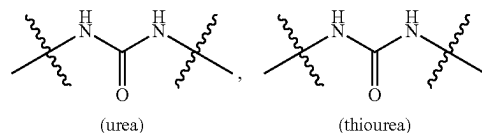
R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade sirtuin activity (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7). In some embodiments, the particular chemical moiety for R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade one or more of SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7. In some embodiments, the particular chemical moiety for R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade SIRT5 activity. In some embodiments, the particular chemical moiety for R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade sirtuin (e.g., SIRT5) related desuccinylase activity. In some embodiments, the particular chemical moiety for R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade sirtuin (e.g., SIRT5) related demalonylase activity. In some embodiments, the particular chemical moiety for R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade sirtuin (e.g., SIRT5) related deglutarylase activity. In some embodiments, the particular chemical moiety for R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound to influence multiple cellular pathways related to sirtuin activity (e.g., SIRT5) such as ammonia detoxification, fatty acid oxidation, cellular respiration, ketone body formation, tricarboxylic acid cycle (TCA), glycolysis and reactive oxygen species (ROS) metabolism. In some embodiments, the particular chemical moiety for R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound to serve as a therapeutic for treating, preventing and/or ameliorating disorders characterized with sirtuin (e.g., SIRT5) activity (e.g., sirtuin related cancer (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer).

[0016] In some embodiments, R¹ is selected from the group consisting of hydrogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, aryl, heteroaryl, arylalkyl, and heteroarylalkyl.

[0017] In some embodiments, R¹ is selected from the group consisting of hydrogen, methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, pentyl,



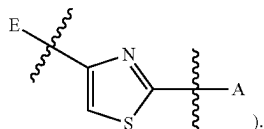
[0018] In some embodiments, A is selected from the group consisting of NH, CH₂, O, S,



amide, and sulfonamide.

[0019] In some embodiments, B is an aromatic ring.

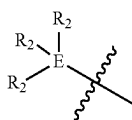
[0020] In some embodiments, B is a thiazole ring (e.g.



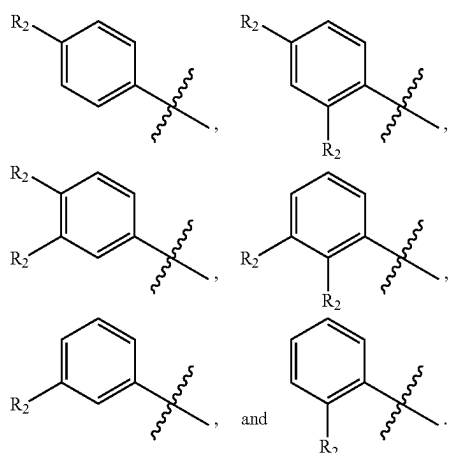
[0021] In some embodiments, E is an aromatic ring.

[0022] In some embodiments, E is a benzene ring.

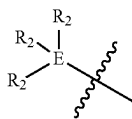
[0023] In some embodiments,



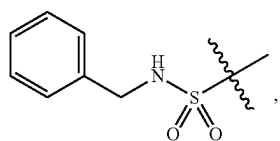
is selected from:



[0024] In some embodiments, each R² within

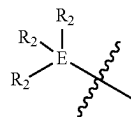


is independently selected from the group consisting of hydrogen, halogen, (e.g., F, Cl, Br, I), alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, halogen, aryl, heteroaryl, arylalkyl, heteroarylalkyl, nitril, cyano, amide or sulfonamide,

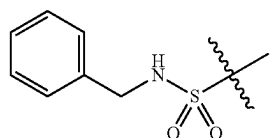


and triphenylphosphine (TPP) attached with linker; and/or

[0025] wherein each R² within



is independently selected from the group consisting of hydrogen, halogen, CF₃, OCH₃, OH, NO₂ (nitril), CN (cyano), amide (e.g., NHC(O)CH₃), and sulfonamide (e.g., S(O₂)NHCH₃),

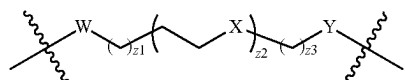


and triphenylphosphine (TPP) group attached with a linker.

[0026] In some embodiments, linker is covalently bonded to connect two parts; the carbon atom in the linear chain can be substituted with oxygen, nitrogen, sulfur, ester, and amide.

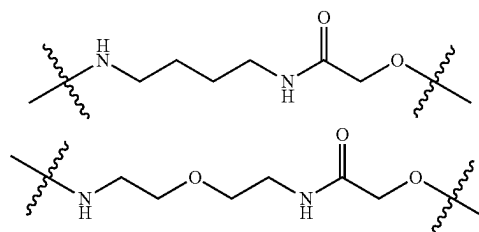
[0027] In some embodiments, linker is of following Formula (L0):

Formula (L0)

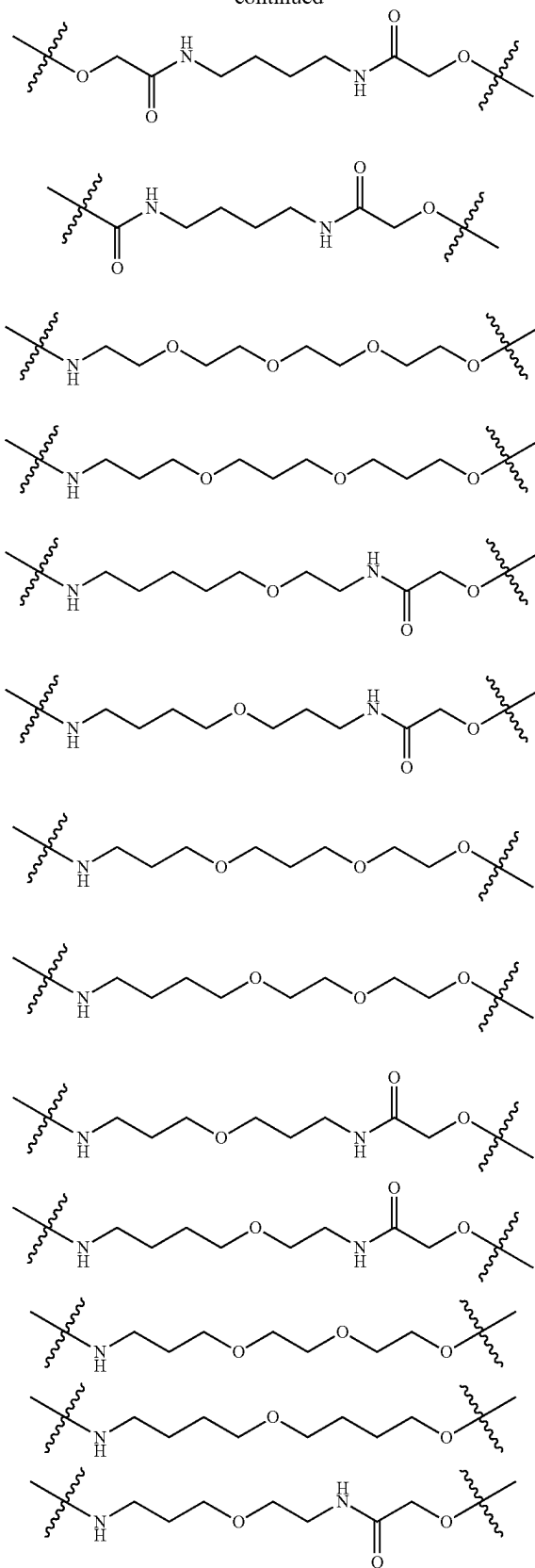


or an enantiomer, diastereomer, or stereoisomer thereof, wherein z₁ is an integer selected from 0 to 10; z₂ is an integer selected from 0 to 10; z₃ is an integer selected from 0 to 10; each X is independently absent, CH₂, O, S, NH, NR₁₃; wherein W is selected from absent, O, NH, NR₁₃, —OCH₂C(O)NH—, —CH₂CH₂C(O)NH—, —CH₂C(O)NH— or —C(O)NH—; wherein Y is absent, O, NH, NR₁₃, —OCH₂C(O)NH—, —CH₂CH₂C(O)NH—, —CH₂C(O)NH— or —C(O)NH—; and each R₁₃ is independently C₁-C₃ alkyl. The linker can be substituted with an alkyl, halide, phenyl, benzyl, aryl, alkylene or heterocycle group.

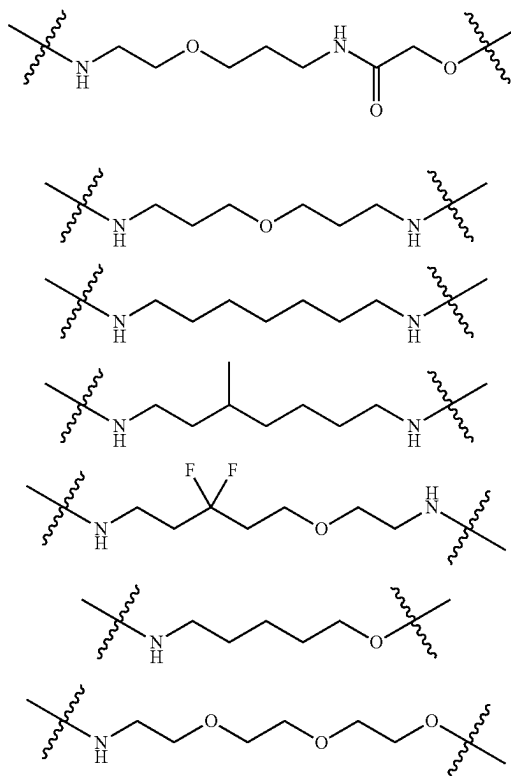
[0028] In some embodiments, linker is one of the following moieties:



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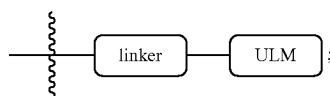


-continued



[0029] In some embodiments, R^2 is represented by Formula (P0):

Formula (P0)



wherein the linker is covalently bonded to connect two parts; wherein the carbon atom in the linear chain can be substituted with oxygen, nitrogen, sulfur, ester, and amide; wherein ULM represents an E3 ubiquitin ligase binding moiety that binds E3 ubiquitin ligase selected from the group consisting of pomalidomide, thalidomide, lenalidomide, Von Hippel-Lindau (VHL), inhibitors of apoptosis proteins (LAP), Cereblon, and mouse double minute 2 (MDM2).

[0030] In some embodiments, R^3 is hydrogen or OH.

[0031] In some embodiments, prodrug is preferably carboxylic acid ester. The ester group masks the negative charge on the carboxylate and thus can increase the cell permeability of the compounds. Inside the cells, the ester group can be hydrolyzed to release the negatively charged carboxylate, which can inhibit SIRT5.

[0032] In some embodiments, compounds shown in Table I are contemplated for Formula (1). Table I additionally shows values for SIRT5 IC_{50} for select compounds.

TABLE I

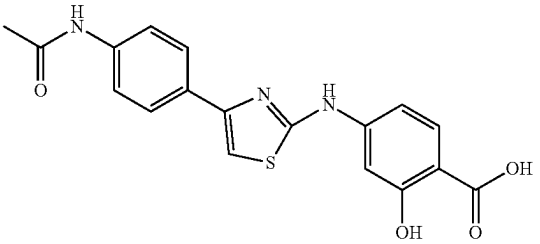
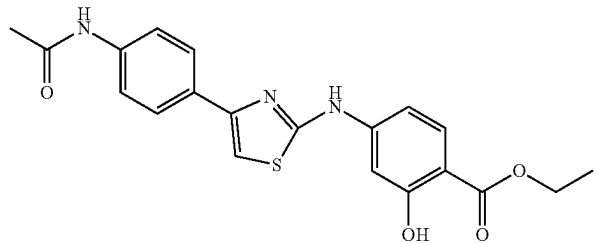
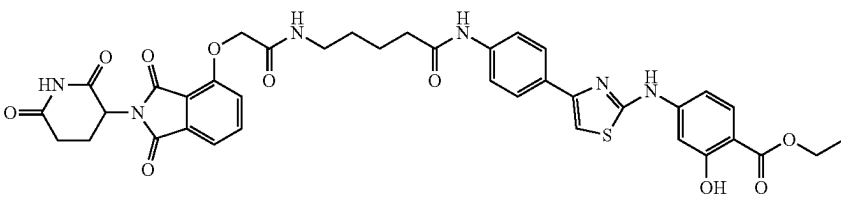
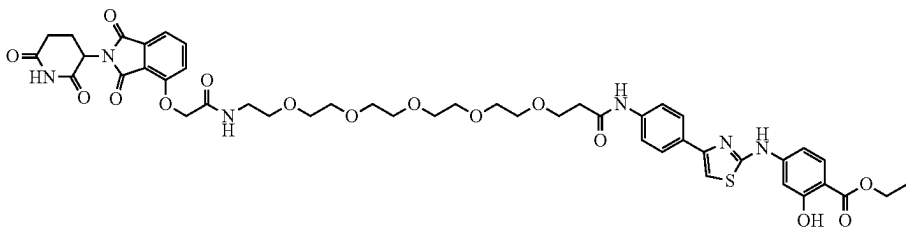
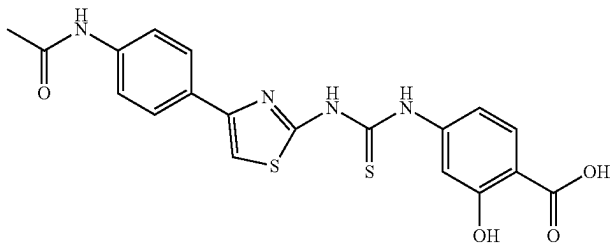
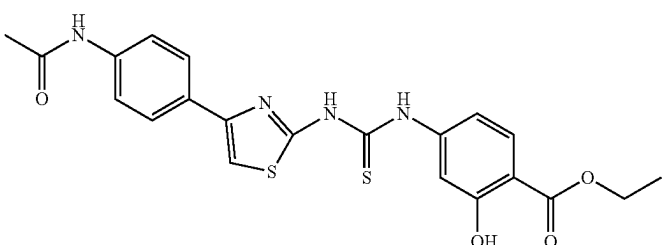
Compound	Structure	SIRT5 IC ₅₀ (μ M)
1-1		26.45 $\pm$ 0.83
1-2		NA
1-3		NA
1-4		NA
1-5		12.43 $\pm$ 0.58
1-6		NA

TABLE I-continued

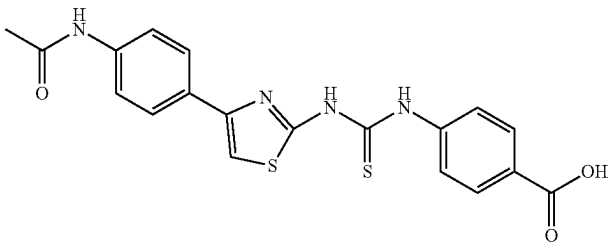
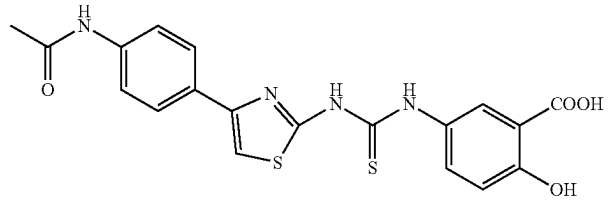
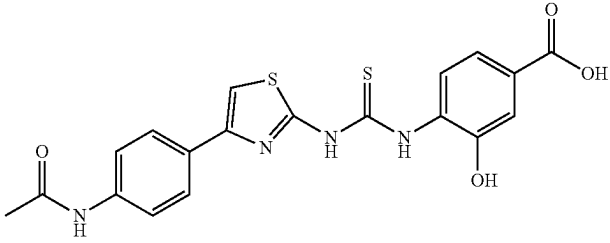
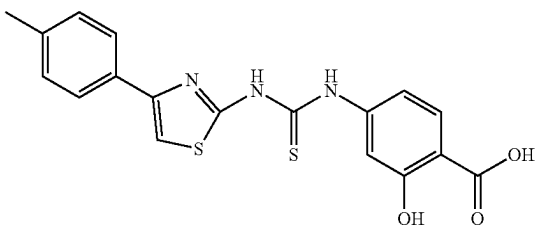
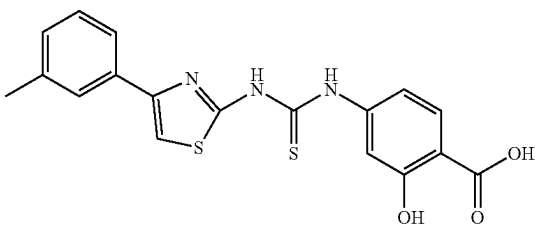
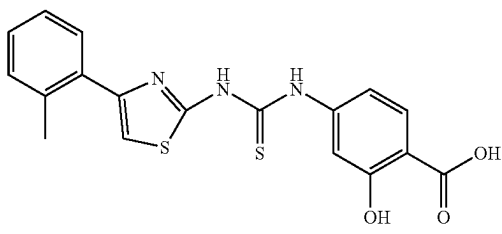
Compound	Structure	SIRT5 IC ₅₀ (μ M)
1-7		101.2 $\pm$ 8.48
1-8		55.94 $\pm$ 5.42
1-9		50.72 $\pm$ 5.64
1-10		23.45 $\pm$ 0.92
1-11		27.74 $\pm$ 1.02
1-12		34.05 $\pm$ 2.18

TABLE I-continued

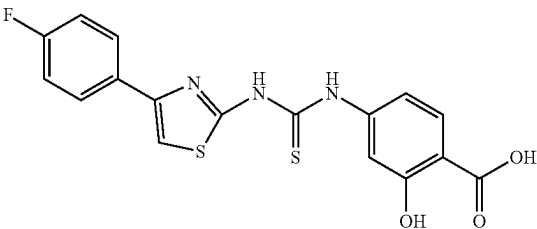
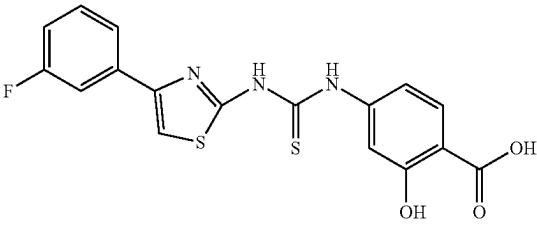
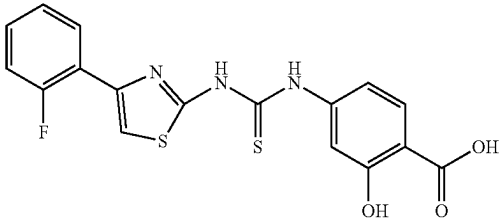
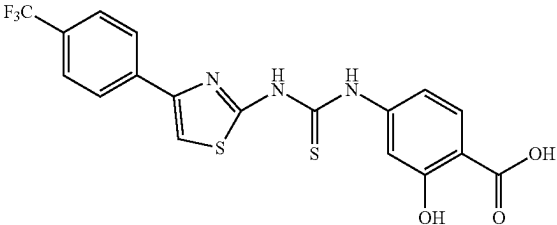
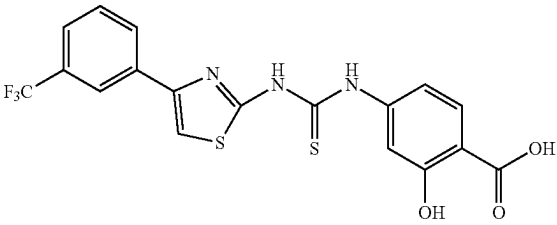
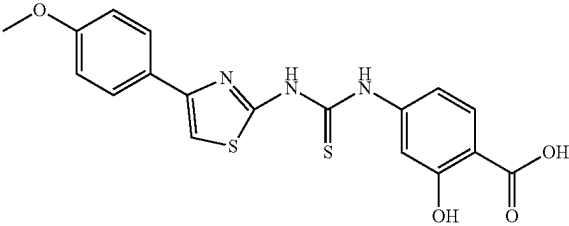
Compound	Structure	SIRT5 IC ₅₀ (μ M)
1-13		17.26 $\pm$ 0.28
1-14		18.34 $\pm$ 0.95
1-15		26.22 $\pm$ 2.54
1-16		26.84 $\pm$ 1.42
1-17		24.27 $\pm$ 2.54
1-18		29.24 $\pm$ 2.88

TABLE I-continued

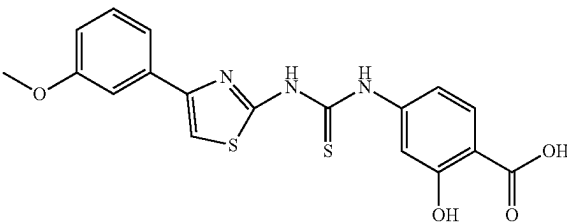
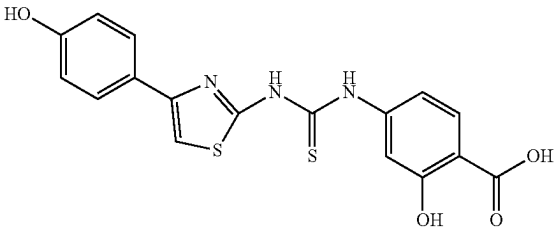
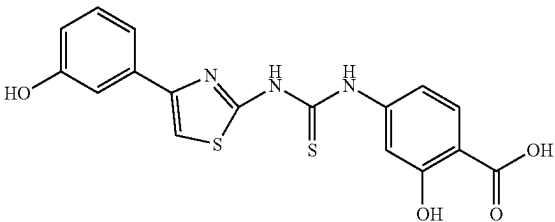
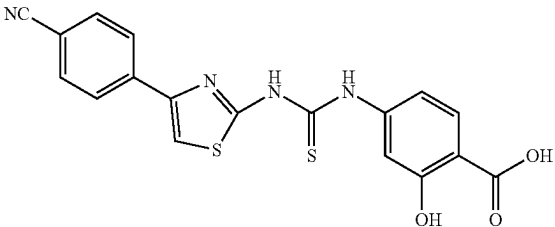
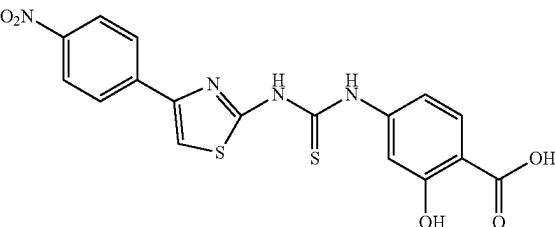
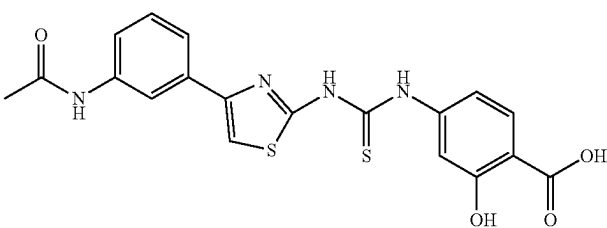
Compound	Structure	SIRT5 IC ₅₀ (μ M)
1-19		41.62 $\pm$ 3.79
1-20		42.54 $\pm$ 3.83
1-21		32.79 $\pm$ 3.87
1-22		11.42 $\pm$ 0.50
1-23		4.3 $\pm$ 0.30
1-24		35.58 $\pm$ 1.30

TABLE I-continued

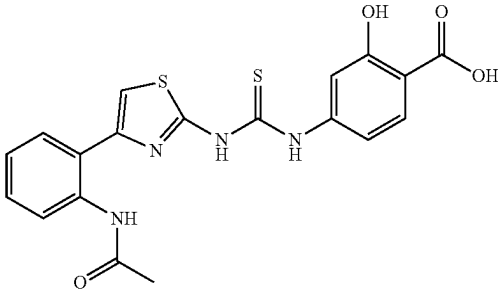
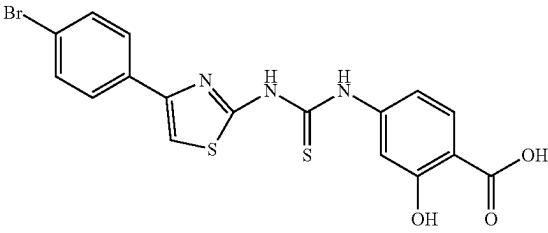
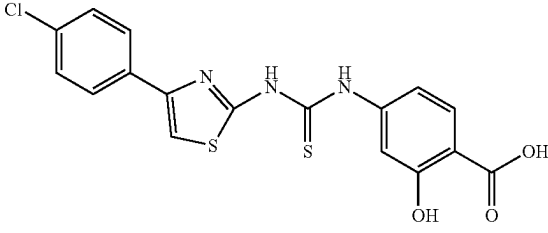
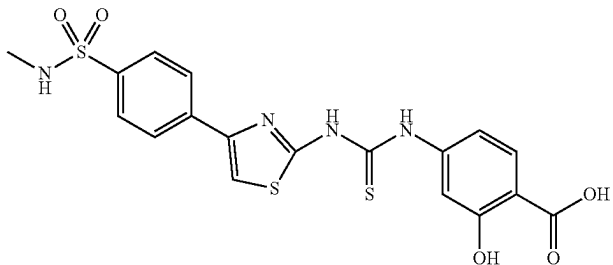
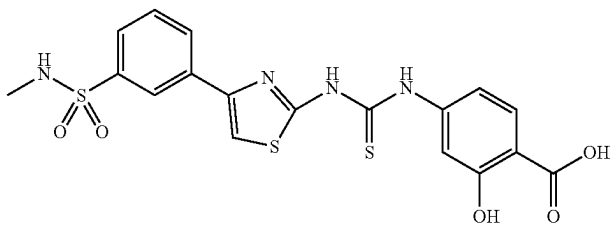
Compound	Structure	SIRT5 IC ₅₀ (μ M)
1-25		55.64 $\pm$ 5.49
1-26		18.56 $\pm$ 0.23
1-27		18.41 $\pm$ 0.28
1-28		38.84 $\pm$ 3.13
1-29		26.14 $\pm$ 1.91

TABLE I-continued

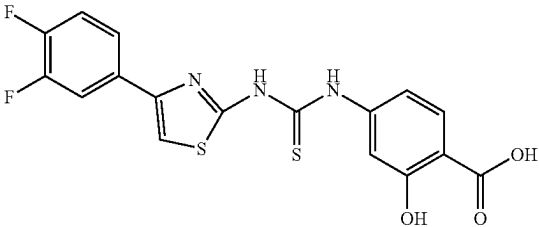
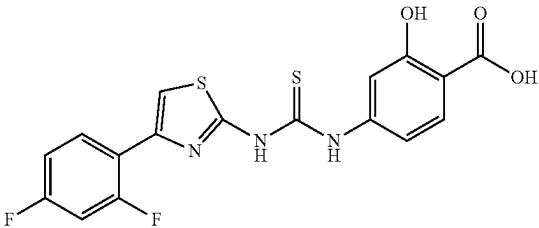
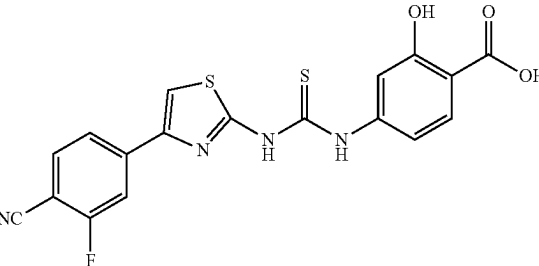
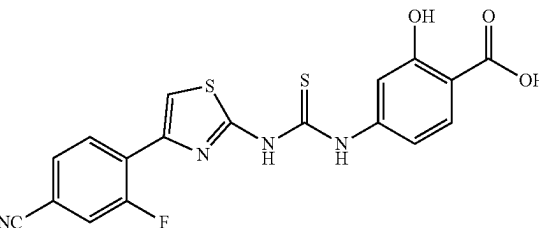
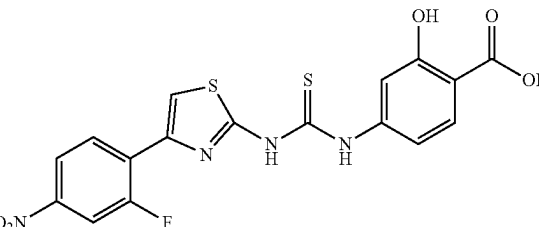
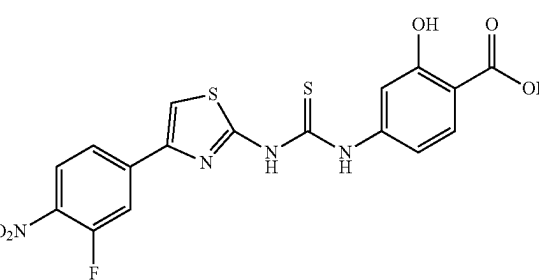
Compound	Structure	SIRT5 IC ₅₀ (μ M)
1-30		16.14 $\pm$ 0.73
1-31		19.32 $\pm$ 0.93
1-32		12.78 $\pm$ 0.48
1-33		14.79 $\pm$ 0.81
1-34		23.76 $\pm$ 2.71
1-35		14.93 $\pm$ 1.35

TABLE I-continued

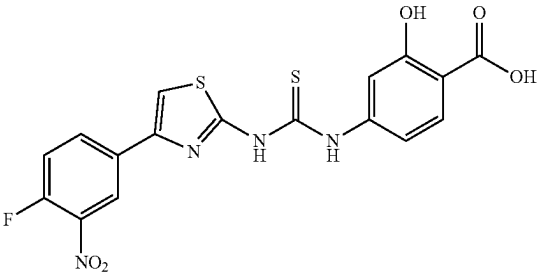
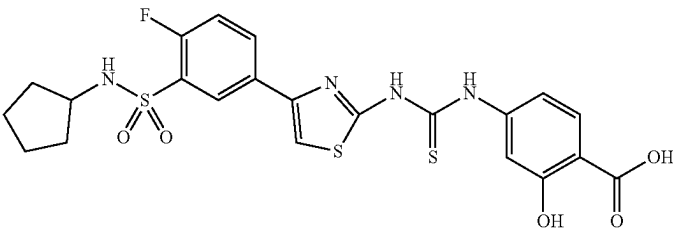
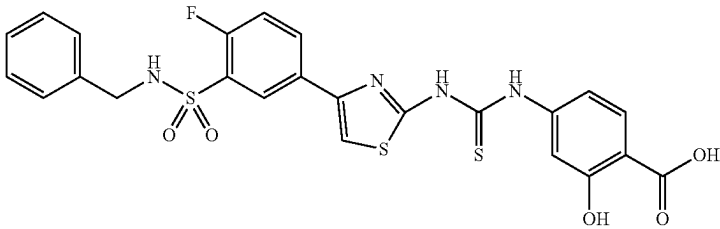
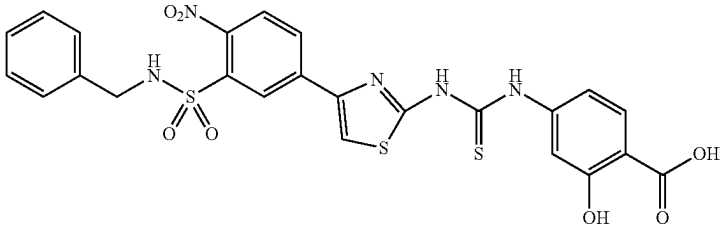
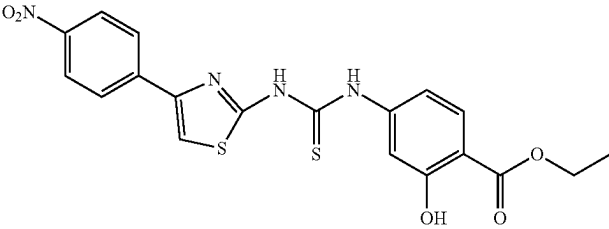
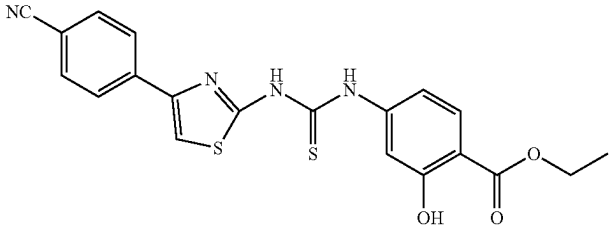
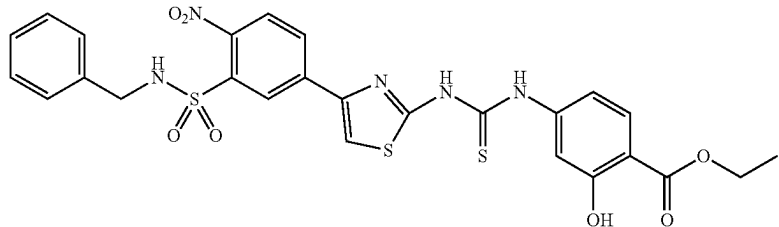
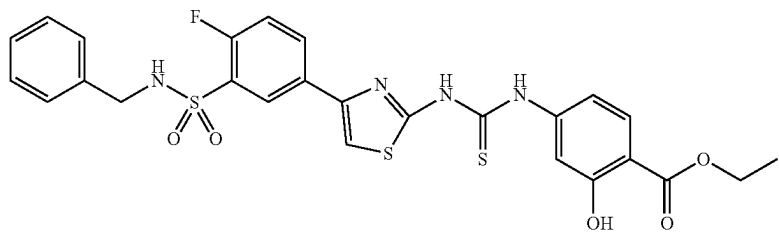
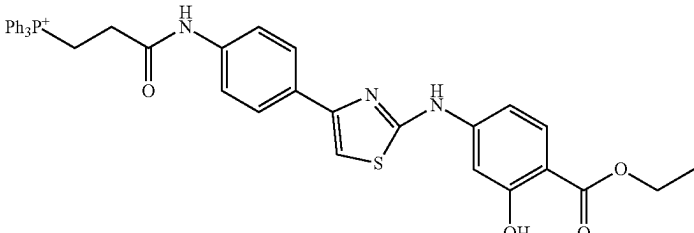
Compound	Structure	SIRT5 IC ₅₀ (μ M)
1-36		11.58 $\pm$ 0.74
1-37		24.86 $\pm$ 0.54
1-38		8.22 $\pm$ 1.3
1-39		2.5 $\pm$ 0.21
1-40		NA
1-41		NA

TABLE I-continued

Compound	Structure	SIRT5 IC ₅₀ (μM)
1-42		NA
1-43		NA
1-44		NA

[0033] The invention further provides processes for preparing any of the compounds of the present invention.

[0034] The invention also provides the use of compounds to not only inhibit sirtuin activity but also signaling pathways dependent upon or related to sirtuins. The invention also relates to the use of compounds for sensitizing cells to additional agent(s), such as agents known to be effective in the treatment of disorders related to sirtuin activity (e.g., cancer).

[0035] The compounds of the invention are useful for the treatment, amelioration, or prevention of disorders associated with sirtuin (e.g., SIRT5) activity (e.g., cancer), such as those responsive to sirtuin (e.g., SIRT5) activity inhibition (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer).

[0036] In certain embodiments, the present invention provides methods of treating, ameliorating, or preventing a disorder related to sirtuin (e.g., SIRT5) activity in a patient comprising administering to said patient a therapeutically effective amount of the pharmaceutical composition comprising a compound encompassed within Formulas 1, 2 or 3, and/or a compound recited in Table.

[0037] The invention also provides kits comprising a compound of the invention and instructions for administering the compound to an animal. The kits may optionally contain other therapeutic agents, e.g., agents useful in treating disorders related to sirtuin (e.g., SIRT5) activity (e.g., cancer).

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] FIG. 1A-B: (A) Structure of compound 1-2; (B) Compound 1-2 inhibited growth of Ewing sarcoma cell—A4573 and A673.

[0039] FIG. 2A-C: (A) Structures of compound 1-3; (B) Compound 1-3 degraded protein SIRT5; (C) Compound 1-3 inhibited growth of Ewing sarcoma cell lines A4573 and A673.

DEFINITIONS

[0040] For convenience, before further description of the present invention, certain terms employed in the specification, examples, and appended claims are described here. These definitions should be read in light of the entire invention and as would be understood by a person skilled in the art.

[0041] As used herein, the term “alkyl” means a saturated hydrocarbon radical having 1 to 15 carbon atoms, preferably 1 to 10 carbon atoms, more preferably 1 to 5 carbon atoms, most preferably 1 to 3 carbon atoms, that may be branched or unbranched. Non-limiting examples of alkyl radicals include methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, pentyl, hexyl, heptyl, octyl and the like, wherein methyl, ethyl, n-propyl and isopropyl represent specifically preferred examples.

[0042] As used herein, the term “cycloalkyl” refers to a saturated hydrocarbon ring that is not aromatic. Cycloalkyl

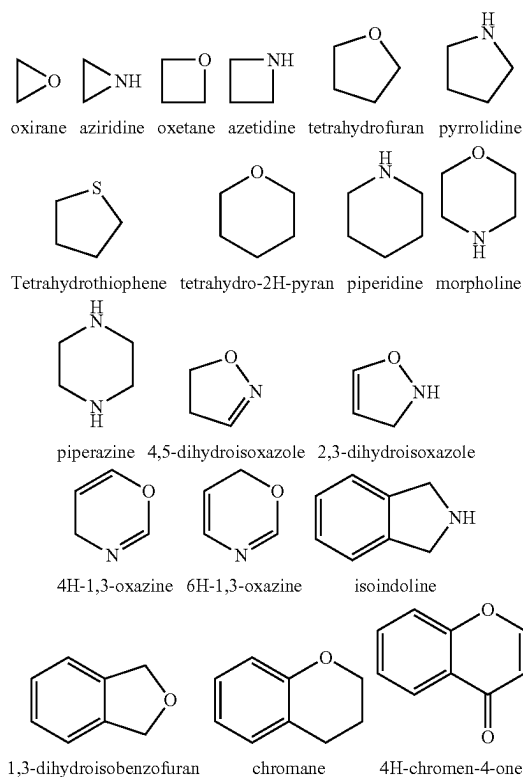
rings are monocyclic, or are fused, spiro, or bridged bicyclic or polycyclic ring systems. Monocyclic cycloalkyl rings contain from about 3 to about 12 carbon atoms, preferably from 3 to 7 carbon atoms, in the ring. Bicyclic cycloalkyl rings contain from 7 to 17 carbon atoms, preferably from 7 to 12 carbon atoms, in the ring. Preferred bicyclic cycloalkyl rings comprise 4-, 5-, 6- or 7-membered rings fused to 5-, 6- or 7-membered rings. Cycloalkyl rings may be unsubstituted or substituted with from 1 to 4 substituents on the ring. Cycloalkyl may be substituted with halo, cyano, alkyl, heteroalkyl, haloalkyl, phenyl, keto, hydroxy, carboxy, amino, acylamino, aryloxy, heteroaryloxy, or any combination thereof. Preferred cycloalkyl rings include cyclopropyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclooctyl and cyclononyl rings.

[0043] As used herein, the term “halo” or “halogen” is fluoro, chloro, bromo or iodo. Preferred halo are fluoro, chloro and bromo.

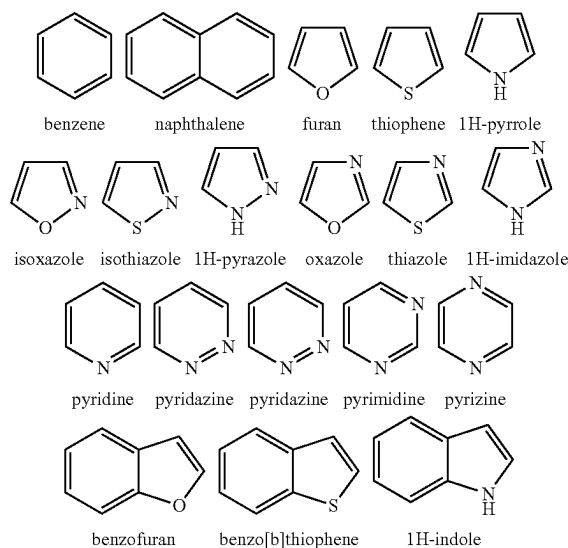
[0044] As used herein, the term “heteroatom” is a nitrogen, sulfur, or oxygen atom. Groups containing more than one heteroatom may contain different heteroatoms.

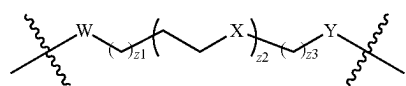
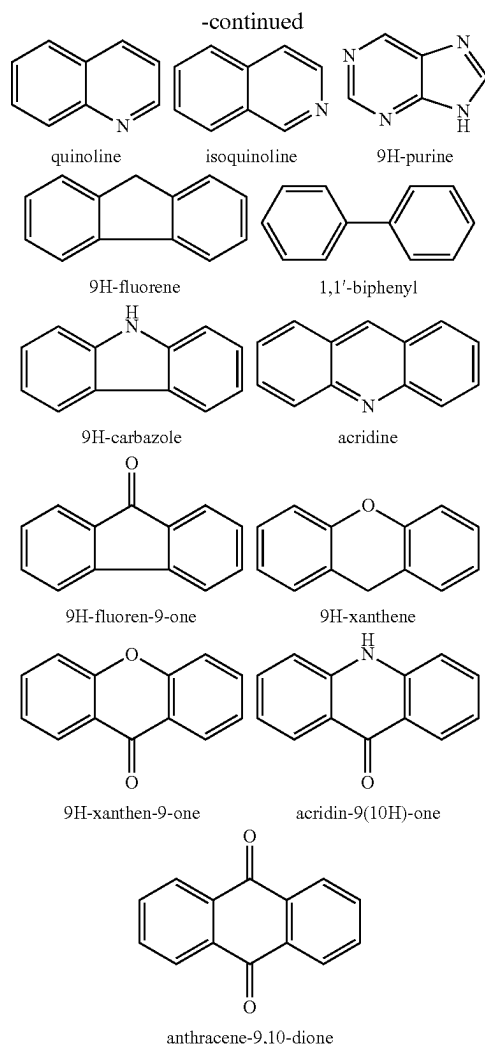
[0045] As used herein, the term “heteroalkyl” is a saturated or unsaturated chain carbon and at least one heteroatom, wherein no two heteroatoms are adjacent. Heteroalkyl chains contain from 2 to 15 member atoms (carbon and heteroatoms) in the chain, preferably 2 to 10, more preferably 2 to 5. For example, alkoxy (i.e., —O-alkyl or —O-heteroalkyl) radicals are included in heteroalkyl. Heteroalkyl chains may be straight or branched. Preferred branched heteroalkyl have one or two branches, preferably one branch. Preferred heteroalkyl are saturated. Unsaturated heteroalkyl have one or more carbon-carbon double bonds and/or one or more carbon-carbon triple bonds. Preferred unsaturated heteroalkyl have one or two carbon-carbon double bonds or one carbon-carbon triple bond, more preferably one double bond. Heteroalkyl chains may be unsubstituted or substituted with from 1 to 4 substituents. Preferred substituted heteroalkyl are mono-, di, or tri-substituted. Heteroalkyl may be substituted with halo, cyano, alkyl, heteroalkyl, haloalkyl, phenyl, keto, hydroxy, carboxy, amino, acylamino, aryloxy, heteroaryloxy, or any combination thereof.

[0046] As used herein, the term “heterocycloalkyl” is a saturated or unsaturated ring containing carbon atoms and from 1 to 4 (preferably 1 to 3) heteroatoms in the ring. Heterocycloalkyl rings are monocyclic, or are fused, spiro, or bridged bicyclic or polycyclic ring systems. Monocyclic heterocycloalkyl rings contain from about 3 to about 9 member atoms (including both carbons and heteroatoms), preferably from 5 to 7 member atoms, in the ring. Bicyclic heterocycloalkyl rings may be fused, spiro, or bridged ring systems. Preferred bicyclic heterocycloalkyl rings comprise 5-, 6- or 7-membered rings fused to 5-, 6- or 7-membered rings. Heterocycloalkyl rings may be unsubstituted (i.e., contain hydrogen) or substituted (on either carbons or heteroatoms or both) with from 1 to 4 substituents selected from halo, cyano, alkyl, heteroalkyl, haloalkyl, phenyl, keto, hydroxy, carboxy, amino, acylamino, aryloxy, heteroaryloxy, or any combination thereof. Preferred heterocycloalkyl rings include, but are not limited to, any of the following:



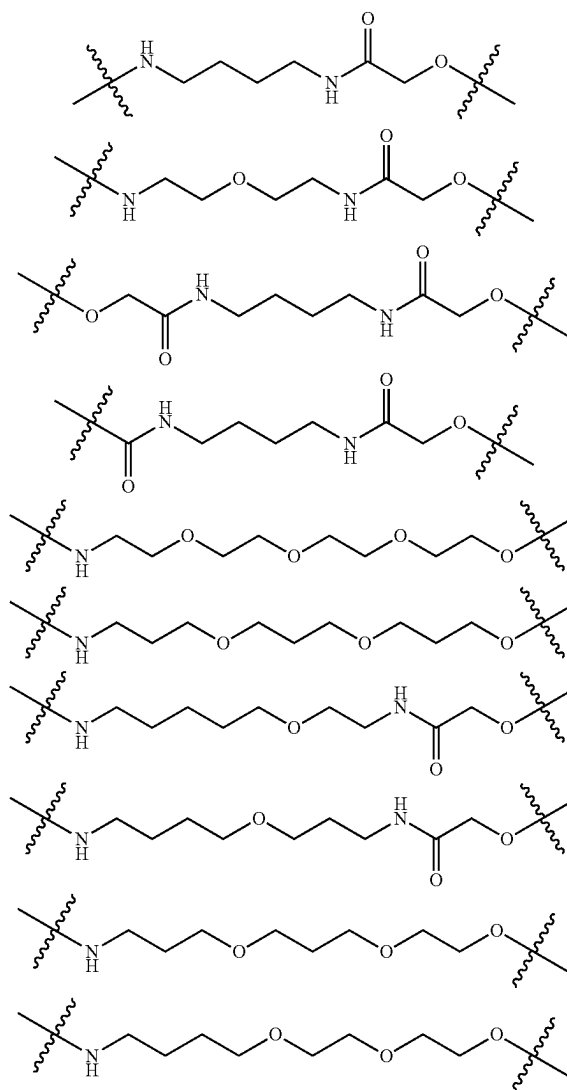
[0047] As used herein, the term “aromatic rings” include aromatic hydrocarbon rings and heteroaromatic rings. Aromatic rings may be unsubstituted or substituted with from 1 to 4 substituents. Preferred substituted aromatic rings are mono-, di, or tri-substituted. Aromatic rings may be substituted with halo, cyano, alkyl, heteroalkyl, haloalkyl, phenyl, keto, hydroxy, carboxy, amino, acylamino, aryloxy, heteroaryloxy, or any combination thereof. Preferred aromatic rings include, but are not limited to, the following:





Formula (L0)

or an enantiomer, diastereomer, or stereoisomer thereof, wherein z_1 is an integer selected from 0 to 10; z_2 is an integer selected from 0 to 10; z_3 is an integer selected from 0 to 10; each X is independently absent, CH_2 , O , S , NH , NR_{13} ; W is selected from absent, O , NH , NR_{13} , $-\text{OCH}_2\text{C}(\text{O})\text{NH}-$, $-\text{CH}_2\text{CH}_2\text{C}(\text{O})\text{NH}-$, $-\text{CH}_2\text{C}(\text{O})\text{NH}-$ or $-\text{C}(\text{O})\text{NH}-$; wherein Y is absent, O , NH , NR_{13} , $-\text{OCH}_2\text{C}(\text{O})\text{NH}-$, $-\text{CH}_2\text{CH}_2\text{C}(\text{O})\text{NH}-$, $-\text{CH}_2\text{C}(\text{O})\text{NH}-$ or $-\text{C}(\text{O})\text{NH}-$; and each R_{13} is independently C_1 - C_3 alkyl. The linker can be substituted with an alkyl, halide, phenyl, benzyl, aryl, alkylene or heterocycle group. Examples of applicable linkers include, but are not limited to, the following:



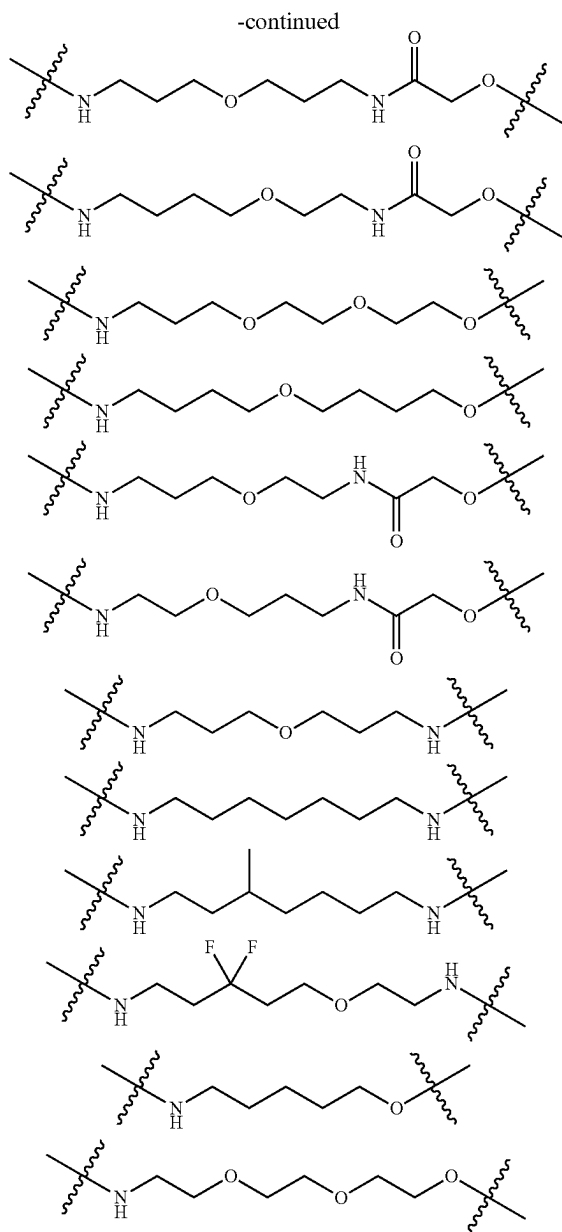
[0048] As used herein, the term “arylalkyl” or “aralkyl” alone or in combination, refers to an alkyl radical in which one hydrogen atom is replaced by an aryl radical, for example, benzyl and the like.

[0049] As used herein, the term “heteroarylalkyl” refers to an alkyl radical in which one hydrogen atom is replaced by a heteroaryl radical.

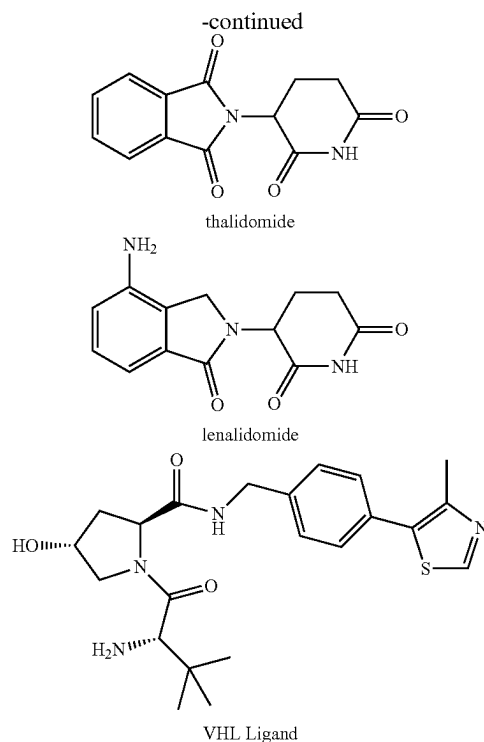
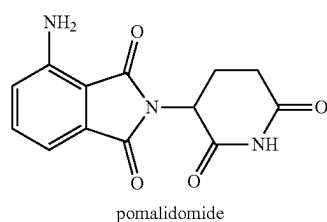
[0050] As used herein, the term “pharmacological composition” refers to a mixture of one or more of the compounds described herein or pharmaceutically acceptable salts thereof, with other chemical components, such as pharmaceutically acceptable carriers and/or excipients. The purpose of a pharmacological composition is to facilitate administration of a compound to an organism.

[0051] As used herein, the term “pharmaceutically acceptable salts” is a cationic salt formed at any acidic (e.g., carboxylic acid) group, or an anionic salt formed at any basic (e.g., amino) group.

[0052] As used herein, the term “linker” is covalently bonded to connect two parts; the carbon atom in the linear chain can be substituted with oxygen, nitrogen, sulfur, ester, and amide. Preferably, linker is of following Formula (L0):



[0053] As used herein, the term “ULM” represents a E3 ubiquitin ligase binding moiety that binds E3 ubiquitin ligase selected from the group consisting of VHL, IAP, Cereblon, and mouse double minute 2 (MDM2). Preferably ULM includes but is not limited to, the following:



[0054] As used herein the term “PROTACS” represents proteolysis-targeting chimeras. Proteolysis-targeting chimeras (PROTACS) are heterobifunctional compounds with two recruiting ligands connected via a linker. One ligand is specific to the protein of interest (POI) while the other moiety specifically recruits an E3 ligase. The PROTAC thus forms a ternary complex upon binding to both its E3 ligase target and the protein of interest. By hijacking the E3 ligase, PROTACS position the POI in a spatially favorable presentation to facilitate substrate poly-ubiquitination, thereby selectively knocking down levels of the targeted protein (e.g., Crews, C. M., et al., *Angewandte Chemie. Int. Ed.* 55, 1966-1973, 2016).

[0055] The terms “optical isomer”, “geometric isomer” (e.g., a cis and/or trans isomer), “tautomers” (e.g., the tautomerism of keto form and enol form), “stereoisomer” and “diastereomer” have the accepted meanings (see, e.g., Hawley’s Condensed Chemical Dictionary, 11th Ed.). The illustration of specific protected forms and other derivatives of the compounds of the instant invention are not intended to be limiting. The application of other useful protecting groups, salt forms, prodrugs etc., is within the ability of the skilled artisan.

[0056] The term “prodrug” is a form of drug that must undergo chemical conversion by metabolic processes before becoming an active, or fully active, pharmacological agent. A prodrug is not active or is less active, in its ingested or absorbed or otherwise administered form. For example, a prodrug may be broken down by bacteria in the digestive system into products, at least one of which will become active as a drug. Alternatively, it may be administered systemically, such as by intravenous injection, and subsequently be metabolized into one or more active molecules.

[0057] The term “solvate” is a complex formed by the combination of a solute (e.g., a metalloprotease inhibitor)

and a solvent (e.g., water). See J. Honig et al., *The Van Nostrand Chemist's Dictionary*, p. 650 (1953).

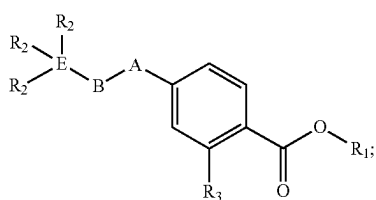
[0058] The term “independently” groups are groups present in the same structure that need not all represent the same substitution.

DETAILED DESCRIPTION OF THE INVENTION

[0059] Experiments conducted during the course of developing embodiments for the present invention involved discovery of a series of 2-hydroxybenzoic acid derivatives as the inhibitors and degraders of sirtuin5 (SIRT5). Sirtuins are a class of nicotinamide adenine dinucleotide (NAD⁺)-dependent deacetylases. There are seven sirtuin members encoded in mammalian genomes, SIRT1 through SIRT7. Unlike other sirtuin family members, SIRT5 is an inefficient deacetylase; instead it possesses very efficient desuccinylase, demalonylase, and deglutarylase activities, thereby influencing multiple cellular pathways such as ammonia detoxification, fatty acid oxidation, cellular respiration, ketone body formation, tricarboxylic acid cycle (TCA), glycolysis and reactive oxygen species (ROS) metabolism. Dysregulation of SIRT5 is found in several human diseases, especially cancer. It was found that SIRT5 depletion in specific cancer types (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, neuroblastoma) induces rapid cell death. Such experiments involved a screening an in-house library of compounds using a thermal shift assay and identified a novel class of 2-hydroxybenzoic acid derivatives as SIRT5 inhibitors. Such experiments also involved a successful development of PROTACs selectively inducing the degradation of SIRT5 protein that mainly resides in the mitochondrial matrix. These inhibitors and degraders are contemplated to be useful in the treatment of disorders associated with sirtuins, for example, cancer (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, neuroblastoma, non-small cell lung cancer).

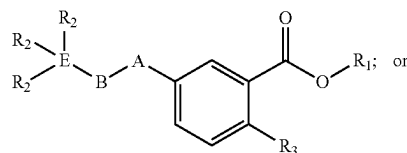
[0060] Accordingly, the present invention relates to a new class of small-molecules having a 2-hydroxybenzoic acid structure which function as sirtuin (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7) inhibitors and/or degraders which function as effective therapeutic agents for treating, ameliorating, and preventing disorders associated with sirtuin activity (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer). In addition, this invention also relates to a new class of proteolysis-targeting chimeras (PROTACs) (as defined herein) which function as sirtuin inhibitors and/or degraders within cancer and/or immune cells. Pharmaceutical compositions comprising said compounds are also within the scope of the present invention.

[0061] In a particular embodiment, compounds encompassed within Formula (1), Formula (2), or Formula (3) are provided:

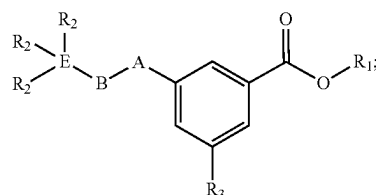


(Formula 1)

-continued



(Formula 2)



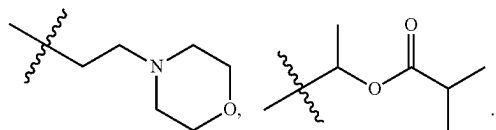
(Formula 3)

including pharmaceutically acceptable salts, solvates, PROTACs, and/or prodrugs thereof.

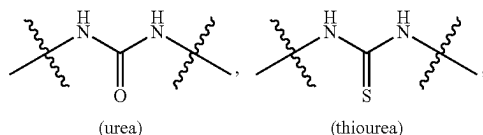
[0062] Formulas (1), (2), and (3) are not limited to a particular chemical moiety for R₁, R₂, R₃, A, B and E. In some embodiments, the particular chemical moiety for R₁, R₂, R₃, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade sirtuin activity (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7). In some embodiments, the particular chemical moiety for R₁, R₂, R₃, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade one or more of SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7. In some embodiments, the particular chemical moiety for R₁, R₂, R₃, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade SIRT5 activity. In some embodiments, the particular chemical moiety for R₁, R₂, R₃, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade sirtuin (e.g., SIRT5) related desuccinylase activity. In some embodiments, the particular chemical moiety for R₁, R₂, R₃, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade sirtuin (e.g., SIRT5) related demalonylase activity. In some embodiments, the particular chemical moiety for R₁, R₂, R₃, A, B and E independently include any chemical moiety that permits the resulting compound to inhibit and/or degrade sirtuin (e.g., SIRT5) related deglutarylase activity. In some embodiments, the particular chemical moiety for R₁, R₂, R₃, A, B and E independently include any chemical moiety that permits the resulting compound to influence multiple cellular pathways related to sirtuin activity (e.g., SIRT5) such as ammonia detoxification, fatty acid oxidation, cellular respiration, ketone body formation, tricarboxylic acid cycle (TCA), glycolysis and reactive oxygen species (ROS) metabolism. In some embodiments, the particular chemical moiety for R₁, R₂, R₃, A, B and E independently include any chemical moiety that permits the resulting compound to serve as a therapeutic for treating, preventing and/or ameliorating disorders characterized with sirtuin (e.g., SIRT5) activity (e.g., sirtuin related cancer (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer).

[0063] In some embodiments, R¹ is selected from the group consisting of hydrogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, aryl, heteroaryl, arylalkyl, and heteroarylalkyl.

[0064] In some embodiments, R^1 is selected from the group consisting of hydrogen, methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, pentyl,



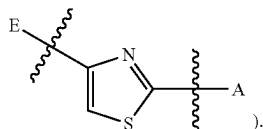
[0065] In some embodiments, A is selected from the group consisting of NH, CH_2 , O, S,



amide, and sulfonamide.

[0066] In some embodiments, B is an aromatic ring.

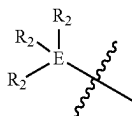
[0067] In some embodiments, B is a thiazole ring (e.g.,



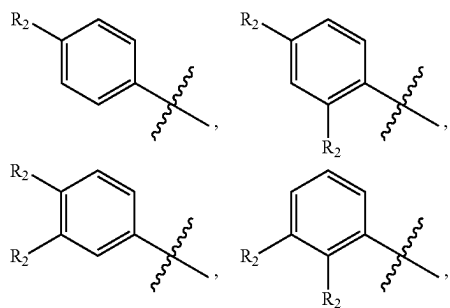
[0068] In some embodiments, E is an aromatic ring.

[0069] In some embodiments, E is a benzene ring.

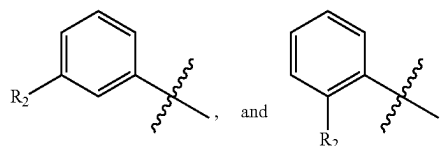
[0070] In some embodiments,



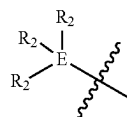
is selected from:



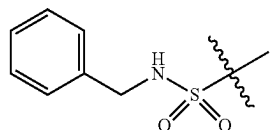
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[0071] In some embodiments, each R^2 within

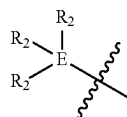


is independently selected from the group consisting of hydrogen, halogen, (e.g., F, Cl, Br, I), alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, halogen, aryl, heteroaryl, arylalkyl, heteroarylalkyl, nitril, cyano, amide or sulfonamide,

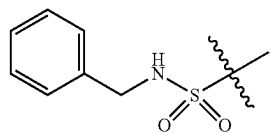


and triphenylphosphine (TPP) attached with linker; and/or

[0072] wherein each R^2 within



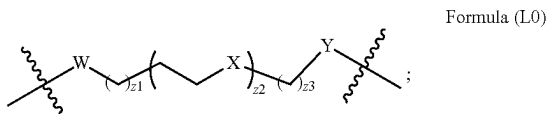
is independently selected from the group consisting of hydrogen, halogen, CF_3 , OCH_3 , OH, NO_2 (nitryl), CN (cyano), amide (e.g., $NHC(O)CH_3$), and sulfonamide (e.g., $S(O_2)NHCH_3$),



and triphenylphosphine (TPP) group attached with a linker.

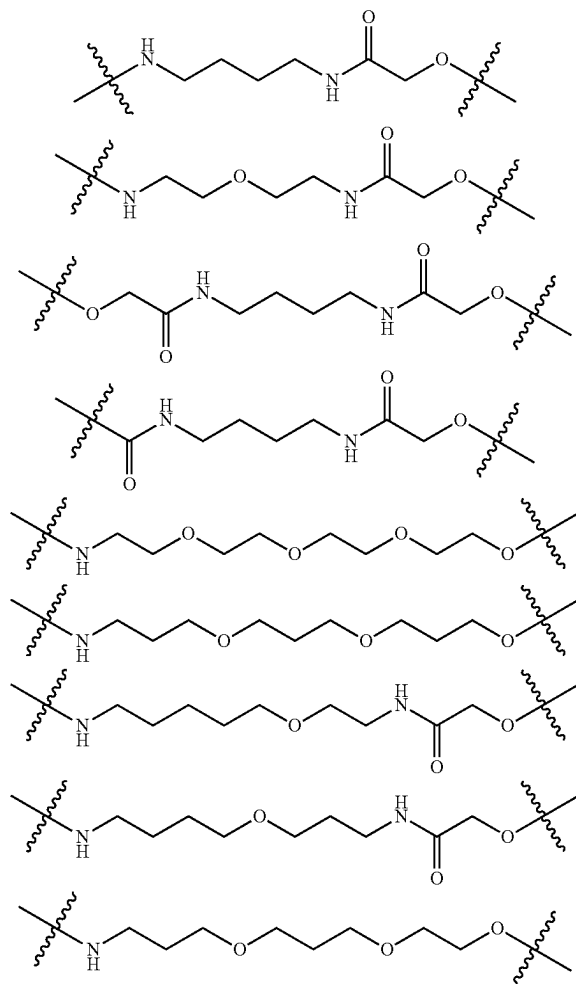
[0073] In some embodiments, linker is covalently bonded to connect two parts; the carbon atom in the linear chain can be substituted with oxygen, nitrogen, sulfur, ester, and amide.

[0074] In some embodiments, linker is of following Formula (L0):

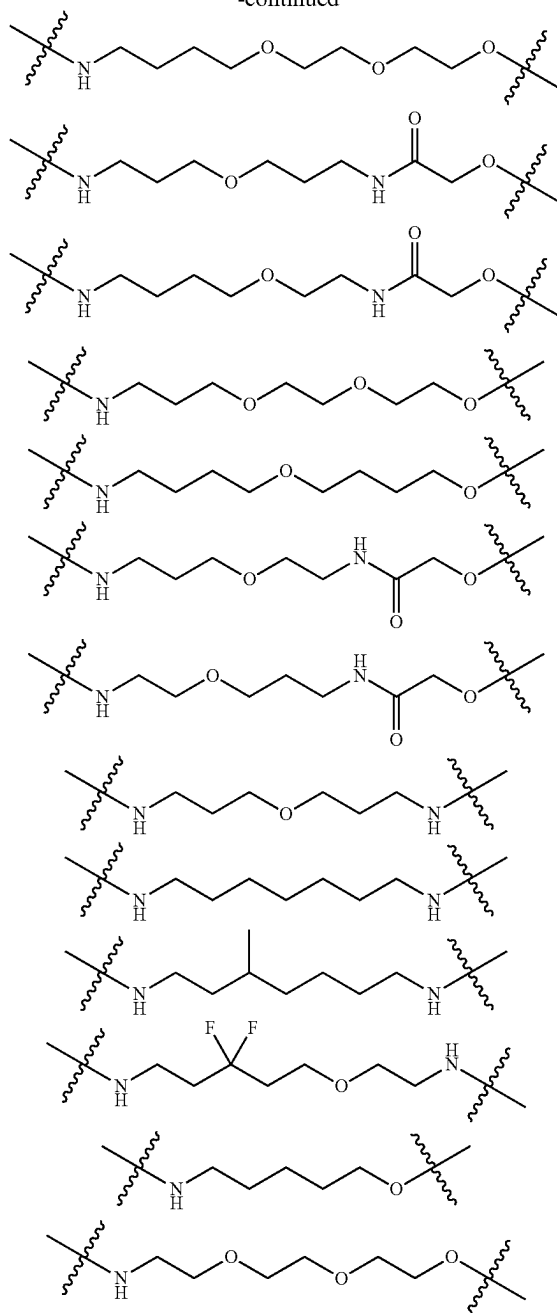


or an enantiomer, diastereomer, or stereoisomer thereof, wherein $Z1$ is an integer selected from 0 to 10; $Z2$ is an integer selected from 0 to 10; $Z3$ is an integer selected from 0 to 10; each X is independently absent, CH_2 , O , S , NH , NR_{13} ; wherein W is selected from absent, O , NH , NR_{13} , $-OCH_2C(O)NH-$, $-CH_2CH_2C(O)NH-$, $-CH_2C(O)NH-$ or $-C(O)NH-$; wherein Y is absent, O , NH , NR_{13} , $-OCH_2C(O)NH-$, $-CH_2CH_2C(O)NH-$, $-CH_2C(O)NH-$ or $-C(O)NH-$; and each R_{13} is independently C_1 - C_3 alkyl. The linker can be substituted with an alkyl, halide, phenyl, benzyl, aryl, alkylene or heterocycle group.

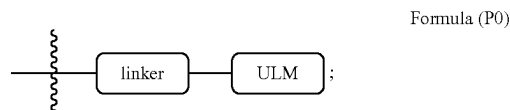
[0075] In some embodiments, linker is one of the following moieties:



-continued



[0076] In some embodiments, R^2 is represented by Formula (P0):



wherein the linker is covalently bonded to connect two parts; wherein the carbon atom in the linear chain can be substi-

tuted with oxygen, nitrogen, sulfur, ester, and amide; wherein ULM represents an E3 ubiquitin ligase binding moiety that binds E3 ubiquitin ligase selected from the group consisting of pomalidomide, thalidomide, lenalidomide, Von Hippel-Lindau (VHL), inhibitors of apoptosis proteins (IAP), Cereblon, and mouse double minute 2 (MDM2).

[0077] In some embodiments, R3 is hydrogen or OH.

[0078] In some embodiments, prodrug is preferably carboxylic acid ester. The ester group masks the negative charge on the carboxylate and thus can increase the cell permeability of the compounds. Inside the cells, the ester group can be hydrolyzed to release the negatively charged carboxylate, which can inhibit SIRT5.

[0079] In some embodiments, compounds shown in Table I are contemplated for Formula (1). Table I additionally shows values for SIRT5 IC₅₀ for select compounds.

[0080] In some embodiments, the compositions and methods of the present invention are used to treat diseased cells, tissues, organs, or pathological conditions and/or disease states in an animal (e.g., a mammalian patient including, but not limited to, humans and veterinary animals). In this regard, various diseases and pathologies are amenable to treatment or prophylaxis using the present methods and compositions. A non-limiting exemplary list of these diseases and conditions includes, but is not limited to, cancer associated with sirtuin (e.g., SIRT5) activity (e.g., melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, non-small cell lung cancer).

[0081] A non-limiting exemplary list of cancers include, but are not limited to, pancreatic cancer, breast cancer, prostate cancer, lymphoma, skin cancer, colon cancer, melanoma, malignant melanoma, ovarian cancer, brain cancer, primary brain carcinoma, head and neck cancer, glioma, glioblastoma, liver cancer, bladder cancer, non-small cell lung cancer, head or neck carcinoma, breast carcinoma, ovarian carcinoma, lung carcinoma, small-cell lung carcinoma, Wilms' tumor, cervical carcinoma, testicular carcinoma, bladder carcinoma, pancreatic carcinoma, stomach carcinoma, colon carcinoma, prostatic carcinoma, genitourinary carcinoma, thyroid carcinoma, esophageal carcinoma, myeloma, multiple myeloma, adrenal carcinoma, renal cell carcinoma, endometrial carcinoma, adrenal cortex carcinoma, malignant pancreatic insulinoma, malignant carcinoid carcinoma, choriocarcinoma, mycosis fungoides, malignant hypercalcemia, cervical hyperplasia, leukemia, acute lymphocytic leukemia, chronic lymphocytic leukemia, acute myelogenous leukemia, chronic myelogenous leukemia, chronic granulocytic leukemia, acute granulocytic leukemia, hairy cell leukemia, neuroblastoma, rhabdomyosarcoma, Kaposi's sarcoma, polycythemia vera, essential thrombocytosis, Hodgkin's disease, non-Hodgkin's lymphoma, soft-tissue sarcoma, osteogenic sarcoma, primary macroglobulinemia, and retinoblastoma, and the like. In some embodiments, the cancer cells being treated are metastatic. In other embodiments, the cancer cells being treated are resistant to anticancer agents.

[0082] Some embodiments of the present invention provide methods for administering an effective amount of a compound of the invention and at least one additional therapeutic agent (including, but not limited to, chemotherapeutic antineoplastics, apoptosis-modulating agents, antimi-

crobials, antivirals, antifungals, and anti-inflammatory agents) and/or therapeutic technique (e.g., surgical intervention, and/or radiotherapies).

[0083] A number of suitable anticancer agents are contemplated for use in the methods of the present invention. Indeed, the present invention contemplates, but is not limited to, administration of numerous anticancer agents such as: agents that induce apoptosis; polynucleotides (e.g., antisense, ribozymes, siRNA); polypeptides (e.g., enzymes and antibodies); biological mimetics; alkaloids; alkylating agents; antitumor antibiotics; antimetabolites; hormones; platinum compounds; monoclonal or polyclonal antibodies (e.g., antibodies conjugated with anticancer drugs, toxins, defensins), toxins; radionuclides; biological response modifiers (e.g., interferons (e.g., IFN- α) and interleukins (e.g., IL-2)); adoptive immunotherapy agents; hematopoietic growth factors; agents that induce tumor cell differentiation (e.g., all-trans-retinoic acid); gene therapy reagents (e.g., antisense therapy reagents and nucleotides); tumor vaccines; angiogenesis inhibitors; proteasome inhibitors; NF-KB modulators; anti-CDK compounds; HDAC inhibitors; and the like. Numerous other examples of chemotherapeutic compounds and anticancer therapies suitable for co-administration with the disclosed compounds are known to those skilled in the art.

[0084] In certain embodiments, anticancer agents comprise agents that induce or stimulate apoptosis. Agents that induce apoptosis include, but are not limited to, radiation (e.g., X-rays, gamma rays, UV); tumor necrosis factor (TNF)-related factors (e.g., TNF family receptor proteins, TNF family ligands, TRAIL, antibodies to TRAIL-R1 or TRAIL-R2); kinase inhibitors (e.g., epidermal growth factor receptor (EGFR) kinase inhibitor, vascular growth factor receptor (VGFR) kinase inhibitor, fibroblast growth factor receptor (FGFR) kinase inhibitor, platelet-derived growth factor receptor (PDGFR) kinase inhibitor, and Bcr-Abl kinase inhibitors (such as GLEEVEC)); antisense molecules; antibodies (e.g., HERCEPTIN, RITUXAN, ZEVALIN, and AVASTIN); anti-estrogens (e.g., raloxifene and tamoxifen); anti-androgens (e.g., flutamide, bicalutamide, finasteride, aminoglutethamide, ketoconazole, and corticosteroids); cyclooxygenase 2 (COX-2) inhibitors (e.g., celecoxib, meloxicam, NS-398, and non-steroidal anti-inflammatory drugs (NSAIDs)); anti-inflammatory drugs (e.g., butazolidin, DECADRON, DELTASONE, dexamethasone, dexamethasone intensol, DEXONE, HEXADROL, hydroxychloroquine, METICORTEN, ORADEXON, ORASONE, oxyphenbutazone, PEDIAPRED, phenylbutazone, PLAQUENIL, prednisolone, prednisone, PRELONE, and TANDEARIL); and cancer chemotherapeutic drugs (e.g., irinotecan (CAMPTOSAR), CPT-11, fludarabine (FLUDARA), dacarbazine (DTIC), dexamethasone, mitoxantrone, MYLOTARG, VP-16, cisplatin, carboplatin, oxaliplatin, 5-FU, doxorubicin, gemcitabine, bortezomib, gefitinib, bevacizumab, TAXOTERE or TAXOL); cellular signaling molecules; ceramides and cytokines; staurosporine, and the like.

[0085] In still other embodiments, the compositions and methods of the present invention provide a compound of the invention and at least one anti-hyperproliferative or antineoplastic agent selected from alkylating agents, antimetabolites, and natural products (e.g., herbs and other plant and/or animal derived compounds).

[0086] Alkylating agents suitable for use in the present compositions and methods include, but are not limited to: 1) nitrogen mustards (e.g., mechlorethamine, cyclophosphamide, ifosfamide, melphalan (L-sarcosine)); and chlorambucil); 2) ethylenimines and methylmelamines (e.g., hexamethylmelamine and thiotepa); 3) alkyl sulfonates (e.g., busulfan); 4) nitrosoureas (e.g., carmustine (BCNU); lomustine (CCNU); semustine (methyl-CCNU); and streptozocin (streptozotocin)); and 5) triazines (e.g., dacarbazine (DTIC; dimethyltriazenoimid-azolecarboxamide).

[0087] In some embodiments, antimetabolites suitable for use in the present compositions and methods include, but are not limited to: 1) folic acid analogs (e.g., methotrexate (amethopterin)); 2) pyrimidine analogs (e.g., fluorouracil (5-fluorouracil; 5-FU), floxuridine (fluorodeoxyuridine; FudR), and cytarabine (cytosine arabinoside)); and 3) purine analogs (e.g., mercaptopurine (6-mercaptopurine; 6-MP), thioguanine (6-thioguanine; TG), and pentostatin (2'-deoxycoformycin)).

[0088] In still further embodiments, chemotherapeutic agents suitable for use in the compositions and methods of the present invention include, but are not limited to: 1) *vinca* alkaloids (e.g., vinblastine (VLB), vincristine); 2) epipodophyllotoxins (e.g., etoposide and teniposide); 3) antibiotics (e.g., dactinomycin (actinomycin D), daunorubicin (daunomycin; rubidomycin), doxorubicin, bleomycin, plicamycin (mithramycin), and mitomycin (mitomycin C)); 4) enzymes

(e.g., L-asparaginase); 5) biological response modifiers (e.g., interferon- α); 6) platinum coordinating complexes (e.g., cisplatin (cis-DDP) and carboplatin); 7) anthracenediones (e.g., mitoxantrone); 8) substituted ureas (e.g., hydroxyurea); 9) methylhydrazine derivatives (e.g., procabazine (N-methylhydrazine; MIH)); 10) adrenocortical suppressants (e.g., mitotane (o,p'-DDD) and aminoglutethimide); 11) adrenocorticosteroids (e.g., prednisone); 12) progestins (e.g., hydroxyprogesterone caproate, medroxyprogesterone acetate, and megestrol acetate); 13) estrogens (e.g., diethylstilbestrol and ethinyl estradiol); 4) antiestrogens (e.g., tamoxifen); 15) androgens (e.g., testosterone propionate and fluoxymesterone); 16) antiandrogens (e.g., flutamide); and 17) gonadotropin-releasing hormone analogs (e.g., leuprolide).

[0089] Any oncolytic agent that is routinely used in a cancer therapy context finds use in the compositions and methods of the present invention. For example, the U.S. Food and Drug Administration maintains a formulary of oncolytic agents approved for use in the United States. International counterpart agencies to the U.S.F.D.A. maintain similar formularies. Table 2 provides a list of exemplary antineoplastic agents approved for use in the U.S. Those skilled in the art will appreciate that the "product labels" required on all U.S. approved chemotherapeutics describe approved indications, dosing information, toxicity data, and the like, for the exemplary agents.

TABLE 2

Aldesleukin (des-alanyl-1,serine-125 human interleukin-2)	Proleukin	Chiron Corp., Emeryville, CA
Alemtuzumab (IgG1k anti CD52 antibody)	Campath	Millennium and ILEX Partners, LP, Cambridge, MA
Alitretinoin (9-cis-retinoic acid)	Panretin	Ligand Pharmaceuticals, Inc., San Diego CA
Allopurinol (1,5-dihydro-4 H -pyrazolo[3,4-d]pyrimidin-4-one monosodium salt)	Zyloprim	GlaxoSmithKline, Research Triangle Park, NC
Altretamine (N,N,N',N'',N'',N''-hexamethyl-1,3,5-triazine-2,4, 6-triamine)	Hexalen	US Bioscience, West Conshohocken, PA
Amifostine (ethanethiol,2-[(3-aminopropyl)amino]-, dihydrogen phosphate (ester))	Ethiol	US Bioscience
Anastrozole (1,3-Benzenediacetonitrile,a,a,a',a'-tetramethyl-5- (1H-1,2,4-triazol-1-ylmethyl))	Arimidex	AstraZeneca Pharmaceuticals, LP, Wilmington, DE
Arsenic trioxide	Trisenox	Cell Therapeutic, Inc., Seattle, WA
Asparaginase (L-asparagine amidohydrolase, type EC-2)	Elspar	Merck & Co., Inc., Whitehouse Station, NJ
BCG Live (lyophilized preparation of an attenuated strain of <i>Mycobacterium bovis</i> (<i>Bacillus Calmette-Guérin</i> [BCG], substrain Montreal)	TICE BCG	Organon Teknika, Corp., Durham, NC
bexarotene capsules (4-[1-(5,6,7,8-tetrahydro-3,5,5,8,8-pentamethyl-2- naphthalenyl) ethenyl] benzoic acid)	Targretin	Ligand Pharmaceuticals
bexarotene gel	Targretin	Ligand Pharmaceuticals
Bleomycin (cytotoxic glycopeptide antibiotics produced by <i>Streptomyces verticillus</i> ; bleomycin A ₂ and bleomycin B ₂)	Blenoxane	Bristol-Myers Squibb Co., NY, NY
Capecitabine (5'-deoxy-5-fluoro-N-[(pentylloxy)carbonyl]-cytidine)	Xeloda	Roche
Carboplatin (platinum, diammine[1,1- cyclobutanedicarboxylato(2-)-0,0']-(SP-4-2))	Paraplatin	Bristol-Myers Squibb
Carmustine (1,3-bis(2-chloroethyl)-1-nitrosourea)	BCNU, BiCNU	Bristol-Myers Squibb

TABLE 2-continued

Carmustine with Polifeprosan 20 Implant	Gliadel Wafer	Guilford Pharmaceuticals, Inc., Baltimore, MD
Celecoxib (as 4-[5-(4-methylphenyl)-3-(trifluoromethyl)-1H-pyrazol-1-yl]benzenesulfonamide)	Celebrex	Searle Pharmaceuticals, England
Chlorambucil (4-[bis(2chloroethyl)amino]benzenebutanoic acid)	Leukeran	GlaxoSmithKline
Cisplatin (PtCl ₂ H ₆ N ₂)	Platinol	Bristol-Myers Squibb
Cladribine (2-chloro-2'-deoxy-b-D-adenosine)	Leustatin, 2-CdA	R.W. Johnson Pharmaceutical Research Institute, Raritan, NJ
Cyclophosphamide (2-[bis(2-chloroethyl)amino]tetrahydro-2H-13,2-oxazaphosphorine 2-oxide monohydrate)	Cytosan, Neosar	Bristol-Myers Squibb
Cytarabine (1-b-D-Arabinofuranosylectosine, C ₉ H ₁₃ N ₃ O ₅)	Cytosar-U	Pharmacia & Upjohn Company
cytarabine liposomal	DepoCyt	Skye Pharmaceuticals, Inc., San Diego, CA
Dacarbazine (5-(3,3-dimethyl-1-triazeno)-imidazole-4-carboxamide (DTIC))	DTIC-Dome	Bayer AG, Leverkusen, Germany
Dactinomycin, actinomycin D (actinomycin produced by <i>Streptomyces parvullus</i> , C ₆₂ H ₈₆ N ₁₂ O ₁₆)	Cosmegen	Merck
Darbepoetin alfa (recombinant peptide)	Aranesp	Amgen, Inc., Thousand Oaks, CA
daunorubicin liposomal ([(8S-cis)-8-acetyl-10-[(3-amino-2,3,6-trideoxy- α -L-lyxo-hexopyranosyl)oxy]-7,8,9,10-tetrahydro-6,8,11-trihydroxy-1-methoxy-5,12-naphthacenedione hydrochloride)	DanuoXome	Nexstar Pharmaceuticals, Inc., Boulder, CO
Daunorubicin HCl, daunomycin ([(1S,3S)-3-Acetyl-1,2,3,4,6,11-hexahydro-3,5,12-trihydroxy-10-methoxy-6,11-dioxo-1-naphthacenyl 3-amino-2,3,6-trideoxy-(α)-L-lyxo-hexopyranoside hydrochloride)	Cerubidine	Wyeth Ayerst, Madison, NJ
Denileukin difitox (recombinant peptide)	Ontak	Seragen, Inc., Hopkinton, MA
Dexrazoxane ([(S)-4,4'-(1-methyl-1,2-ethanediyl)bis-2,6-piperazinedione)	Zinecard	Pharmacia & Upjohn Company
Docetaxel ([(2R,3S)-N-carboxy-3-phenylisoserine,N-tert-butyl ester, 13-ester with 5b-20-epoxy-12a,4,7b,10b,13a-hexahydroxytax-11-en-9-one 4-acetate 2-benzoate, trihydrate)	Taxotere	Aventis Pharmaceuticals, Inc., Bridgewater, NJ
Doxorubicin HCl (8S,10S)-10-[(3-amino-2,3,6-trideoxy-a-L-lyxo-hexopyranosyl)oxy]-8-glycolyl-7,8,9,10-tetrahydro-6,8,11-trihydroxy-1-methoxy-5,12-naphthacenedione hydrochloride)	Adriamycin, Rubex	Pharmacia & Upjohn Company
doxorubicin	Adriamycin PFS Intravenous injection	Pharmacia & Upjohn Company
doxorubicin liposomal	Doxil	Sequus Pharmaceuticals, Inc., Menlo park, CA
dromostanolone propionate (17b-Hydroxy-2a-methyl-5a-androstan-3-one propionate)	Dromostanolone	Eli Lilly & Company, Indianapolis, IN
dromostanolone propionate	Masterone injection	Syntex, Corp., Palo Alto, CA
Elliott's B Solution	Elliott's B Solution	Orphan Medical, Inc
Epirubicin ([(8S-cis)-10-[(3-amino-2,3,6-trideoxy-a-L-arabino-hexopyranosyl)oxy]-7,8,9,10-tetrahydro-6,8,11-trihydroxy-8-(hydroxyacetyl)-1-methoxy-5,12-naphthacenedione hydrochloride)	Ellence	Pharmacia & Upjohn Company
Epoetin alfa (recombinant peptide)	Epogen	Amgen, Inc

TABLE 2-continued

Estramustine (estra-1,3,5(10)-triene-3,17-diol(17(beta))-3-[bis(2-chloroethyl)carbamate] 17-(dihydrogen phosphate), disodium salt, monohydrate, or estradiol 3-[bis(2-chloroethyl)carbamate] 17-(dihydrogen phosphate), disodium salt, monohydrate)	Emcyt	Pharmacia & Upjohn Company
Etoposide phosphate (4'-Demethylepipodophyllotoxin 9-[4,6-O-(R)-ethylidene-(beta)-D-glucopyranoside],4'-(dihydrogen phosphate))	Etopophos	Bristol-Myers Squibb
etoposide, VP-16 (4'-demethylepipodophyllotoxin 9-[4,6-O-(R)-ethylidene-(beta)-D-glucopyranoside])	Vepesid	Bristol-Myers Squibb
Exemestane (6-methylenandrosta-1,4-diene-3,17-dione)	Aromasin	Pharmacia & Upjohn Company
Filgrastim (r-metHuG-CSF)	Neupogen	Amgen, Inc
floxuridine (intraarterial) (2'-deoxy-5-fluorouridine)	FUDR	Roche
Fludarabine (fluorinated nucleotide analog of the antiviral agent vidarabine, 9-b-D-arabinofuranosyladenine (ara-A))	Fludara	Berlex Laboratories, Inc., Cedar Knolls, NJ
Fluorouracil, 5-FU (5-fluoro-2,4(1H,3H)-pyrimidinedione)	Adrucil	ICN Pharmaceuticals, Inc., Humacao, Puerto Rico
Fulvestrant (7-alpha-[9-(4,4,5,5,5-penta fluoropentylsulphinyl) nonyl]estra-1,3,5-(10)-triene-3,17-beta-diol)	Faslodex	IPR Pharmaceuticals, Guayama, Puerto Rico
Gemcitabine (2'-deoxy-2',2'-difluorocytidine monohydrochloride (b-isomer))	Gemzar	Eli Lilly
Gemtuzumab Ozogamicin (anti-CD33 hP67.6)	Mylotarg	Wyeth Ayerst
Goserelin acetate	Zoladex Implant	AstraZeneca Pharmaceuticals
Hydroxyurea	Hydrea	Bristol-Myers Squibb
Ibritumomab Tiuxetan (immunoconjugate resulting from a thiourea covalent bond between the monoclonal antibody Ibritumomab and the linker-chelator tiuxetan [N-[2-bis(carboxymethyl)amino]-3-(p-isothiocyanatophenyl)-propyl]-[N-[2-bis(carboxymethyl)amino]-2-(methyl)-ethyl]glycine)	Zevalin	Biogen IDEC, Inc., Cambridge MA
Idarubicin (5,12-Naphthacenedione,9-acetyl-7-[(3-amino-2,3,6-trideoxy-(alpha)-L-lyxo-hexopyranosyl)oxy]-7,8,9,10-tetrahydro-6,9,11-trihydroxyhydrochloride, (7S- cis))	Idamycin	Pharmacia & Upjohn Company
Ifosfamide (3-(2-chloroethyl)-2-[(2-chloroethyl)amino]tetrahydro-2H-1,3,2-oxazaphosphorine 2-oxide)	IFEX	Bristol-Myers Squibb
Imatinib Mesilate (4-[(4-Methyl-1-piperazinyl)methyl]-N-[4-methyl-3-[[4-(3-pyridinyl)-2-pyrimidinyl]amino]-phenyl]benzamide methanesulfonate)	Gleevec	Novartis AG, Basel, Switzerland
Interferon alfa-2a (recombinant peptide)	Roferon-A	Hoffmann-La Roche, Inc., Nutley, NJ
Interferon alfa-2b (recombinant peptide)	Intron A (Lyophilized Betaseron)	Schering AG, Berlin, Germany
Irinotecan HCl ((4S)-4,11-diethyl-4-hydroxy-9-[(4-piperidinopiperidino)carbonyloxy]-1H-pyrano[3',4': 6,7] indolizino[1,2-b] quinoline-3,14(4H,12H) dione hydrochloride trihydrate)	Camptosar	Pharmacia & Upjohn Company
Letrozole (4,4'-(1H-1,2,4-Triazol-1-ylmethylene) dibenzonitrile)	Femara	Novartis
Leucovorin (L-Glutamic acid, N[4][(2-amino-5-formyl-1,4,5,6,7,8 hexahydro-4oxo-6-pteridinyl)methyl]amino]benzoyl], calcium salt (1:1))	Wellcovorin, Leucovorin	Immunex, Corp., Seattle, WA

TABLE 2-continued

Levamisole HCl ((—)-(S)-2,3,5,6-tetrahydro-6-phenylimidazo [2,1-b] thiazole monohydrochloride C ₁₁ H ₁₂ N ₂ S•HCl)	Ergamisol	Janssen Research Foundation, Titusville, NJ
Lomustine (1-(2-chloro-ethyl)-3-cyclohexyl-1-nitrosourea)	CeeNU	Bristol-Myers Squibb
Meclorethamine, nitrogen mustard (2-chloro-N-(2-chloroethyl)-N-methylethanamine hydrochloride)	Mustargen	Merck
Megestrol acetate 17 α (acetyloxy)-6-methylpregna-4,6-diene-3,20-dione	Megace	Bristol-Myers Squibb
Melphalan, L-PAM (4-[bis(2-chloroethyl) amino]-L-phenylalanine)	Alkeran	GlaxoSmithKline
Mercaptopurine, 6-MP (1,7-dihydro-6 H-purine-6-thione monohydrate)	Purinethol	GlaxoSmithKline
Mesna (sodium 2-mercaptoethane sulfonate)	Mesnex	Asta Medica
Methotrexate (N-[4-[[[(2,4-diamino-6- pteridinyl)methyl]methylamino]benzoyl]-L- glutamic acid)	Methotrexate	Lederle Laboratories
Methoxsalen (9-methoxy-7H-furo[3,2-g][1]-benzopyran-7-one)	Uvadex	Therakos, Inc., Way Exton, Pa
Mitomycin C	Mutamycin	Bristol-Myers Squibb
mitomycin C	Mitozytrex	SuperGen, Inc., Dublin, CA
Mitotane (1,1-dichloro-2-(o-chlorophenyl)-2-(p-chlorophenyl) ethane)	Lysodren	Bristol-Myers Squibb
Mitoxantrone (1,4-dihydroxy-5,8-bis[[2-[(2- hydroxyethyl)amino]ethyl]amino]-9,10- anthracenedione dihydrochloride)	Novantrone	Immunex Corporation
Nandrolone phenpropionate	Durabolin-50	Organon, Inc., West Orange, NJ
Nofetumomab	Verluma	Boehringer Ingelheim Pharma KG, Germany
Oprelvekin (IL-11)	Neumega	Genetics Institute, Inc., Alexandria, VA
Oxaliplatin (cis-[(1R,2R)-1,2-cyclohexanediamine-N,N'] [oxalato(2-)-O,O'] platinum)	Eloxatin	Sanofi Synthelabo, Inc., NY, NY
Paclitaxel (5 β ,20-Epoxy-1,2a,4,7 β ,10 β ,13a-hexahydroxytax- 11-en-9-one 4,10-diacetate 2-benzoate 13-ester with (2R,3S)-N-benzoyl-3-phenylisoserine)	TAXOL	Bristol-Myers Squibb
Pamidronate (phosphonic acid (3-amino-1-hydroxypropylidene) bis-, disodium salt, pentahydrate, (APD))	Aredia	Novartis
Pegademase (monomethoxypolyethylene glycol succinimidyl) 11-17-adenosine deaminase)	Adagen (Pegademase Bovine)	Enzon Pharmaceuticals, Inc., Bridgewater, NJ
Pegasparase (monomethoxypolyethylene glycol succinimidyl L- asparaginase)	Oncaspar	Enzon
Pegfilgrastim (covalent conjugate of recombinant methionyl human G-CSF (Filgrastim) and monomethoxypolyethylene glycol)	Neulasta	Amgen, Inc
Pentostatin	Nipent	Parke-Davis Pharmaceutical Co., Rockville, MD
Pipobroman	Vercyte	Abbott Laboratories, Abbott Park, IL
Plicamycin, Mithramycin (antibiotic produced by <i>Streptomyces plicatus</i>)	Mithracin	Pfizer, Inc., NY, NY
Porfimer sodium	Photofrin	QLT Phototherapeutics, Inc., Vancouver, Canada
Procarbazine (N-isopropyl- μ -(2-methylhydrazino)-p-tolamide monohydrochloride)	Matulane	Sigma Tau Pharmaceuticals, Inc., Gaithersburg, MD

TABLE 2-continued

Quinacrine (6-chloro-9-(1-methyl-4-diethyl-amine) butylamino-2-methoxyacridine)	Atabrine	Abbott Labs
Rasburicase (recombinant peptide)	Elitek	Sanofi-Synthelabo, Inc.,
Rituximab (recombinant anti-CD20 antibody)	Rituxan	Genentech, Inc., South San Francisco, CA
Sargramostim (recombinant peptide)	Prokine	Immunex Corp
Streptozocin (streptozocin 2-deoxy-2- [[[(methylnitrosoamino)carbonyl]amino]-a(and b)- D-glucopyranose and 220 mg citric acid anhydrous)	Zanosar	Pharmacia & Upjohn Company
Talc (Mg ₃ Si ₄ O ₁₀ (OH) ₂)	Sclerosol	Bryan, Corp., Woburn, MA
Tamoxifen ([(Z)-2-[4-(1,2-diphenyl-1-butenyl)phenoxy]-N,N- dimethylethanamine 2-hydroxy-1,2,3- propanetricarboxylate (1:1)])	Nolvadex	AstraZeneca Pharmaceuticals
Temozolomide (3,4-dihydro-3-methyl-4-oxoimidazo[5,1-d]-as- tetrazine-8-carboxamide)	Temodar	Schering
teniposide, VM-26 (4'-demethylepipodophyllotoxin 9-[4,6-O-(R)-2- thenylidene-(beta)-D-glucopyranoside])	Vumon	Bristol-Myers Squibb
Testolactone (13-hydroxy-3-oxo-13,17-secoandrosta-1,4-dien-17- oic acid [dgr]-lactone)	Teslac	Bristol-Myers Squibb
Thioguanine, 6-TG (2-amino-1,7-dihydro-6H-purine-6-thione)	Thioguanine	GlaxoSmithKline
Thiotepa (Aziridine,1,1',1''-phosphinothioylidynetris-, or Tris (1-aziridinyl) phosphine sulfide)	Thioplex	Immunex Corporation
Topotecan HCl ([(S)-10-[(dimethylamino)methyl]-4-ethyl-4,9- dihydroxy-1H-pyranol[3',4': 6,7] indolizino [1,2-b] quinoline-3,14-(4H,12H)-dione monohydrochloride)	Hycamtin	GlaxoSmithKline
Toremifene (2-(p-[(Z)-4-chloro-1,2-diphenyl-1-butenyl]- phenoxy)-N,N-dimethylethylamine citrate (1:1))	Fareston	Roberts Pharmaceutical Corp., Eatontown, NJ
Tositumomab, I 131 Tositumomab (recombinant murine immunotherapeutic monoclonal IgG _{2a} lambda anti-CD20 antibody (I 131 is a radioimmunotherapeutic antibody))	Bexxar	Corixa Corp., Seattle, WA
Trastuzumab (recombinant monoclonal IgG ₁ kappa anti-HER2 antibody)	Herceptin	Genentech, Inc
Tretinoin, ATRA (all-trans retinoic acid)	Vesanoid	Roche
Uracil Mustard	Uracil Mustard Capsules	Roberts Labs
Valrubicin, N-trifluoroacetyladiamycin-14-valerate ([(2S-cis)-2-[1,2,3,4,6,11-hexahydro-2,5,12- trihydroxy-7 methoxy-6,11-dioxo-[[4 2,3,6- trideoxy-3-[(trifluoroacetyl)-amino-α-L-lyxo- hexopyranosyl]oxy]]-2-naphthaceny]]-2-oxoethyl pentanoate)	Valstar	Anthra --> Medeva
Vinblastine, Leurocristine (C ₄₆ H ₅₆ N ₄ O ₁₀ •H ₂ SO ₄)	Velban	Eli Lilly
Vincristine (C ₄₆ H ₅₆ N ₄ O ₁₀ •H ₂ SO ₄)	Oncovin	Eli Lilly
Vinorelbine (3',4'-didehydro-4'-deoxy-C'-norvincaleukoblastine [R-(R*,R*)-2,3-dihydroxybutanedioate (1:2)(salt)])	Navelbine	GlaxoSmithKline
Zoledronate, Zoledronic acid ([(1-Hydroxy-2-imidazol-1-yl-phosphonoethyl) phosphonic acid monohydrate])	Zometa	Novartis

[0090] Anticancer agents further include compounds which have been identified to have anticancer activity. Examples include, but are not limited to, 3-AP, 12-O-tetradecanoylphorbol-13-acetate, 17AAG, 852A, ABI-007, ABR-217620, ABT-751, ADI-PEG 20, AE-941,

AG-013736, AGRO100, alanosine, AMG 706, antibody G250, antineoplastons, AP23573, apaziquone, APC8015, atiprimod, ATN-161, atrasenten, azacitidine, 1BB-10901, BCX-1777, bevacizumab, BG00001, bicalutamide, BMS 247550, bortezomib, bryostatin-1, buserelin, calcitriol, CCI-

779, CDB-2914, cefixime, cetuximab, CG0070, cilengitide, clofarabine, combretastatin A4 phosphate, CP-675,206, CP-724,714, CpG 7909, curcumin, decitabine, DENSPM, doxercalciferol, E7070, E7389, ecteinascidin 743, efaproxiral, eflornithine, EKB-569, enzastaurin, erlotinib, exisulind, fenretinide, flavopiridol, fludarabine, flutamide, fotemustine, FR901228, G17DT, galiximab, gefitinib, genistein, glufosfamide, GTI-2040, histrelin, HKI-272, homoharringtonine, HSPPC-96, hu14.18-interleukin-2 fusion protein, HuMax-CD4, iloprost, imiquimod, infliximab, interleukin-12, IPI-504, irofulven, ixabepilone, lapatinib, lenalidomide, lestaurtinib, leuprolide, LMB-9 immunotoxin, lonafamib, luniliximab, mafosfamide, M1B07133, MDX-010, MLN2704, monoclonal antibody 3F8, monoclonal antibody J591, motexafin, MS-275, MVA-MUC1-IL2, nilutamide, nitrocamptothecin, nolatrexed dihydrochloride, nolvadex, NS-9, O6-benzylguanine, oblimersen sodium, ONYX-015, oregovomab, OSI-774, panitumumab, paraplatin, PD-0325901, pemetrexed, PHY906, pioglitazone, pirfenidone, pixantrone, PS-341, PSC 833, PXD101, pyrazoloacridine, R115777, RAD001, ranpirnase, rebeccamycin analogue, rhuAngiostatin protein, rhuMab 2C4, rosiglitazone, rubitecan, S-1, S-8184, satraplatin, SB-, 15992, SGN-0010, SGN-40, sorafenib, SR31747A, ST1571, SU011248, suberoylanilide hydroxamic acid, suramin, talabostat, talampal, tariquidar, temsirolimus, TGF α -PE38 immunotoxin, thalidomide, thymalfasin, tipifarnib, tirapazamine, TLK286, trabectedin, trimetrexate glucuronate, TroVax, UCN-1, valproic acid, vinflunine, VNP40101M, volociximab, vorinostat, VX-680, ZD1839, ZD6474, zileuton, and zosuquidar trihydrochloride.

[0091] The present invention provides methods for administering a compound of the invention with radiation therapy. The invention is not limited by the types, amounts, or delivery and administration systems used to deliver the therapeutic dose of radiation to an animal. For example, the animal may receive photon radiotherapy, particle beam radiation therapy, other types of radiotherapies, and combinations thereof. In some embodiments, the radiation is delivered to the animal using a linear accelerator. In still other embodiments, the radiation is delivered using a gamma knife.

[0092] The source of radiation can be external or internal to the animal. External radiation therapy is most common and involves directing a beam of high-energy radiation to a tumor site through the skin using, for instance, a linear accelerator. While the beam of radiation is localized to the tumor site, it is nearly impossible to avoid exposure of normal, healthy tissue. However, external radiation is usually well tolerated by animals. Internal radiation therapy involves implanting a radiation-emitting source, such as beads, wires, pellets, capsules, particles, and the like, inside the body at or near the tumor site including the use of delivery systems that specifically target cancer cells (e.g., using particles attached to cancer cell binding ligands). Such implants can be removed following treatment, or left in the body inactive. Types of internal radiation therapy include, but are not limited to, brachytherapy, interstitial irradiation, intracavity irradiation, radioimmunotherapy, and the like.

[0093] The animal may optionally receive radiosensitizers (e.g., metronidazole, misonidazole, intra-arterial Budr, intravenous iododeoxyuridine (IudR), nitroimidazole, 5-substituted-4-nitroimidazoles, 2H-isoindolediones, [[(2-bromoethyl)-amino]methyl]-nitro-1H-imidazole-1-ethanol,

nitroaniline derivatives, DNA-affinic hypoxia selective cytotoxins, halogenated DNA ligand, 1,2,4 benzotriazine oxides, 2-nitroimidazole derivatives, fluorine-containing nitroazole derivatives, benzamide, nicotinamide, acridine-intercalator, 5-thiotetrazole derivative, 3-nitro-1,2,4-triazole, 4,5-dinitroimidazole derivative, hydroxylated texaphrins, cisplatin, mitomycin, tiripazamine, nitrosourea, mercaptopurine, methotrexate, fluorouracil, bleomycin, vincristine, carboplatin, epirubicin, doxorubicin, cyclophosphamide, vindesine, etoposide, paclitaxel, heat (hyperthermia), and the like), radioprotectors (e.g., cysteamine, aminoalkyl dihydrogen phosphorothioates, amifostine (WR 2721), IL-1, IL-6, and the like). Radiosensitizers enhance the killing of tumor cells. Radioprotectors protect healthy tissue from the harmful effects of radiation.

[0094] Any type of radiation can be administered to an animal, so long as the dose of radiation is tolerated by the animal without unacceptable negative side-effects. Suitable types of radiotherapy include, for example, ionizing (electromagnetic) radiotherapy (e.g., X-rays or gamma rays) or particle beam radiation therapy (e.g., high linear energy radiation). Ionizing radiation is defined as radiation comprising particles or photons that have sufficient energy to produce ionization, i.e., gain or loss of electrons (as described in, for example, U.S. Pat. No. 5,770,581 incorporated herein by reference in its entirety). The effects of radiation can be at least partially controlled by the clinician. In one embodiment, the dose of radiation is fractionated for maximal target cell exposure and reduced toxicity.

[0095] In one embodiment, the total dose of radiation administered to an animal is about 0.01 Gray (Gy) to about 100 Gy. In another embodiment, about 10 Gy to about 65 Gy (e.g., about 15 Gy, 20 Gy, 25 Gy, 30 Gy, 35 Gy, 40 Gy, 45 Gy, 50 Gy, 55 Gy, or 60 Gy) are administered over the course of treatment. While in some embodiments a complete dose of radiation can be administered over the course of one day, the total dose is ideally fractionated and administered over several days. Desirably, radiotherapy is administered over the course of at least about 3 days, e.g., at least 5, 7, 10, 14, 17, 21, 25, 28, 32, 35, 38, 42, 46, 52, or 56 days (about 1-8 weeks). Accordingly, a daily dose of radiation will comprise approximately 1-5 Gy (e.g., about 1 Gy, 1.5 Gy, 1.8 Gy, 2 Gy, 2.5 Gy, 2.8 Gy, 3 Gy, 3.2 Gy, 3.5 Gy, 3.8 Gy, 4 Gy, 4.2 Gy, or 4.5 Gy), or 1-2 Gy (e.g., 1.5-2 Gy). The daily dose of radiation should be sufficient to induce destruction of the targeted cells. If stretched over a period, in one embodiment, radiation is not administered every day, thereby allowing the animal to rest and the effects of the therapy to be realized. For example, radiation desirably is administered on 5 consecutive days, and not administered on 2 days, for each week of treatment, thereby allowing 2 days of rest per week. However, radiation can be administered 1 day/week, 2 days/week, 3 days/week, 4 days/week, 5 days/week, 6 days/week, or all 7 days/week, depending on the animal's responsiveness and any potential side effects. Radiation therapy can be initiated at any time in the therapeutic period. In one embodiment, radiation is initiated in week 1 or week 2, and is administered for the remaining duration of the therapeutic period. For example, radiation is administered in weeks 1-6 or in weeks 2-6 of a therapeutic period comprising 6 weeks for treating, for instance, a solid tumor. Alternatively, radiation is administered in weeks 1-5 or weeks 2-5 of a therapeutic period comprising 5 weeks. These exem-

plary radiotherapy administration schedules are not intended, however, to limit the present invention.

[0096] Antimicrobial therapeutic agents may also be used as therapeutic agents in the present invention. Any agent that can kill, inhibit, or otherwise attenuate the function of microbial organisms may be used, as well as any agent contemplated to have such activities. Antimicrobial agents include, but are not limited to, natural and synthetic antibiotics, antibodies, inhibitory proteins (e.g., defensins), anti-sense nucleic acids, membrane disruptive agents and the like, used alone or in combination. Indeed, any type of antibiotic may be used including, but not limited to, antibacterial agents, antiviral agents, antifungal agents, and the like.

[0097] In some embodiments of the present invention, a compound of the invention and one or more therapeutic agents or anticancer agents are administered to an animal under one or more of the following conditions: at different periodicities, at different durations, at different concentrations, by different administration routes, etc. In some embodiments, the compound is administered prior to the therapeutic or anticancer agent, e.g., 0.5, 1, 2, 3, 4, 5, 10, 12, or 18 hours, 1, 2, 3, 4, 5, or 6 days, or 1, 2, 3, or 4 weeks prior to the administration of the therapeutic or anticancer agent. In some embodiments, the compound is administered after the therapeutic or anticancer agent, e.g., 0.5, 1, 2, 3, 4, 5, 10, 12, or 18 hours, 1, 2, 3, 4, 5, or 6 days, or 1, 2, 3, or 4 weeks after the administration of the anticancer agent. In some embodiments, the compound and the therapeutic or anticancer agent are administered concurrently but on different schedules, e.g., the compound is administered daily while the therapeutic or anticancer agent is administered once a week, once every two weeks, once every three weeks, or once every four weeks. In other embodiments, the compound is administered once a week while the therapeutic or anticancer agent is administered daily, once a week, once every two weeks, once every three weeks, or once every four weeks.

[0098] Compositions within the scope of this invention include all compositions wherein the compounds of the present invention are contained in an amount which is effective to achieve its intended purpose. While individual needs vary, determination of optimal ranges of effective amounts of each component is within the skill of the art. Typically, the compounds may be administered to mammals, e.g. humans, orally at a dose of 0.0025 to 50 mg/kg, or an equivalent amount of the pharmaceutically acceptable salt thereof, per day of the body weight of the mammal being treated for disorders responsive to induction of apoptosis. In one embodiment, about 0.01 to about 25 mg/kg is orally administered to treat, ameliorate, or prevent such disorders. For intramuscular injection, the dose is generally about one-half of the oral dose. For example, a suitable intramuscular dose would be about 0.0025 to about 25 mg/kg, or from about 0.01 to about 5 mg/kg.

[0099] The unit oral dose may comprise from about 0.01 to about 1000 mg, for example, about 0.1 to about 100 mg of the compound. The unit dose may be administered one or more times daily as one or more tablets or capsules each containing from about 0.1 to about 10 mg, conveniently about 0.25 to 50 mg of the compound or its solvates.

[0100] In a topical formulation, the compound may be present at a concentration of about 0.01 to 100 mg per gram of carrier. In a one embodiment, the compound is present at

a concentration of about 0.07-1.0 mg/ml, for example, about 0.1-0.5 mg/ml, and in one embodiment, about 0.4 mg/ml.

[0101] In addition to administering the compound as a raw chemical, the compounds of the invention may be administered as part of a pharmaceutical preparation containing suitable pharmaceutically acceptable carriers comprising excipients and auxiliaries which facilitate processing of the compounds into preparations which can be used pharmaceutically. The preparations, particularly those preparations which can be administered orally or topically and which can be used for one type of administration, such as tablets, dragees, slow release lozenges and capsules, mouth rinses and mouth washes, gels, liquid suspensions, hair rinses, hair gels, shampoos and also preparations which can be administered rectally, such as suppositories, as well as suitable solutions for administration by intravenous infusion, injection, topically or orally, contain from about 0.01 to 99 percent, in one embodiment from about 0.25 to 75 percent of active compound(s), together with the excipient.

[0102] The pharmaceutical compositions of the invention may be administered to any patient which may experience the beneficial effects of the compounds of the invention. Foremost among such patients are mammals, e.g., humans, although the invention is not intended to be so limited. Other patients include veterinary animals (cows, sheep, pigs, horses, dogs, cats and the like).

[0103] The compounds and pharmaceutical compositions thereof may be administered by any means that achieve their intended purpose. For example, administration may be by parenteral, subcutaneous, intravenous, intramuscular, intraperitoneal, transdermal, buccal, intrathecal, intracranial, intranasal or topical routes. Alternatively, or concurrently, administration may be by the oral route. The dosage administered will be dependent upon the age, health, and weight of the recipient, kind of concurrent treatment, if any, frequency of treatment, and the nature of the effect desired.

[0104] The pharmaceutical preparations of the present invention are manufactured in a manner which is itself known, for example, by means of conventional mixing, granulating, dragee-making, dissolving, or lyophilizing processes. Thus, pharmaceutical preparations for oral use can be obtained by combining the active compounds with solid excipients, optionally grinding the resulting mixture and processing the mixture of granules, after adding suitable auxiliaries, if desired or necessary, to obtain tablets or dragee cores.

[0105] Suitable excipients are, in particular, fillers such as saccharides, for example lactose or sucrose, mannitol or sorbitol, cellulose preparations and/or calcium phosphates, for example tricalcium phosphate or calcium hydrogen phosphate, as well as binders such as starch paste, using, for example, maize starch, wheat starch, rice starch, potato starch, gelatin, tragacanth, methyl cellulose, hydroxypropylmethylcellulose, sodium carboxymethylcellulose, and/or polyvinyl pyrrolidone. If desired, disintegrating agents may be added such as the above-mentioned starches and also carboxymethyl-starch, cross-linked polyvinyl pyrrolidone, agar, or alginic acid or a salt thereof, such as sodium alginate. Auxiliaries are, above all, flow-regulating agents and lubricants, for example, silica, talc, stearic acid or salts thereof, such as magnesium stearate or calcium stearate, and/or polyethylene glycol. Dragee cores are provided with suitable coatings which, if desired, are resistant to gastric juices. For this purpose, concentrated saccharide solutions

may be used, which may optionally contain gum arabic, talc, polyvinyl pyrrolidone, polyethylene glycol and/or titanium dioxide, lacquer solutions and suitable organic solvents or solvent mixtures. In order to produce coatings resistant to gastric juices, solutions of suitable cellulose preparations such as acetylcellulose phthalate or hydroxypropylmethylcellulose phthalate, are used. Dye stuffs or pigments may be added to the tablets or dragee coatings, for example, for identification or in order to characterize combinations of active compound doses.

[0106] Other pharmaceutical preparations which can be used orally include push-fit capsules made of gelatin, as well as soft, sealed capsules made of gelatin and a plasticizer such as glycerol or sorbitol. The push-fit capsules can contain the active compounds in the form of granules which may be mixed with fillers such as lactose, binders such as starches, and/or lubricants such as talc or magnesium stearate and, optionally, stabilizers. In soft capsules, the active compounds are in one embodiment dissolved or suspended in suitable liquids, such as fatty oils, or liquid paraffin. In addition, stabilizers may be added.

[0107] Possible pharmaceutical preparations which can be used rectally include, for example, suppositories, which consist of a combination of one or more of the active compounds with a suppository base. Suitable suppository bases are, for example, natural or synthetic triglycerides, or paraffin hydrocarbons. In addition, it is also possible to use gelatin rectal capsules which consist of a combination of the active compounds with a base. Possible base materials include, for example, liquid triglycerides, polyethylene glycols, or paraffin hydrocarbons.

[0108] Suitable formulations for parenteral administration include aqueous solutions of the active compounds in water-soluble form, for example, water-soluble salts and alkaline solutions. In addition, suspensions of the active compounds as appropriate oily injection suspensions may be administered. Suitable lipophilic solvents or vehicles include fatty oils, for example, sesame oil, or synthetic fatty acid esters, for example, ethyl oleate or triglycerides or polyethylene glycol-400. Aqueous injection suspensions may contain substances which increase the viscosity of the suspension include, for example, sodium carboxymethyl cellulose, sorbitol, and/or dextran. Optionally, the suspension may also contain stabilizers.

[0109] The topical compositions of this invention are formulated in one embodiment as oils, creams, lotions, ointments and the like by choice of appropriate carriers. Suitable carriers include vegetable or mineral oils, white petrolatum (white soft paraffin), branched chain fats or oils, animal fats and high molecular weight alcohol (greater than C₁₂). The carriers may be those in which the active ingredient is soluble. Emulsifiers, stabilizers, humectants and antioxidants may also be included as well as agents imparting color or fragrance, if desired. Additionally, transdermal penetration enhancers can be employed in these topical formulations. Examples of such enhancers can be found in U.S. Pat. Nos. 3,989,816 and 4,444,762; each herein incorporated by reference in its entirety.

[0110] Ointments may be formulated by mixing a solution of the active ingredient in a vegetable oil such as almond oil with warm soft paraffin and allowing the mixture to cool. A typical example of such an ointment is one which includes about 30% almond oil and about 70% white soft paraffin by weight. Lotions may be conveniently prepared by dissolving

the active ingredient, in a suitable high molecular weight alcohol such as propylene glycol or polyethylene glycol.

[0111] One of ordinary skill in the art will readily recognize that the foregoing represents merely a detailed description of certain preferred embodiments of the present invention. Various modifications and alterations of the compositions and methods described above can readily be achieved using expertise available in the art and are within the scope of the invention.

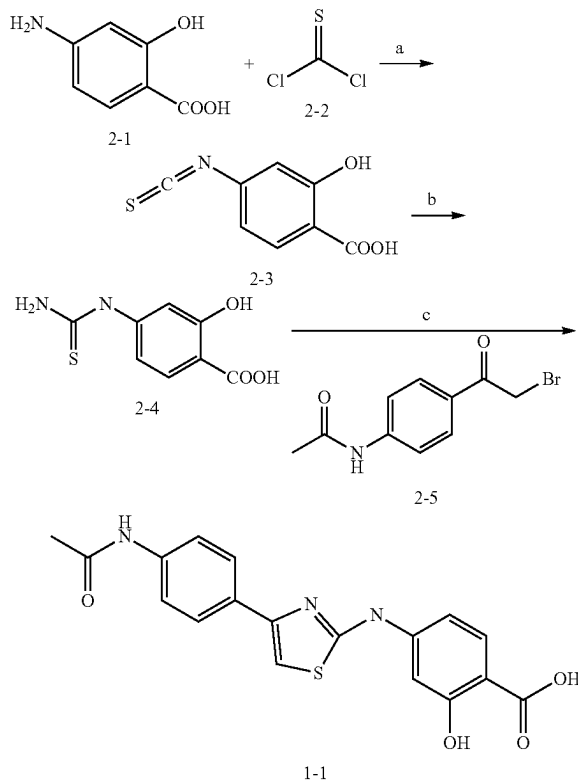
EXPERIMENTAL

[0112] The following examples are illustrative, but not limiting, of the compounds, compositions, and methods of the present invention. Other suitable modifications and adaptations of the variety of conditions and parameters normally encountered in clinical therapy and which are obvious to those skilled in the art are within the spirit and scope of the invention. As used herein, the use of personal pronouns such as “we”, “I”, and/or “our” refers to the inventors.

Example 1

Preparation of 4-((4-(4-acetamidophenyl)thiazol-2-yl)amino)-2-hydroxybenzoic acid (Compound 1-1)

Scheme 1. Synthesis of 1-1^a



^aReaction conditions: (a) HCl, H₂O, rt; (b) aqueous ammonia, rt; N-(4-(2-bromoacetyl)phenyl)acetamide, EtOH, reflux.

[0113] 2-hydroxy-4-isothiocyanatobenzoic acid (2-3). 4-Aminosalicylic acid (3.06 g, 0.02 mol) was suspended in

35 mL of water and 6.7 mL of concentrated hydrochloric acid added with mechanical stirring. To this suspension another 35 mL portion of water was added and finally thiophosgene (2.7 g, 0.024 mol) was added in one portion. After 3 hours of stirring, the orange color of the thiophosgene was dissipated indicating completion of the reaction. The solid product was filtered; the filter cake was washed several times with water and dried over phosphorus pentoxide. This product (3.35 g, 86% yield) of white solid that was recrystallized from toluene. The pure product was slightly unstable to light. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 7.83 (d, $J=8.4$ Hz, 1H), 7.00 (d, $J=2.0$ Hz, 1H), 6.96 (dd, $J=8.4$, 2.1 Hz, 1H).

[0114] 2-hydroxy-4-thioureidobenzoic acid (2-4). Compound 2-3 (3.756 g, 0.019 mol) was dissolved in 70 mL of aqueous ammonia (28%) and the solution stirred overnight. This product (2.98 g, 73% yield) of white solid was recrystallized from aqueous ethanol. ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 11.38 (s, 1H), 9.97 (s, 1H), 8.07 (s, 1H), 7.70 (d, $J=8.7$ Hz, 1H), 7.38 (d, $J=2.1$ Hz, 1H), 6.96 (dd, $J=8.7$, 2.2 Hz, 1H).

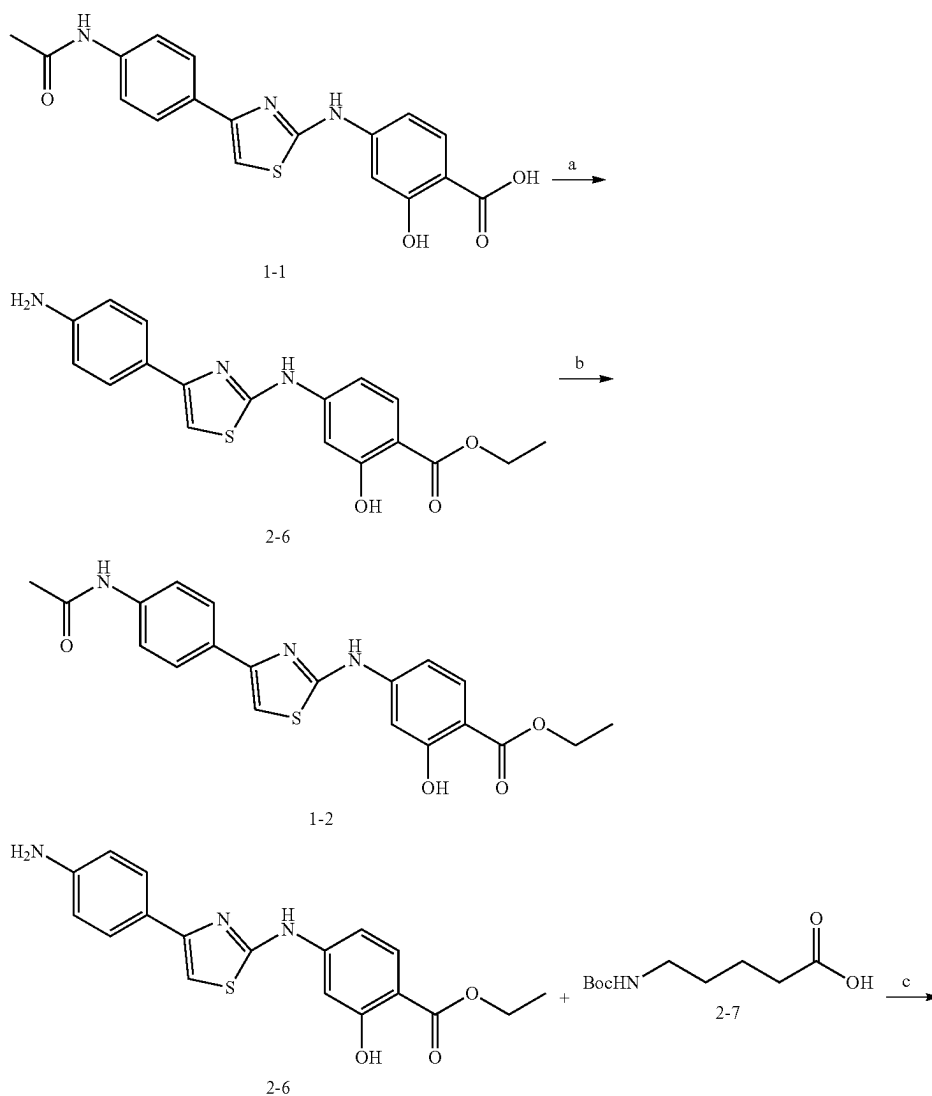
δ 11.38 (s, 1H), 9.97 (s, 1H), 8.07 (s, 1H), 7.70 (d, $J=8.7$ Hz, 1H), 7.38 (d, $J=2.1$ Hz, 1H), 6.96 (dd, $J=8.7$, 2.2 Hz, 1H).

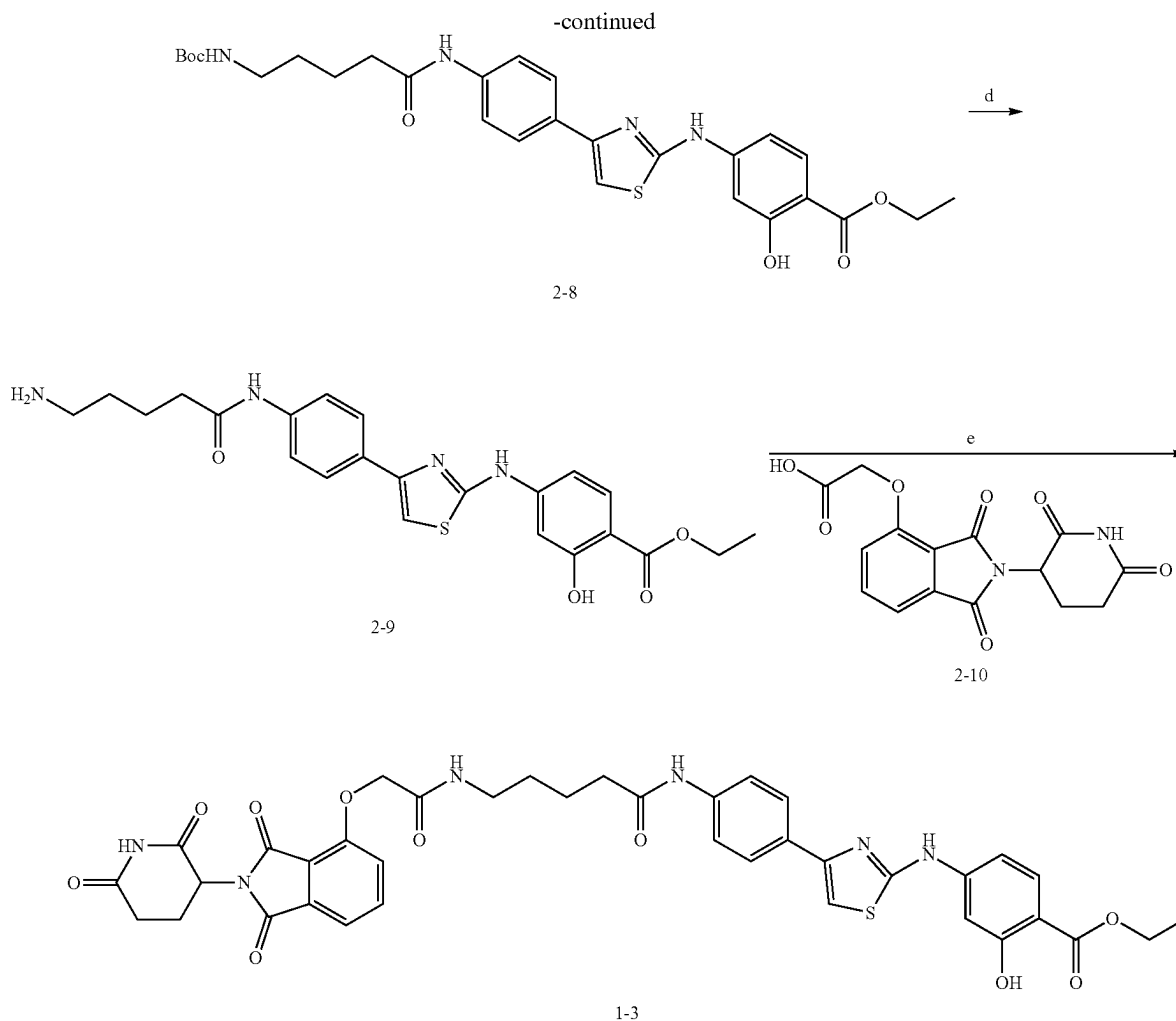
[0115] 4-((4-(4-acetamidophenyl)thiazol-2-yl)amino)-2-hydroxybenzoic acid (1-1). To a solution of compound 2-4 (424 mg, 2.0 mmol) in ethanol was added N-(4-(2-bromoacetyl)phenyl)acetamide 2-5 (512 mg, 2.0 mmol). The reaction mixture was refluxed for 3 h. Preparative HPLC purification afforded the pure product 1-1 as a white solid (400 mg, 54% yield). ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 10.72 (s, 1H), 10.10 (s, 1H), 7.84 (d, $J=8.5$ Hz, 2H), 7.74 (d, $J=8.7$ Hz, 1H), 7.67 (d, $J=8.5$ Hz, 2H), 7.56 (d, $J=2.1$ Hz, 1H), 7.34 (s, 1H), 7.10 (dd, $J=8.9$, 2.1 Hz, 1H), 2.07 (s, 3H). LCMS (ESI) 370 $[\text{M}+\text{H}]^+$, 368 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 99.5%.

Example 2

Preparation of ethyl4-((4-(4-acetamidophenyl)thiazol-2-yl)amino)-2-hydroxybenzoate (Compound 1-2)

Scheme 2. Synthesis of 1-2 and 1-3^a





^aReaction conditions: (a) H₂SO₄, EtOH, reflux; (b) acetylchloride, DMF, 0° C. to rt; (c) HATU, DIPEA, DMF, rt; (d) TFA, DCM, rt; (e) 2-((2-(2,6-dioxopiperidin-3-yl)-1,2-dioxoisindolin-4-yl)oxy)acetic acid, HATU, DIPEA, DMF, rt.

[0116] Ethyl 4-((4-(4-aminophenyl)thiazol-2-yl)amino)-2-hydroxybenzoate (2-6). To a solution of compound 1-1 (370 mg, 1 mmol) in ethanol (3 mL) was added conc. H₂SO₄ (0.2 mL). The reaction mixture was refluxed for 8 h. Preparative HPLC purification afforded the pure product 2-6 as a white solid (200 mg, 56% yield).

[0117] ¹H NMR (400 MHz, DMSO-d₆) δ 10.88 (s, 1H), 10.70 (s, 1H), 7.84-7.80 (m, 2H), 7.77 (d, J=8.8 Hz, 1H), 7.59 (d, J=2.1 Hz, 1H), 7.30 (s, 1H), 7.12 (dd, J=8.8, 2.2 Hz, 1H), 7.02 (d, J=8.5 Hz, 2H), 4.35 (q, J=7.1 Hz, 2H), 1.35 (t, J=7.1 Hz, 3H). MS (ESI) 356 [M+H]⁺.

[0118] Ethyl 4-((4-(4-acetamidophenyl)thiazol-2-yl)amino)-2-hydroxybenzoate (1-2). Compound 2-6 (30 mg, 0.085 mmol) was dissolved in anhydrous DMF (2 mL). Acetylchloride (10 μL, 0.141 mmol) and DIPEA (40 μL, 0.242 mmol) were added sequentially at 0° C. The reaction mixture was then allowed to reach ambient temperature and stirred for 2 h. Preparative HPLC purification afforded the pure product 1-2 as a white solid (29 mg, 86% yield). ¹H NMR (400 MHz, DMSO-d₆) δ 10.87 (s, 1H), 10.71 (s, 1H), 10.06 (s, 1H), 7.86 (d, J=8.7 Hz, 2H), 7.78 (d, J=8.8 Hz, 1H),

7.67 (d, J=8.8 Hz, 2H), 7.57 (d, J=2.1 Hz, 1H), 7.36 (s, 1H), 7.15 (dd, J=8.8, 2.2 Hz, 1H), 4.35 (q, J=7.1 Hz, 2H), 2.07 (s, 3H), 1.35 (t, J=7.1 Hz, 3H). LCMS (ESI) 398 [M+H]⁺, 396 [M-H]⁻. HPLC purity at 254 nm, 99.6%.

Example 3

Preparation of ethyl 4-((4-(4-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetamido)pentanamido)phenyl)thiazol-2-yl)amino)-2-hydroxybenzoate (compound 1-3)

[0119] Ethyl 4-((4-(4-(5-((tert-butoxycarbonyl)amino)pentanamido)phenyl)thiazol-2-yl)amino)-2-hydroxybenzoate (2-8). Compound 2-6 (35 mg, 0.1 mmol) was dissolved in anhydrous DMF (2 mL). 5-((tert-butoxycarbonyl)amino)pentanoic acid 2-7 (22 mg, 0.1 mmol), HATU (42 mg, 0.11 mmol), and DIPEA (50 μL, 0.3 mmol) were added sequentially. The reaction mixture was stirred at room temperature for 2 h and then purified by HPLC to isolate the product as a slightly yellow solid (40 mg, 72% yield). ¹H NMR (400 MHz, DMSO-d₆) δ 10.87 (s, 1H), 10.71 (s, 1H), 9.99 (s,

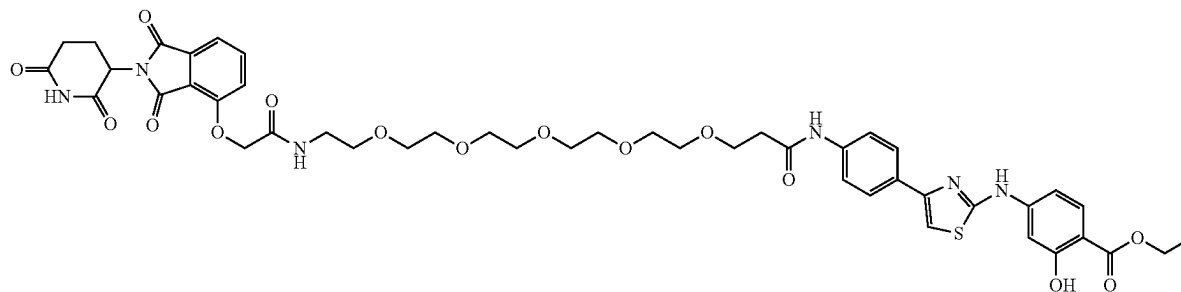
1H), 7.89-7.82 (m, 2H), 7.78 (d, J=8.8 Hz, 1H), 7.68 (d, J=8.8 Hz, 2H), 7.56 (d, J=2.1 Hz, 1H), 7.37 (s, 1H), 7.15 (dd, J=8.8, 2.2 Hz, 1H), 6.83 (t, J=5.6 Hz, 1H), 4.35 (q, J=7.1 Hz, 2H), 2.95 (q, J=6.7 Hz, 2H), 2.33 (t, J=7.3 Hz, 2H), 1.59 (p, J=7.6 Hz, 2H), 1.43 (q, J=7.5 Hz, 2H), 1.38 (s, 9H), 1.35 (t, J=7.1 Hz, 3H). MS (ESI) 555 [M+H]⁺.

[0120] Intermediate 2-8 (30 mg, 0.054 mmol) was dissolved in DCM (2 mL) and TFA (1 mL). After stirring for 2 h, the solvent was evaporated and the residue 2-9 was used in the next step without further purification.

[0121] Ethyl 4-((4-(4-(5-(2-((2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-acetamido)pentanamido)phenyl)thiazol-2-yl)amino)-2-hydroxybenzoate (1-3). The amine intermediate 2-9 (25 mg, 0.055 mmol), HATU (25 mg, 0.065 mmol) and DIPEA (30 μ L, 0.18 mmol) were added sequentially to a stirred solution of compound 2-((2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)acetic acid 2-10 (18 mg, 0.055 mmol) in DMF (2 mL). The reaction mixture was stirred at room temperature for 2 h and then purified by HPLC to get the product as a white solid (32 mg, 75% yield). ¹H NMR (400 MHz, DMSO-d₆) δ 11.13 (s, 1H), 10.88 (s, 1H), 10.72 (s, 1H), 10.01 (s, 1H), 8.03 (t, J=5.7 Hz, 1H), 7.89-7.75 (m, 4H), 7.68 (d, J=8.5 Hz, 2H), 7.57 (d, J=2.1 Hz, 1H), 7.49 (d, J=7.2 Hz, 1H), 7.41 (d, J=8.6 Hz, 1H), 7.36 (s, 1H), 7.15 (dd, J=8.8, 2.1 Hz, 1H), 5.13 (dd, J=12.9, 5.4 Hz, 1H), 4.80 (s, 2H), 4.35 (q, J=7.1 Hz, 2H), 3.20 (q, J=6.5 Hz, 2H), 2.90 (ddd, J=18.0, 14.0, 5.5 Hz, 1H), 2.64-2.55 (m, 2H), 2.36 (t, J=7.2 Hz, 2H), 2.05 (ddt, J=12.0, 7.0, 3.9 Hz, 1H), 1.63 (dq, J=12.5, 7.2 Hz, 2H), 1.51 (p, J=6.9 Hz, 2H), 1.35 (t, J=7.1 Hz, 3H). LCMS (ESI) 769 [M+H]⁺, 767 [M-H]⁻. HPLC purity at 254 nm, 99.5%.

Example 4

Preparation of ethyl 4-((4-(4-(1-((2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)oxy)-2-oxo-6,9,12,15,18-pentaoxa-3-azahenicosan-21-amido)phenyl)thiazol-2-yl)amino)-2-hydroxybenzoate (compound 1-4)



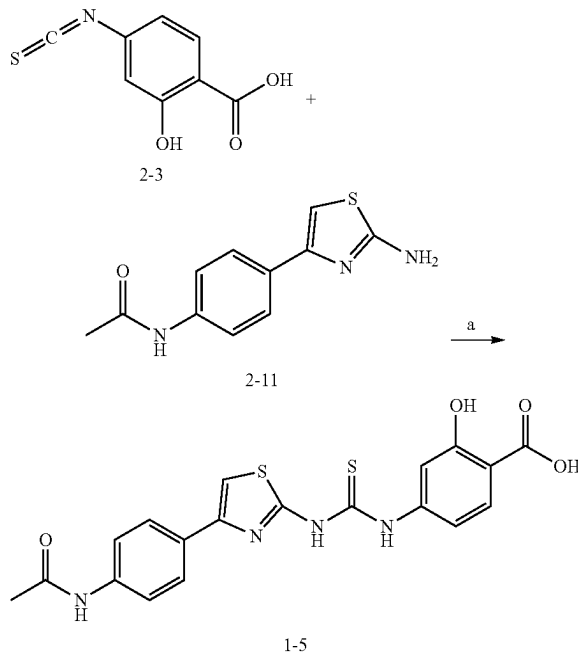
[0122] Following similar procedure used to prepare compound 1-3, we synthesized compound 1-4. ¹H NMR (400 MHz, DMSO-d₆) δ 11.13 (s, 1H), 10.87 (s, 1H), 10.70 (s, 1H), 10.06 (s, 1H), 8.01 (t, J=5.7 Hz, 1H), 7.90-7.75 (m, 4H), 7.71-7.64 (m, 2H), 7.56 (d, J=2.1 Hz, 1H), 7.50 (d, J=7.2 Hz, 1H), 7.40 (dd, J=8.6, 0.6 Hz, 1H), 7.36 (s, 1H), 7.14 (dd, J=8.8, 2.2 Hz, 1H), 5.12 (dd, J=12.9, 5.4 Hz, 1H), 4.79 (s, 2H), 4.35 (q, J=7.1 Hz, 2H), 3.72 (t, J=6.2 Hz, 2H), 3.56-3.42 (m, 18H), 3.31 (q, J=5.7 Hz, 2H), 2.90 (ddd,

J=17.3, 14.0, 5.4 Hz, 1H), 2.65-2.54 (m, 4H), 2.10-1.98 (m, 1H), 1.35 (t, J=7.1 Hz, 3H). LCMS (ESI) 961 [M+H]⁺, 959 [M-H]⁻. HPLC purity at 254 nm, 99.5%.

Example 5

Preparation of 4-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-5)

Scheme 3. Synthesis of 1-5^a



^aReaction conditions: (a) pyridine, 60° C.

1-4

4-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (1-5)

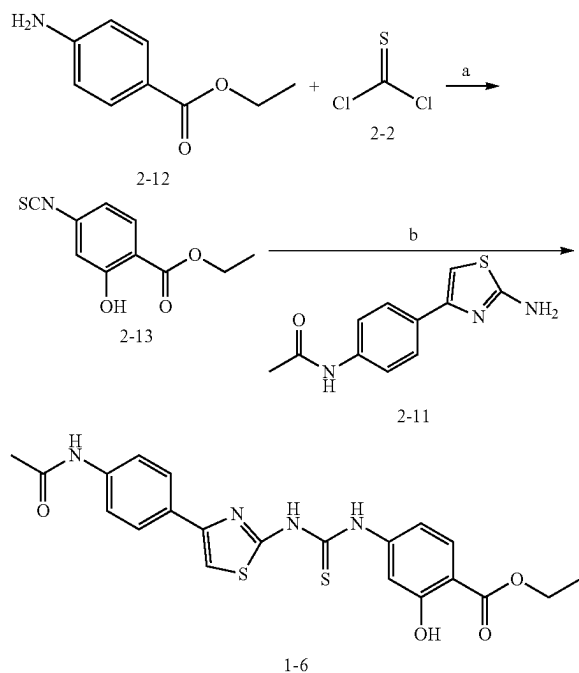
[0123] Intermediate 2-3 (20 mg, 0.1 mmol) and amine 2-11 (23 mg, 0.1 mmol) were dissolved in anhydrous pyridine (2 mL). The reaction mixture was stirred at 60° C. overnight and then purified by HPLC to get the product as a slightly yellow solid (26 mg, 60% yield). ¹H NMR (400 MHz, DMSO-d₆) δ 11.40 (s, 1H), 10.08 (s, 1H), 7.99-7.03

(m, 8H), 2.08 (d, $J=2.4$ Hz, 3H). LCMS (ESI) 429 $[M+H]^+$, 427 $[M-H]^-$. HPLC purity at 254 nm, 99.5%.

Example 6

Preparation of ethyl 4-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoate (compound 1-6)

Scheme 4. Synthesis of 1-6^a



^aReaction conditions: (a) CaCO_3 , DCM, H_2O , reflux; (b) pyridine, 60°C .

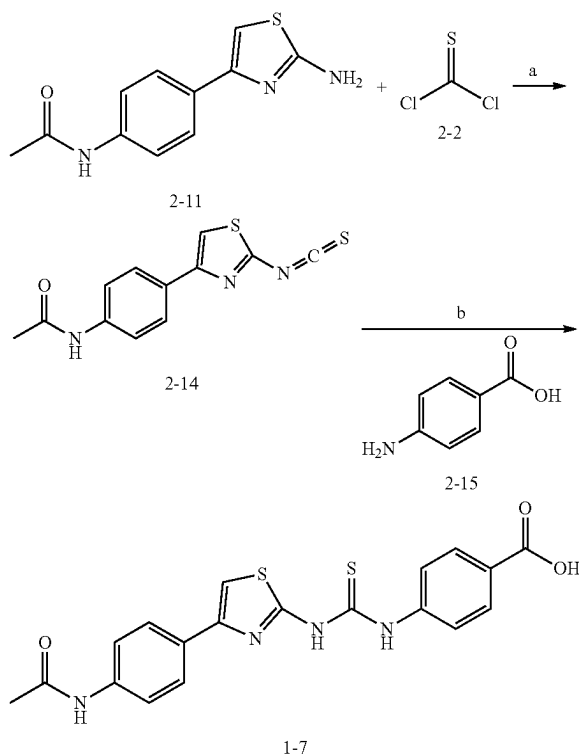
[0124] Ethyl 2-hydroxy-4-isothiocyanatobenzoate (2-13). A mixture of thiophosgene (228 μL , 3 mmol), CaCO_3 (175 mg, 1.75 mmol), DCM (2 mL) and H_2O (4 mL) was stirred at rt, then a solution of amine 2-12 (181 mg, 1 mmol) in DCM (2 mL) was added dropwise. The reaction mixture was refluxed for 48 h. The solid was suction-filtered and washed with CHCl_3 many times, the resulting mixture was extracted with CHCl_3 , then the organic layer was washed with saturated aqueous NaCl, the residue was purified by flash silica chromatography to give this product as a slightly yellow solid (90 mg, 40% yield). ^1H NMR (400 MHz, Chloroform- d_3) δ 11.00 (s, 1H), 7.85 (dd, $J=8.5$, 0.4 Hz, 1H), 6.86-6.81 (m, 1H), 6.75 (dd, $J=8.5$, 2.0 Hz, 1H), 4.44 (q, $J=7.1$ Hz, 2H), 1.44 (t, $J=7.2$ Hz, 3H).

[0125] Ethyl 4-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoate (1-6). Intermediate ethyl 2-hydroxy-4-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioamido)benzoate 2-13 (22 mg, 0.1 mmol) and amine 2-11 (23 mg, 0.1 mmol) were dissolved in anhydrous pyridine (2 mL). The reaction mixture was stirred at 60°C overnight and then purified by HPLC to obtain the product as a white solid (27 mg, 60% yield). ^1H NMR (400 MHz, DMSO- d_6) δ 10.75 (s, 1H), 10.10 (s, 1H), 7.85-7.23 (m, 8H), 4.36 (q, $J=7.1$ Hz, 2H), 2.08 (s, 3H), 1.35 (t, $J=7.1$ Hz, 3H). LCMS (ESI) 457 $[M+H]^+$, 455 $[M-H]^-$. HPLC purity at 254 nm, 99.2%.

Example 7

Preparation of 4-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-7)

Scheme 5. Synthesis of 1-7^a



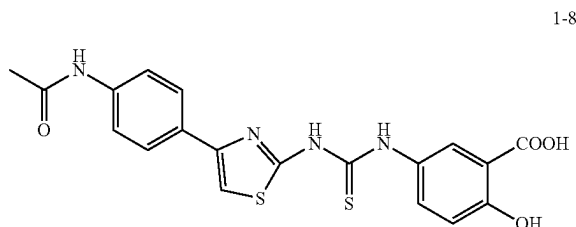
^aReaction conditions: (a) CaCO_3 , DCM, H_2O , rt; (b) pyridine, 60°C .

[0126] N-(4-(2-isothiocyanatothiazol-4-yl)phenyl)acetamide (2-14). A mixture of thiophosgene (114 μL , 1.5 mmol), CaCO_3 (175 mg, 1.75 mmol), DCM (2 mL) and H_2O (4 mL) was stirred at 0°C , then a solution of amine 2-11 (233 mg, 1 mmol) in DCM (2 mL) was added dropwise. The reaction mixture was stirred at room temperature overnight. The solid was suction-filtered and washed with CHCl_3 many times, the resulting mixture was extracted with CHCl_3 , then the organic layer was washed with saturated aqueous NaCl, the residue was purified by flash silica chromatography to obtain the product as a slightly yellow solid (22 mg, 8% yield). ^1H NMR (400 MHz, Chloroform- d_3) δ 7.84-7.78 (m, 2H), 7.59 (d, $J=8.5$ Hz, 2H), 7.32 (s, 1H), 7.26 (s, 1H), 2.22 (s, 3H). MS (ESI) 276 $[M+H]^+$.

[0127] 4-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioureido)benzoic acid (1-7). Intermediate 2-14 (28 mg, 0.1 mmol) and amine 2-15 (14 mg, 0.1 mmol) were dissolved in anhydrous pyridine (2 mL). The reaction mixture was stirred at 60°C overnight and then purified by HPLC to obtain the product as a white solid (26 mg, 62% yield). ^1H NMR (400 MHz, DMSO- d_6) δ 10.09 (s, 1H), 7.92 (q, $J=8.7$ Hz, 4H), 7.84-7.77 (m, 2H), 7.68 (d, $J=8.7$ Hz, 2H), 7.39 (s, 1H), 2.08 (s, 3H). LCMS (ESI) 413 $[M+H]^+$, 411 $[M-H]^-$. HPLC purity at 254 nm, 99.5%.

Example 8

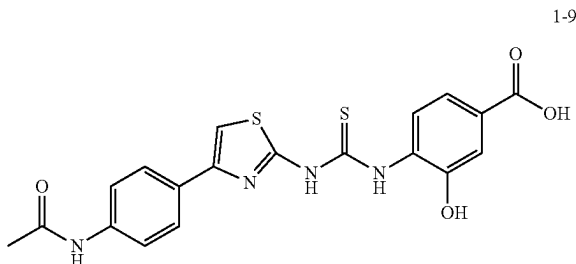
Preparation of 5-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-8)



[0128] Compound 1-8 was obtained following the procedure used to prepare compound 1-7. ¹H NMR (400 MHz, DMSO-d₆) δ 11.21 (s, 1H), 10.72 (s, 1H), 10.05 (s, 1H), 8.08-7.30 (m, 7H), 6.96 (m, 1H), 2.07 (s, 3H). LCMS (ESI) 429 [M+H]⁺, 427 [M-H]⁻. HPLC purity at 254 nm, 97.1%.

Example 9

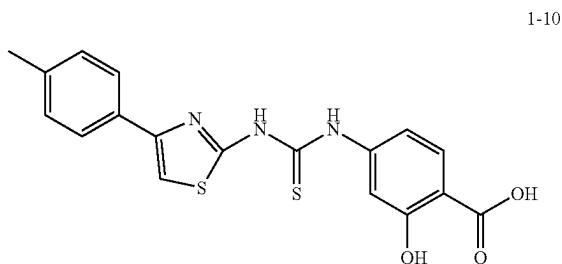
Preparation of 4-(3-(4-(4-acetamidophenyl)thiazol-2-yl)thioureido)-3-hydroxybenzoic acid (Compound 1-9)



[0129] Compound 1-9 was obtained following the procedure used to prepare compound 1-7. ¹H NMR (400 MHz, DMSO-d₆) δ 12.79 (s, 1H), 12.22 (s, 1H), 10.78 (s, 1H), 10.06 (s, 1H), 8.72 (s, 1H), 8.05-7.18 (m, 8H), 2.08 (s, 3H). LCMS (ESI) 429 [M+H]⁺, 427 [M-H]⁻. HPLC purity at 254 nm, 99.7%.

Example 10

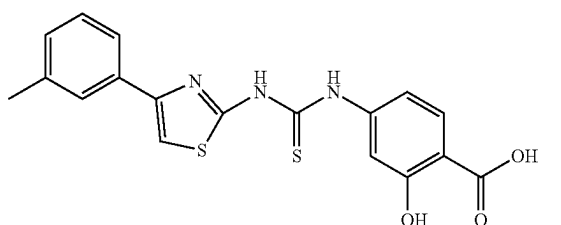
Preparation of 2-hydroxy-4-(3-(4-(p-tolyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-10)



[0130] Compound 1-10 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 7.81-7.60 (m, 4H), 7.46 (d, J=2.1 Hz, 1H), 7.29 (t, J=7.5 Hz, 3H), 2.34 (s, 3H). LCMS (ESI) 386 [M+H]⁺, 384 [M-H]⁻. HPLC purity at 254 nm, 99.5%.

Example 11

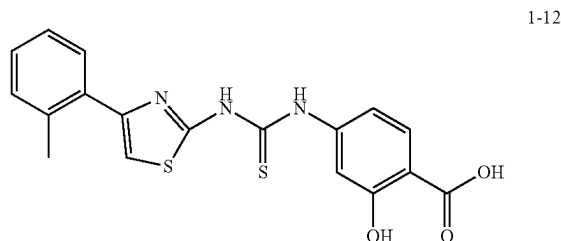
Preparation of 2-hydroxy-4-(3-(4-(m-tolyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-11)



[0131] Compound 1-11 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 7.83-7.02 (m, 8H), 2.37 (s, 3H). LCMS (ESI) 386 [M+H]⁺, 384 [M-H]⁻. HPLC purity at 254 nm, 99.5%.

Example 12

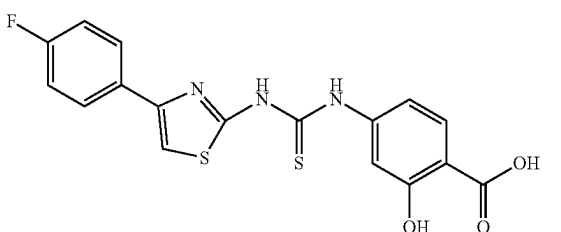
Preparation of 2-hydroxy-4-(3-(4-(o-tolyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-12)



[0132] Compound 1-12 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 11.37 (s, 1H), 7.73-7.60 (m, 2H), 7.53-7.45 (m, 1H), 7.40-7.24 (m, 4H), 7.07 (s, 1H), 2.38 (s, 3H). LCMS (ESI) 386 [M+H]⁺, 384 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 13

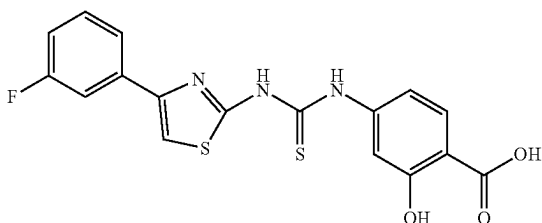
Preparation of 4-(3-(4-(4-fluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-13)



[0133] Compound 1-13 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 11.41 (s, 1H), 10.80 (s, 1H), 7.93 (ddd, J=9.1, 5.2, 2.2 Hz, 2H), 7.74 (dd, J=18.1, 8.6 Hz, 1H), 7.55 (m, 2H), 7.42-7.15 (m, 3H). LCMS (ESI) 390 [M+H]⁺, 388 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 14

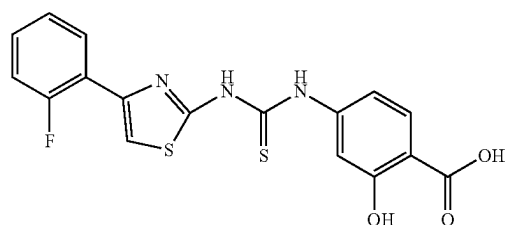
Preparation of 4-(3-(4-(3-fluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-14)



[0134] Compound 1-14 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.02 (s, 1H), δ 11.43 (s, 1H), 10.73 (s, 1H), δ 7.75 (q, J=11.0, 9.9 Hz, 4H), 7.63-7.42 (m, 2H), 7.21 (d, J=9.8 Hz, 2H). LCMS (ESI) 390 [M+H]⁺, 388 [M-H]⁻. HPLC purity at 254 nm, 99.4%.

Example 15

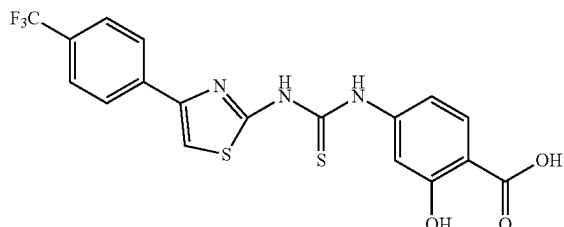
Preparation of 4-(3-(4-(2-fluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-15)



[0135] Compound 1-15 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.05 (s, 1H), δ 11.41 (s, 1H), 10.89 (s, 1H), 7.99 (s, 1H), 7.75 (dd, J=17.9, 8.8 Hz, 1H), 7.63 (d, J=2.1 Hz, 1H), 7.54-7.30 (m, 4H), 7.20 (s, 1H). LCMS (ESI) 390 [M+H]⁺, 388 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 16

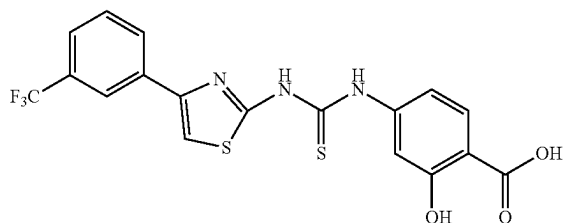
Preparation of 2-hydroxy-4-(3-(4-(4-(trifluoromethyl)phenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-16)



[0136] Compound 1-16 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.12 (s, 1H), δ 11.42 (s, 1H), 10.77 (s, 1H), δ 8.12 (d, J=8.0 Hz, 2H), 7.81 (dd, J=19.1, 8.4 Hz, 4H), 7.63 (d, J=2.1 Hz, 1H), 7.19 (s, 1H). LCMS (ESI) 440 [M+H]⁺, 438 [M-H]⁻. HPLC purity at 254 nm, 96.4%.

Example 17

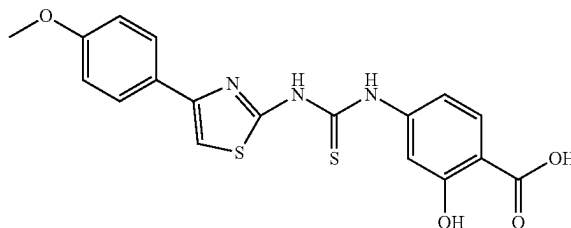
Preparation of 2-hydroxy-4-(3-(4-(3-(trifluoromethyl)phenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-17)



[0137] Compound 1-17 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.12 (s, 1H), δ 11.41 (s, 1H), 10.80 (s, 1H), δ 8.28-8.19 (m, 2H), 7.87-7.69 (m, 4H), 7.63 (d, J=2.1 Hz, 1H), 7.18 (s, 1H). LCMS (ESI) 440 [M+H]⁺, 438 [M-H]⁻. HPLC purity at 254 nm, 98.9%.

Example 18

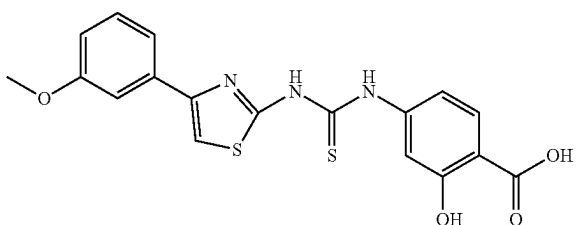
Preparation of 2-hydroxy-4-(3-(4-(4-methoxyphenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-18)



[0138] Compound 1-18 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 11.39 (s, 1H), 7.86-7.71 (m, 3H), 7.63 (s, 1H), 7.33 (s, 2H), 7.04 (dd, J=8.6, 6.6 Hz, 2H), 3.82 (s, 3H). LCMS (ESI) 402 [M+H]⁺, 400 [M-H]⁻. HPLC purity at 254 nm, 99.8%.

Example 19

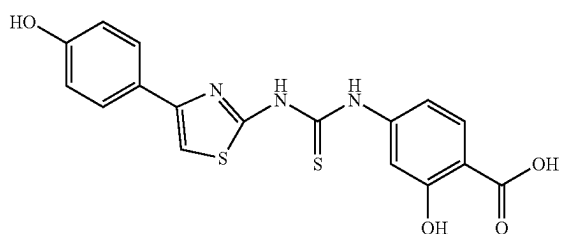
Preparation of 2-hydroxy-4-(3-(4-(3-methoxyphenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-19)



[0139] Compound 1-19 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 11.40 (s, 1H), 10.81 (s, 1H), 7.76 (d, J=8.7 Hz, 1H), 7.64-7.52 (m, 2H), 7.49-7.35 (m, 3H), 7.28 (s, 1H), 6.99-6.91 (m, 1H), 3.84 (s, 3H). LCMS (ESI) 402 [M+H]⁺, 400 [M-H]⁻. HPLC purity at 254 nm, 99.8%.

Example 20

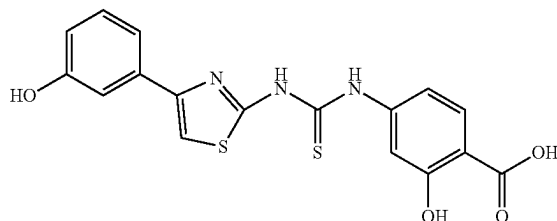
Preparation of 2-hydroxy-4-(3-(4-(4-hydroxyphenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-20)



[0140] Compound 1-20 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 11.38 (s, 1H), 9.73 (s, 1H), 7.80-7.56 (m, 4H), 7.50-7.12 (m, 2H), 6.85 (dd, J=8.6, 6.5 Hz, 2H). LCMS (ESI) 388 [M+H]⁺, 386 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 21

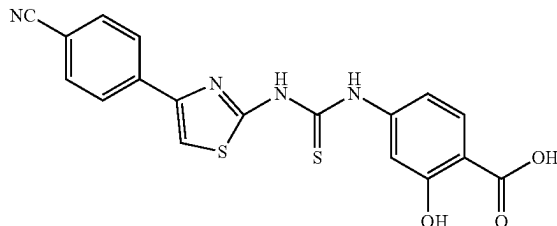
Preparation of 2-hydroxy-4-(3-(4-(3-hydroxyphenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-21)



[0141] Compound 1-21 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 11.39 (s, 1H), 9.61 (s, 1H), 7.76 (d, J=8.7 Hz, 1H), 7.63 (d, J=2.1 Hz, 1H), 7.50-7.37 (m, 1H), 7.35-7.18 (m, 4H), 6.80 (d, J=7.7 Hz, 1H). LCMS (ESI) 388 [M+H]⁺, 386 [M-H]⁻. HPLC purity at 254 nm, 99.5%.

Example 22

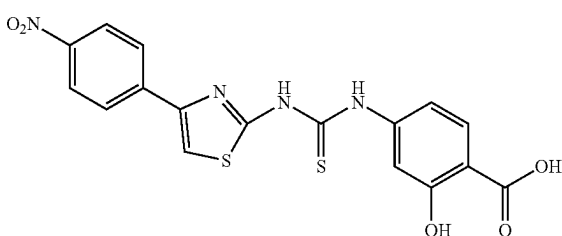
Preparation of 4-(3-(4-(4-cyanophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-22)



[0142] Compound 1-22 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.07 (s, 1H), 11.36 (d, J=34.0 Hz, 1H), 10.76 (s, 1H), 8.13-8.05 (m, 2H), 7.93 (dd, J=8.2, 5.5 Hz, 2H), 7.85 (d, J=6.5 Hz, 1H), 7.79 (d, J=8.6 Hz, 1H), 7.62 (d, J=2.1 Hz, 1H), 7.18 (s, 1H). LCMS (ESI) 397 [M+H]⁺, 395 [M-H]⁻. HPLC purity at 254 nm, 98.5%.

Example 23

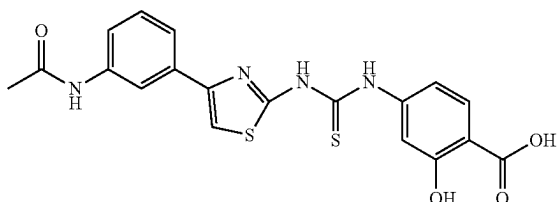
Preparation of 2-hydroxy-4-(3-(4-(4-nitrophenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-23)



[0143] Compound 1-23 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 12.12 (s, 1H), 11.41 (s, 1H), 10.75 (s, 1H), 8.39-8.30 (m, 2H), 8.21-8.15 (m, 2H), 7.95 (s, 1H), 7.80 (d, $J=8.6$ Hz, 1H), 7.62 (d, $J=2.1$ Hz, 1H), 7.17 (s, 1H). LCMS (ESI) 417 $[\text{M}+\text{H}]^+$, 415 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 99.2%.

Example 24

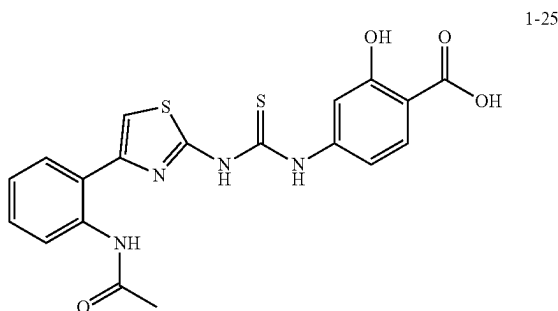
Preparation of 4-(3-(4-(3-acetamidophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-24)



[0144] Compound 1-24 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 10.05 (s, 1H), 8.26-7.10 (m, 8H), 2.08 (s, 3H). LCMS (ESI) 429 $[\text{M}+\text{H}]^+$, 427 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 99.9%.

Example 25

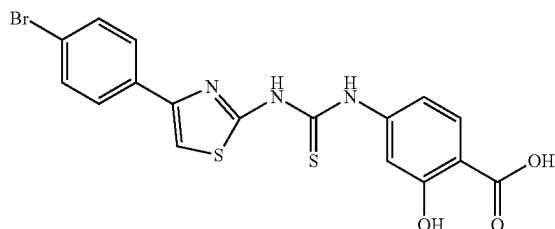
Preparation of 4-(3-(4-(2-acetamidophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-25)



[0145] Compound 1-25 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 13.57 (s, 1H), 12.07 (s, 1H), 11.35 (s, 1H), 8.29-6.87 (m, 8H), 2.16-2.02 (m, 3H). LCMS (ESI) 429 $[\text{M}+\text{H}]^+$, 427 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 98.2%.

Example 26

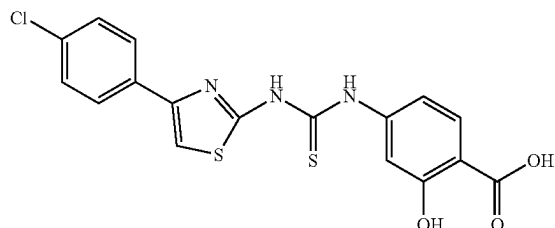
Preparation of 4-(3-(4-(4-bromophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-26)



[0146] Compound 1-26 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 12.05 (s, 1H), 11.40 (s, 1H), 10.81 (s, 1H), 7.89-7.83 (m, 2H), 7.77 (d, $J=8.7$ Hz, 1H), 7.72-7.54 (m, 4H), 7.21 (s, 1H). LCMS (ESI) 450 $[\text{M}+\text{H}]^+$. HPLC purity at 254 nm, 99.9%.

Example 27

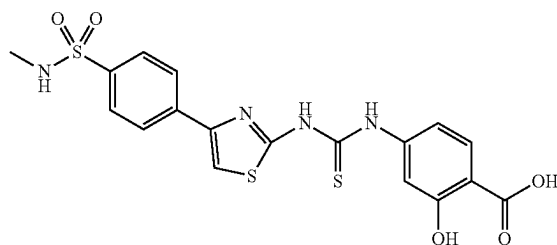
Preparation of 4-(3-(4-(4-chlorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-27)



[0147] Compound 1-27 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 12.03 (s, 1H), δ 11.40 (s, 1H), 10.81 (s, 1H), 7.98-7.88 (m, 2H), 7.77 (d, $J=8.7$ Hz, 1H), 7.69-7.49 (m, 4H), 7.21 (s, 1H). LCMS (ESI) 406 $[\text{M}+\text{H}]^+$, 404 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 99.0%.

Example 28

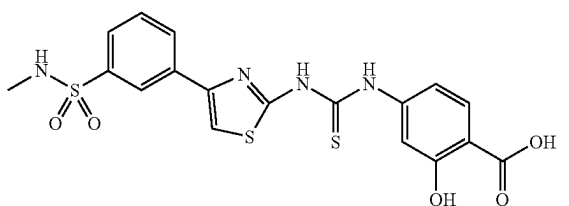
Preparation of 2-hydroxy-4-(3-(4-(4-(N-methylsulfonyl)phenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-28)



[0148] Compound 1-28 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 12.06 (s, 1H), 11.41 (s, 1H), 10.79 (s, 1H), 8.11 (d, $J=8.2$ Hz, 2H), 7.92-7.72 (m, 4H), 7.63 (d, $J=2.1$ Hz, 1H), 7.54-7.05 (m, 2H), 2.45 (dd, $J=5.0, 2.6$ Hz, 3H). LCMS (ESI) 465 $[\text{M}+\text{H}]^+$, 463 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 98.9%.

Example 29

Preparation of 2-hydroxy-4-(3-(4-(3-(N-methylsulfonyl)phenyl)thiazol-2-yl)thioureido)benzoic acid (Compound 1-29)

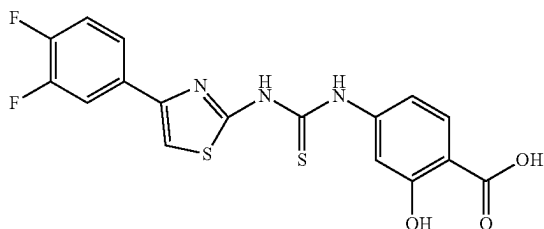


1-29

[0149] Compound 1-29 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 12.06 (s, 1H), 11.41 (s, 1H), 10.72 (s, 1H), 8.34 (s, 1H), 8.16 (d, $J=7.6$ Hz, 1H), 7.87-7.66 (m, 5H), 7.49 (dd, $J=7.4, 3.4$ Hz, 1H), 7.16 (d, $J=20.1$ Hz, 1H), 2.46 (dd, $J=5.0, 1.7$ Hz, 3H). LCMS (ESI) 465 $[\text{M}+\text{H}]^+$, 463 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 98.8%.

Example 30

Preparation of 4-(3-(4-(3,4-difluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-30)

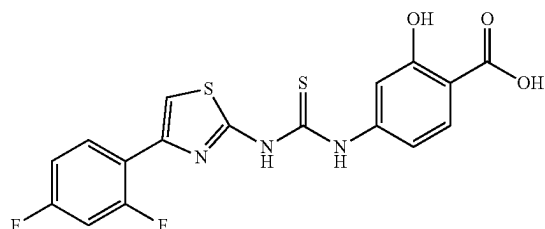


1-30

[0150] Compound 1-30 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 12.03 (s, 1H), 11.40 (s, 1H), 10.79 (s, 1H), 7.94 (t, $J=9.9$ Hz, 1H), 7.84-7.40 (m, 5H), 7.18 (s, 1H). LCMS (ESI) 408 $[\text{M}+\text{H}]^+$, 406 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 99.5%.

Example 31

Preparation of 4-(3-(4-(2,4-difluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-31)

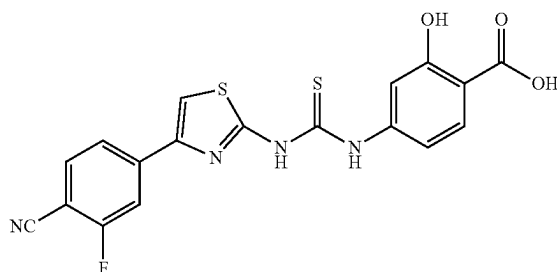


1-31

[0151] Compound 1-31 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 11.39 (s, 1H), 8.02 (s, 1H), 7.75 (dd, $J=19.7, 8.5$ Hz, 1H), 7.62 (d, $J=2.1$ Hz, 1H), 7.41 (d, $J=13.1$ Hz, 2H), 7.20 (d, $J=31.9$ Hz, 2H). LCMS (ESI) 408 $[\text{M}+\text{H}]^+$, 406 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 99.0%.

Example 32

Preparation of 4-(3-(4-(4-cyano-3-fluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-32)

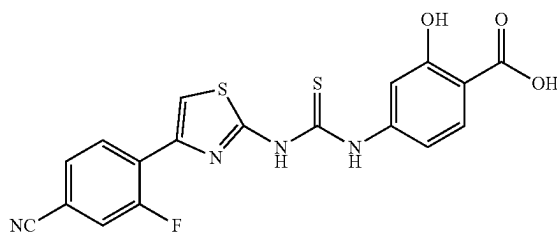


1-32

[0152] Compound 1-32 was obtained following the procedure used to prepare compound 1-5. ^1H NMR (400 MHz, DMSO-d_6) δ 12.14 (s, 1H), 11.39 (s, 1H), 10.73 (s, 1H), 8.01-7.93 (m, 4H), 7.76 (dd, $J=18.3, 8.6$ Hz, 1H), 7.59 (d, $J=2.1$ Hz, 1H), 7.20-7.13 (m, 1H). LCMS (ESI) 415 $[\text{M}+\text{H}]^+$, 413 $[\text{M}-\text{H}]^-$. HPLC purity at 254 nm, 98.8%.

Example 33

Preparation of 4-(3-(4-(4-cyano-2-fluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-33)

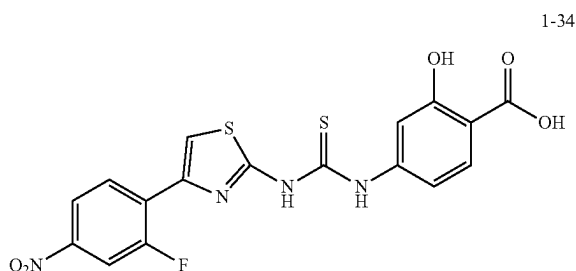


1-33

[0153] Compound 1-33 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.08 (s, 1H), 11.40 (s, 1H), 10.77 (s, 1H), 8.19 (q, J=7.8 Hz, 1H), 8.05-7.94 (m, 1H), 7.87-7.68 (m, 3H), 7.60 (d, J=2.1 Hz, 1H), 7.15 (d, J=9.2 Hz, 1H). LCMS (ESI) 415 [M+H]⁺, 413 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 34

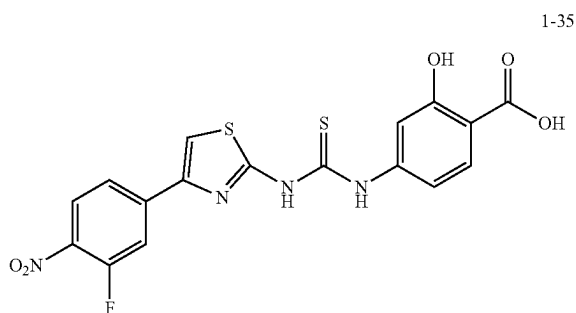
Preparation of 4-(3-(4-(2-fluoro-4-nitrophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-34)



[0154] Compound 1-34 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.15 (s, 1H), 11.41 (s, OH), 10.74 (s, 1H), 8.40-8.00 (m, 4H), 7.85-7.75 (m, 1H), 7.60 (d, J=2.1 Hz, 1H), 7.15 (d, J=9.0 Hz, 1H). LCMS (ESI) 435 [M+H]⁺, 433 [M-H]⁻. HPLC purity at 254 nm, 99.8%.

Example 35

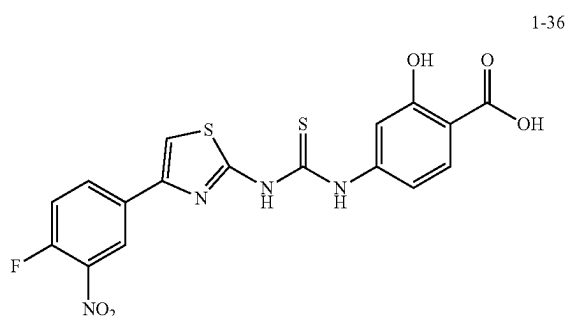
Preparation of 4-(3-(4-(3-fluoro-4-nitrophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-35)



[0155] Compound 1-35 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.09 (s, 1H), 11.40 (s, 1H), 10.73 (s, 1H), 8.33-8.22 (m, 1H), 8.14-7.91 (m, 3H), 7.77 (dd, J=18.4, 8.5 Hz, 1H), 7.60 (d, J=2.1 Hz, 1H), 7.25-7.03 (m, 1H). LCMS (ESI) 435 [M+H]⁺, 433 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 36

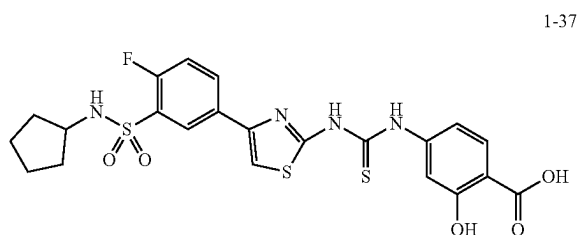
Preparation of 4-(3-(4-(4-fluoro-3-nitrophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-36)



[0156] Compound 1-36 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.10 (s, 1H), 11.41 (s, 1H), 10.75 (s, 1H), 8.64 (dt, J=7.3, 2.3 Hz, 1H), 8.32 (ddt, J=8.8, 4.5, 2.3 Hz, 1H), 7.89-7.58 (m, 4H), 7.12 (s, 1H). LCMS (ESI) 435 [M+H]⁺, 433 [M-H]⁻. HPLC purity at 254 nm, 97.2%.

Example 37

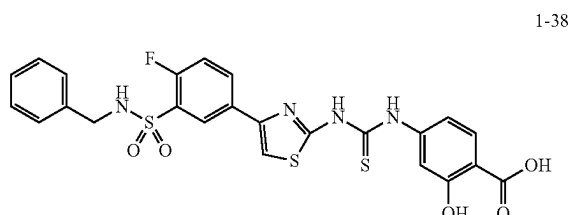
Preparation of 4-(3-(4-(3-(N-cyclopentylsulfamoyl)-4-fluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-37)



[0157] Compound 1-37 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.13 (s, 1H), 11.43 (s, 1H), 10.78 (s, 1H), 8.36 (d, J=6.8 Hz, 1H), 8.19 (ddd, J=8.8, 4.6, 2.4 Hz, 1H), 8.03 (t, J=7.6 Hz, 1H), 7.82-7.65 (m, 3H), 7.61-7.46 (m, 1H), 7.14 (s, 1H), 3.55 (q, J=7.3 Hz, 1H), 1.69-1.49 (m, 4H), 1.45-1.30 (m, 4H). LCMS (ESI) 537 [M+H]⁺, 535 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 38

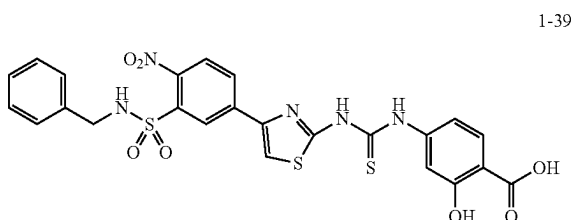
Preparation of 4-(3-(4-(3-(N-benzylsulfamoyl)-4-fluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-38)



[0158] Compound 1-38 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.12 (s, 1H), 11.42 (s, 1H), 10.80 (s, 1H), 8.57 (q, J=6.2 Hz, 1H), 8.28 (s, 1H), 8.13 (s, 1H), 7.82-7.64 (m, 3H), 7.51-7.41 (m, 1H), 7.27-7.03 (m, 6H), 4.15 (d, J=6.2 Hz, 2H). LCMS (ESI) 559 [M+H]⁺, 557 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 39

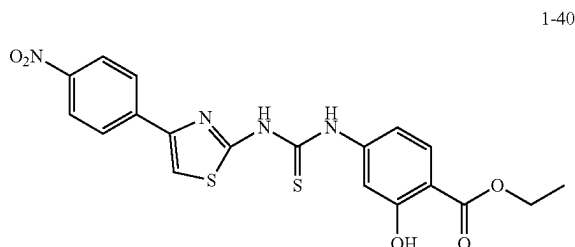
Preparation of 4-(3-(4-(3-(N-benzylsulfamoyl)-4-nitrophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoic acid (Compound 1-39)



[0159] Compound 1-39 was obtained following the procedure used to prepare compound 1-5. ¹H NMR (400 MHz, DMSO-d₆) δ 12.17 (s, 1H), 11.41 (s, 1H), 10.77 (s, 1H), 8.57 (t, J=6.2 Hz, 1H), 8.36 (d, J=1.9 Hz, 1H), 8.24 (dd, J=8.3, 2.0 Hz, 1H), 8.03 (dd, J=8.3, 4.3 Hz, 1H), 7.88 (s, 1H), 7.81-7.63 (m, 2H), 7.28-7.06 (m, 6H), 4.24 (d, J=6.2 Hz, 2H). LCMS (ESI) 586 [M+H]⁺, 584 [M-H]⁻. HPLC purity at 254 nm, 97.7%.

Example 40

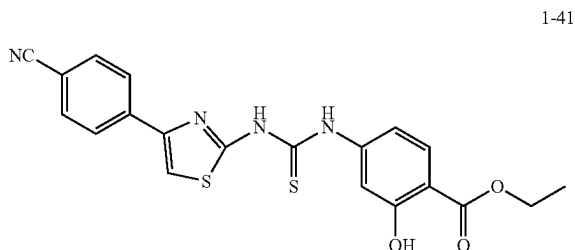
Preparation of ethyl 2-hydroxy-4-(3-(4-(4-nitrophenyl)thiazol-2-yl)thioureido)benzoate (Compound 1-40)



[0160] Compound 1-40 was obtained following the procedure used to prepare compound 1-6. ¹H NMR (400 MHz, DMSO-d₆) δ 10.74 (s, 1H), 8.38-8.28 (m, 2H), 8.21-8.11 (m, 2H), 7.92 (s, 1H), 7.79 (d, J=8.7 Hz, 1H), 7.68 (s, 1H), 7.20 (s, 1H), 4.37 (q, J=7.1 Hz, 2H), 1.35 (t, J=7.1 Hz, 3H). LCMS (ESI) 445 [M+H]⁺, 443 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 41

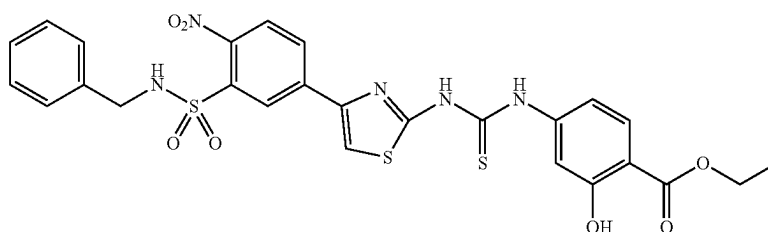
Preparation of ethyl 4-(3-(4-(4-cyanophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoate (Compound 1-41)



[0161] Compound 1-41 was obtained following the procedure used to prepare compound 1-6. ¹H NMR (400 MHz, DMSO-d₆) δ 12.15 (s, 1H), 11.01 (s, 1H), 10.74 (s, 1H), 8.13-8.05 (m, 2H), 8.00-7.73 (m, 4H), 7.68 (s, 1H), 7.20 (s, 1H), 4.37 (q, J=7.0 Hz, 2H), 1.35 (t, J=7.1 Hz, 3H). LCMS (ESI) 425 [M+H]⁺, 423 [M-H]⁻. HPLC purity at 254 nm, 99.6%.

Example 42

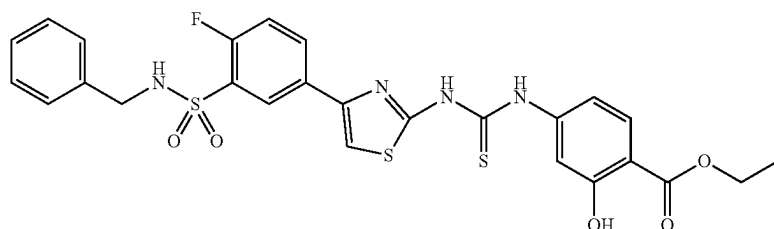
Preparation of ethyl 4-(3-(4-(3-(N-benzylsulfamoyl)-4-nitrophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoate (compound 1-42)



[0162] Compound 1-42 was obtained following the procedure used to prepare compound 1-6. ¹H NMR (400 MHz, DMSO-d₆) δ 12.16 (s, 1H), 10.74 (s, 1H), 8.57 (t, J=6.2 Hz, 1H), 8.36 (s, 1H), 8.25 (dd, J=8.4, 1.9 Hz, 1H), 8.03 (d, J=8.4 Hz, 1H), 7.90 (s, 1H), 7.79 (d, J=8.7 Hz, 1H), 7.70 (s, 1H), 7.28-7.08 (m, 6H), 4.37 (q, J=7.1 Hz, 2H), 4.24 (d, J=6.1 Hz, 2H), 1.35 (t, J=7.1 Hz, 3H). LCMS (ESI) 614 [M+H]⁺, 612 [M-H]⁻. HPLC purity at 254 nm, 99.7%.

Example 43

Preparation of ethyl 4-(3-(4-(3-(N-benzylsulfamoyl)-4-fluorophenyl)thiazol-2-yl)thioureido)-2-hydroxybenzoate (Compound 1-43)

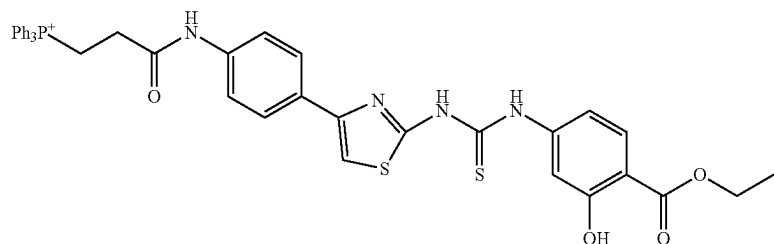


1-43

[0163] Compound 1-43 was obtained following the procedure used to prepare compound 1-6. ¹H NMR (400 MHz, DMSO-d₆) δ 12.20 (s, 1H), 10.96 (s, 1H), 10.75 (s, 1H), 8.56 (t, J=6.3 Hz, 1H), 8.28 (s, 1H), 8.13 (ddd, J=7.9, 4.3, 2.2 Hz, 1H), 7.86-7.65 (m, 3H), 7.46 (t, J=9.3 Hz, 1H), 7.30-7.04 (m, 6H), 4.37 (q, J=7.1 Hz, 2H), 4.16 (d, J=6.2 Hz, 2H), 1.35 (t, J=7.1 Hz, 3H). LCMS (ESI) 587 [M+H]⁺, 585 [M-H]⁻. HPLC purity at 254 nm, 99.9%.

Example 44

Preparation of (3-((4-(2-((4-(ethoxycarbonyl)-3-hydroxyphenyl)amino)thiazol-4-yl)phenyl)amino)-3-oxopropyl)triphenylphosphonium (Compound 1-44)



1-44

[0164] Compound 1-44 was obtained following the procedure used to prepare compound 1-6. ¹H NMR (400 MHz, DMSO-d₆) δ 10.87 (s, 1H), 10.71 (s, 1H), 10.19 (s, 1H), 7.98-7.73 (m, 18H), 7.64-7.52 (m, 3H), 7.37 (s, 1H), 7.14 (dd, J=8.8, 2.2 Hz, 1H), 4.35 (q, J=7.1 Hz, 2H), 4.00-3.83 (m, 2H), 2.84-2.71 (m, 2H), 1.35 (t, J=7.1 Hz, 3H). LCMS (ESI) 673 [M+H]⁺, 671 [M-H]⁻. HPLC purity at 254 nm, 99.8%.

INCORPORATION BY REFERENCE

[0165] The entire disclosure of each of the patent documents and scientific articles referred to herein is incorporated by reference for all purposes. The following references are herein incorporated by reference in their entireties:

[0166] Dankort, D., Curley, D. P., Cartledge, R. A., Nelson, B., Karnezis, A. N., Damsky, W. E., Jr., You, M. J., DePinho, R. A., McMahon, M., and Bosenberg, M. (2009). Braf(V600E) cooperates with Pten loss to induce metastatic melanoma. *Nat Genet* 41, 544-552.

[0167] Du, J., Zhou, Y., Su, X., Yu, J. J., Khan, S., Jiang, H., Kim, J., Woo, J., Kim, J. H., Choi, B. H., et al. (2011).

SIRT5 is a NAD-dependent protein lysine demalonylase and desuccinylase. *Science* 334, 806-809.

[0168] Gaspar, N., Hawkins, D. S., Dirksen, U., Lewis, I. J., Ferrari, S., Le Deley, M. C., Kovar, H., Grimer, R., Whelan, J., Claude, L., et al. (2015). Ewing Sarcoma: Current Management and Future Approaches Through Collaboration. *J Clin Oncol* 33, 3036-3046.

[0169] Giblin, W., Skinner, M. E., and Lombard, D. B. (2014). Sirtuins: guardians of mammalian healthspan. *Trends Genet* 30, 271-286.

[0170] Guetschow, E. D., Kumar, S., Lombard, D. B., and Kennedy, R. T. (2016). Identification of sirtuin 5 inhibitors

by ultrafast microchip electrophoresis using nanoliter volume samples. *Analytical and bioanalytical chemistry* 408, 721-731.

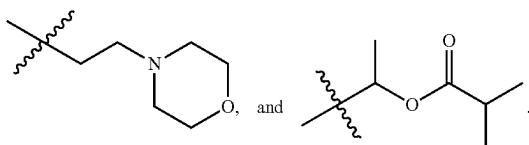
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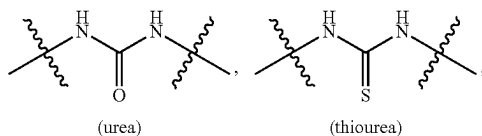
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- the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.
- What is claimed is:
1. A compound described by Formula (1), Formula (2), or Formula (3) are provided:
- (Formula 1)
-
- (Formula 2)
-
- or
- (Formula 3)
-
- including pharmaceutically acceptable salts, solvates, PROTACS, and/or prodrugs thereof; wherein for R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound capable of inhibiting and/or degrading sirtuin activity.
2. The compound of claim 1, wherein the sirtuin activity is one or more of SIRT1 activity, SIRT2 activity, SIRT3 activity, SIRT4 activity, SIRT5 activity, SIRT6 activity, and SIRT7 activity.
3. The compound of claim 1, wherein R1, R2, R3, A, B and E independently include any chemical moiety that permits the resulting compound capable of one or more of:
- inhibiting and/or degrading sirtuin (e.g., SIRT5) related desuccinylase activity;
 - inhibiting and/or degrading sirtuin (e.g., SIRT5) related demalonylase activity;
 - inhibiting and/or degrading sirtuin (e.g., SIRT5) related deglutarylase activity; and
 - influencing multiple cellular pathways related to sirtuin activity (e.g., SIRT5) such as ammonia detoxification, fatty acid oxidation, cellular respiration, ketone body formation, tricarboxylic acid cycle (TCA), glycolysis and reactive oxygen species (ROS) metabolism.
4. The compound of claim 1, wherein R¹ is selected from the group consisting of hydrogen, alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, aryl, heteroaryl, arylalkyl, and heteroarylalkyl.
5. The compound of claim 1, wherein R¹ is selected from the group consisting of hydrogen, methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, pentyl,

EQUIVALENTS

[0207] The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims rather than by



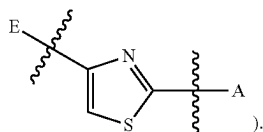
6. The compound of claim 1, wherein A is selected from the group consisting of NH, CH₂, O, S,



amide, and sulfonamide.

7. The compound of claim 1, wherein B is an aromatic ring.

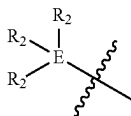
8. The compound of claim 1, wherein B is a thiazole ring (e.g.



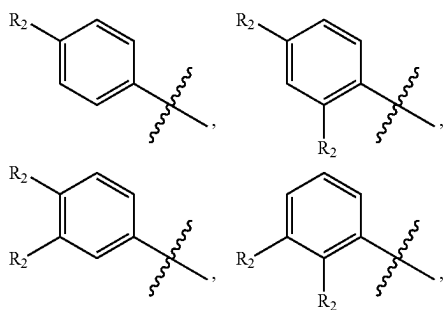
9. The compound of claim 1, wherein E is an aromatic ring.

10. The compound of claim 1, wherein E is a benzene ring.

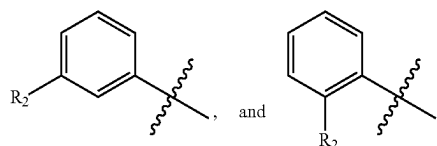
11. The compound of claim 1, wherein



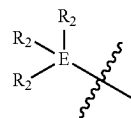
is selected from:



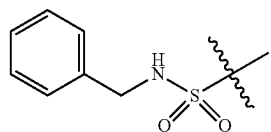
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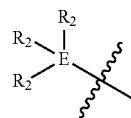
12. The compound of claim 1, wherein each R² within



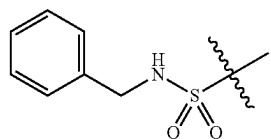
is independently selected from the group consisting of hydrogen, halogen (e.g., F, Cl, Br, I), alkyl, cycloalkyl, heteroalkyl, heterocycloalkyl, halogen, aryl, heteroaryl, arylalkyl, heteroarylalkyl, nitril, cyano, amide or sulfonamide,



and triphenylphosphine (TPP) attached with linker; and/or wherein each R² within



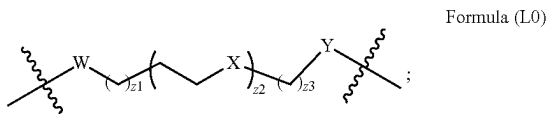
is independently selected from the group consisting of hydrogen, halogen, CF₃, OCH₃, OH, NO₂ (nitryl), CN (cyano), amide (e.g., NHC(O)CH₃), and sulfonamide (e.g., S(O₂)NHCH₃),



and triphenylphosphine (TPP) group attached with a linker.

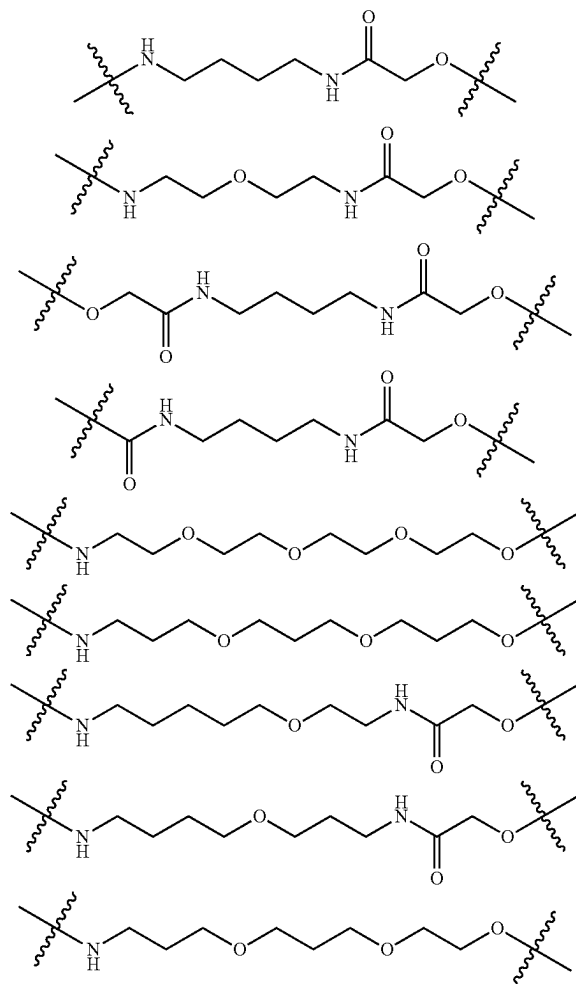
13. The compound of claim 12, wherein the linker is covalently bonded to connect two parts; wherein the carbon atom in the linear chain can be substituted with oxygen, nitrogen, sulfur, ester, and amide.

14. The compound of claim 12, wherein the linker is of following Formula (L0):

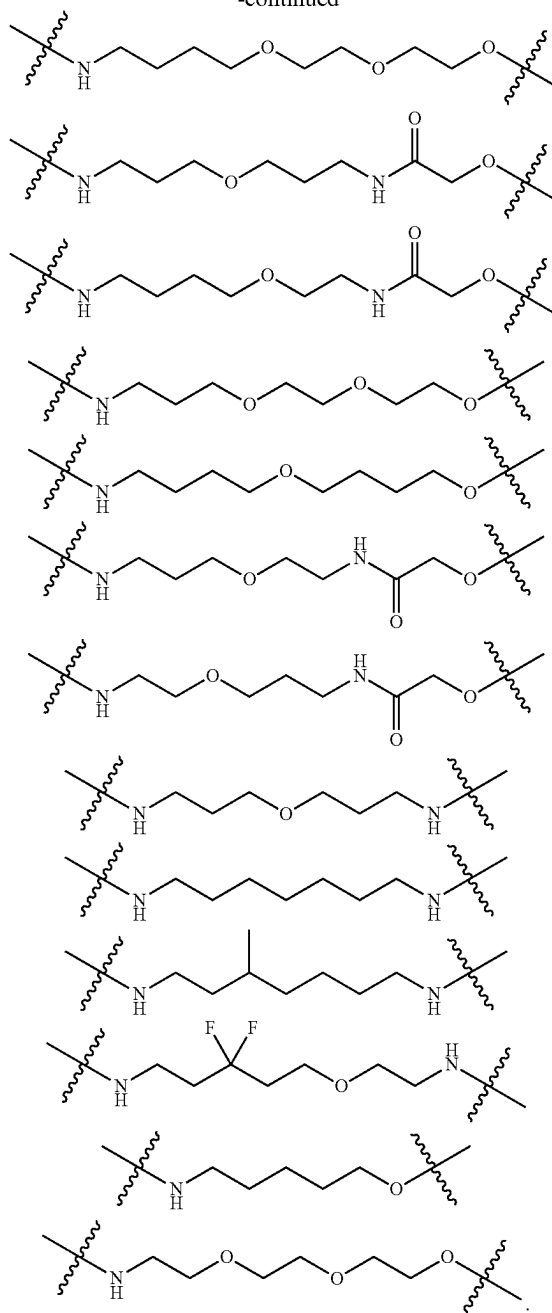


or an enantiomer, diastereomer, or stereoisomer thereof, wherein z1 is an integer selected from 0 to 10; z2 is an integer selected from 0 to 10; z3 is an integer selected from 0 to 10; each X is independently absent, CH₂, O, S, NH, NR₁₃; wherein W is selected from absent, O, NH, NR₁₃, —OCH₂C(O)NH—, —CH₂CH₂C(O)NH—, —CH₂C(O)NH— or —C(O)NH—; wherein Y is absent, O, NH, NR₁₃, —OCH₂C(O)NH—, —CH₂CH₂C(O)NH—, —CH₂C(O)NH— or —C(O)NH—; and each R₁₃ is independently C₁-C₃ alkyl. The linker can be substituted with an alkyl, halide, phenyl, benzyl, aryl, alkylene or heterocycle group.

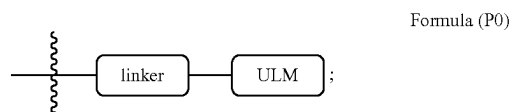
15. The compound of claim 12, wherein the linker is one of the following moieties:



-continued



16. The compound of claim 1, wherein R² is represented by Formula (P0):



wherein the linker is covalently bonded to connect two parts; wherein the carbon atom in the linear chain can be substi-

tuted with oxygen, nitrogen, sulfur, ester, and amide; wherein ULM represents an E3 ubiquitin ligase binding moiety that binds E3 ubiquitin ligase selected from the group consisting of pomalidomide, thalidomide, lenalidomide, Von Hippel-Lindau (VHL), inhibitors of apoptosis proteins (IAP), Cereblon, and mouse double minute 2 (MDM2).

17. The compound of claim **1**, wherein R3 is hydrogen or OH.

18. The compound of claim **1**, wherein the compound is selected from one of the compounds recited in Table I.

19. A pharmaceutical composition comprising a compound recited in claim **1**.

20. A method of treating, ameliorating, or preventing a disorder related to sirtuin (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7) activity in a patient comprising administering to said patient a therapeutically effective amount of the pharmaceutical composition of claim **18**.

21. The method of claim **20**, wherein said disorder related to sirtuin (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7) activity is cancer.

22. The method of claim **21**, wherein the cancer is one or more of melanoma, Ewing sarcoma, malignant peripheral nerve sheath tumor, and non-small cell lung cancer.

23. The method of claim **20**, wherein said patient is a human patient.

24. The method of claim **20**, further comprising administering to said patient one or more agents for treating the disorder related to sirtuin (e.g., SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7) activity.

25. The method of claim **24**, wherein the agents comprise anticancer agents, wherein said anticancer agent one or more of a chemotherapeutic agent, and radiation therapy.

* * * * *